# What Connects Local Housing Markets? Regulation, Migration and the National Cycle

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**Abstract**

Housing cannot be traded across regions, yet U.S. house prices have become far more synchronized. I estimate directed networks of house-price forecast-error dependence for the 48 contiguous states and the 100 largest metropolitan areas, 1985–2025, with and without the common factor of each ten-year window. The rise in connectedness is almost entirely that factor, which has grown from a fifth to two-thirds of the variance of state price growth, and dependence beyond it has not grown. Beyond the national cycle, land-use regulation lines up with how much dependence runs through a market and migration with where it comes from. A one-standard-deviation higher regulation index is associated with 7.5 log points more of other states' forecast-error variance attributed to a state. Regulated metropolitan areas take more of their variance from other areas where their own labor demand is rising, as a binding supply constraint implies, and which out-of-state areas that dependence comes from lines up with migration links.

**JEL:** R31, R23, C32, C33 **Keywords:** house prices, connectedness, migration networks, spatial equilibrium, housing supply

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# 1 Introduction

Housing is the textbook nontraded good. A house in Phoenix cannot be shipped to San Diego, and each market's prices are set by its own buyers and its own supply of homes. Yet U.S. house prices have become far more synchronized. In the ten-year windows ending in the late 1980s a single common factor accounts for about a fifth of the variance of a typical state's monthly house-price growth. In those ending since 2010 it accounts for two-thirds. Factor and comovement studies of state and metropolitan prices document the same rise (Del Negro and Otrok, 2007; Kallberg et al., 2014; Cotter et al., 2015; Fairchild et al., 2015; Tidwell et al., 2023). A common factor of that size can hide very different structures underneath it. Markets may still be tied to particular other markets, through the households who move between them, or they may answer to one national cycle and little else. This paper asks what connects local housing markets once their common cycle is taken out.

I measure those connections as a directed network of forecast-error dependence and ask which characteristics of a market they line up with. For every pair of markets the network records how much of the target's house-price forecast-error variance is attributed to price surprises in the source. Each market therefore has two sides. Its receiving side is its *external connectedness* (FROM), the share of its own forecast-error variance attributed to other markets. Its sending side is its TO connectedness, the share of other markets' forecast-error variance attributed to it. How its external connectedness divides across particular sources is its *allocation*. The national cycle complicates all three. A market whose price moves closely with a common factor shares variation with every other market, and the decomposition credits that shared variation to both sides of its position. I therefore estimate every network twice. The *raw* network is estimated on house-price growth as observed. The *factor-adjusted* network is estimated on growth from which each window's first principal component has been removed, and it measures dependence beyond the national cycle.

The networks are Diebold–Yilmaz connectedness networks estimated in rolling windows (Diebold and Yilmaz, 2012, 2014). In each 120-month window an elastic-net vector autoregression of order two in monthly house-price growth supplies the joint dynamics and the innovation covariance. The generalized forecast-error variance decomposition of its moving-average representation at a twelve-month horizon gives the connectedness matrix, with sources in columns, targets in rows and each row normalized to 100. I estimate one system for the 48 contiguous states and one for the 100 largest metropolitan areas, which are closer to the markets in which supply constraints bind. No migration, distance or regulation measure enters the autoregression. Those variables meet the estimated networks in a second stage. The second stage uses the annual networks from 1985 through 2025 for a market's position and those from 2005 for the allocation of its dependence, because migration exposure is pooled from IRS flows observed before each window begins.

The raw networks describe a strongly and increasingly connected system. Systemwide connectedness, the average share of a state's forecast-error variance attributed to other states, rises from about 68 percent in the late 1980s to about 90 percent since 2020. States differ far more on the

sending side than on the receiving side. Between the 10th and 90th percentiles, TO connectedness runs from 32 to 135 percentage points and external connectedness only from 60 to 93. Bilateral dependence falls with distance, by less over time. Almost all of this is the common factor. With each window's first principal component removed, systemwide connectedness stays near 65 percent throughout, the distance gradient no longer flattens, and the spread of TO connectedness across states falls by about two-fifths. Dependence orthogonal to the national cycle has not grown. The purge cannot say whether the factor's growth reflects common shocks or propagation among markets. What remains beyond it is uneven across markets and still falls with distance, and the question becomes what it lines up with.

Housing supply is the first place to look. Where building responds little, a demand shift raises prices rather than the stock (Glaeser et al., 2008; Saiz, 2010; Paciorek, 2013). The location-choice model of Schubert (2024) organizes the comparisons. Households choose among destinations, so a price rise in one market redirects demand toward the others its households consider. The weight a target's price places on a source's price is then the product of two terms. A supply multiplier sets how much the target's price moves with other markets, and it rises as the target's housing supply becomes less elastic. Migration exposure sets which markets it moves with, and it is built from where the target's residents came from and where each origin's residents move. The model has two predictions. A constrained target should take more of its dependence from elsewhere. Migration should order a target's sources while its regulation should not, because the multiplier scales every weight in the target's row and cancels from the shares. Durable housing sharpens the first prediction. A supply constraint binds when demand rises, while durable housing limits the response to falling demand in every market (Glaeser and Gyourko, 2005). A constrained target should therefore take more of its dependence from elsewhere where its own demand rises and not where it falls. A constraint may also put a market's own demand shocks into its price, so that other markets respond to it. The model does not derive this sending side, and I read it informally.

Beyond the national cycle, more of other states' forecast uncertainty is traced to tightly regulated states. A one-standard-deviation higher Wharton Residential Land Use Regulatory Index (WRLURI, 2006 survey) is associated with 7.5 log points more TO connectedness in the factor-adjusted network, with a 95-percent interval of $[2.7, 12.7]$, against 10.5 in the raw network. A state at the 90th percentile of regulation accounts for about 20 percent more of other states' forecast-error variance than one at the 10th, and regulated states sit closer to being net sources. Size and natural amenities are associated with a larger sending side in the raw network but not in the factor-adjusted one. Regulation is associated with it in both. The association holds under alternative treatments of the autoregression's own lags, with Census-region effects, with the 2018 vintage of the index, before and after 2005, and without any single state or region. It is weakest in the windows since 2020, when the national cycle leaves least room for differences between states. It survives removing two common factors but not the larger number an information criterion selects, so part of it may be regulated states' position within regional clusters.

The receiving side separates the national cycle from a binding constraint. In the raw network regulated states take more of their forecast-error variance from elsewhere, partly because their prices track the national factor more closely. The first principal component accounts for 8 percentage points more of their growth variance per standard deviation of the index. That is what one expects where national conditions show up in prices rather than in building (Glaeser et al., 2008; Guren et al., 2021). Whether an average receiving-side gradient remains beyond the cycle depends on how the autoregression shrinks each state's own lags, and I leave it open. The timing of demand gives a sharper test. Across the 100 metropolitan areas I measure demand without the price data, as labor demand predicted from each area's industry mix (Bartik, 1991). Beyond the national cycle the regulation gradient of external connectedness rises with that shock, by 0.76 log points per standard deviation (0.21). Larger own shocks are the alternative for regulated places that are also productive and high-amenity (Davidoff, 2016; Diamond, 2016). They would lower external connectedness whatever their sign, so the slopes above and below the year's average shock would sum to zero. They sum to 1.42 (0.43). The slope is 1.24 (0.37) above the average and 0.19 (0.41) below it. The data cannot tell a kink at the average from a gradient that rises steadily with relative demand. The interaction keeps its sign within areas over time, 0.69 (0.28), and on an apportioned node set that gives each of the 48 states at least one of 100 areas, where it loses precision. It rests on regulation. The share of land made unavailable by steep slopes, the steep land share, shows no such ordering.

Migration lines up with which markets a market's dependence comes from. I build migration exposure from IRS flows measured before each window begins, as the model defines it for the states and from flows among movers only for the metropolitan areas. Among metropolitan areas in different states, beyond the national cycle and with geography and source-by-year and target-by-year effects held fixed, a pair whose exposure is one percentage point above an equal split carries 0.06 points more of the target's external connectedness. That is about 8 percent of the average share of such pairs. The direct part of exposure carries the association. The part that runs through shared origins is too imprecise to include or exclude. The estimate loses precision when nearby pairs are dropped, and it holds on the apportioned node set. The target's regulation does not change the ordering, as the model predicts. Across states migration adds nothing to geography. At that scale exposure is close to a gravity function of distance, contiguity and region, and once geography and source-by-year effects are held fixed together the association disappears. Within pairs the data cannot say whether migration makes dependence run more in one direction than the other.

These results connect two regularities about supply-constrained housing markets. Their prices move more with common cycles. A city's sensitivity to its region's house-price cycle is loosely correlated with inelastic supply (Guren et al., 2021), cities with more migration exposure to constrained markets have higher house-price betas (Schubert, 2024), and the national factor explains more of prices in high-wage, high-price states (Tidwell et al., 2023). And price shocks in them are followed by larger movements elsewhere. In a global VAR of 31 states, shocks in land-scarce California spread across the country while those in Texas do not (Vansteenkiste, 2007). West Coast

markets recur as leading sources in connectedness rankings (Miao et al., 2011; Chiang and Tsai, 2016; Tsai and Lin, 2019; Roy Chowdhury et al., 2023). Both regularities rest on the finding that inelastic supply amplifies local booms and busts (Glaeser et al., 2008; Huang and Tang, 2012; Paciorek, 2013). In an estimated spatial-equilibrium model a California productivity shock raises local prices more and draws fewer migrants than one in Texas (Cun and Pesaran, 2022). The designs behind them do not separate a directed dependence from a shared cycle, and the distinction turns out to matter. The raw receiving side of constrained markets is partly the common cycle and depends on how the network treats own lags. Their sending side survives its removal in every own-lag network.

Most evidence on what links housing markets either relates market characteristics to symmetric comovement or fixes the links and estimates how strongly prices move along them. Comovement has been related to regulation, climate, income and supply (Kim and Rous, 2012; Cohen et al., 2024; Hernández-Murillo et al., 2017). Pair correlations and price responses have been related to banking integration, economic proximity, social ties, corporate networks and migration (Landier et al., 2017; Zhu et al., 2013; Bailey et al., 2018; Song, 2025; Sinai and Souleles, 2013; Cohen et al., 2016). Spatial models impose contiguity and find diffusion that survives common factors and lasts for years (Holly et al., 2010; Brady, 2014). Schubert (2024) is the sharpest case. His instrument, wage shocks in other cities interacted with their land constraints, identifies how strongly a destination's prices respond along migration links fixed at a 1990–1995 baseline. Diebold–Yilmaz applications to housing estimate directed structure without imposing it. Most of them map where dependence is high or how its level moves with the economy, not what it lines up with across markets (Antonakakis et al., 2018; Tsai and Lin, 2019; Payne and Sun, 2023; Gabauer et al., 2024; Antonakakis et al., 2021; Lee and Ma, 2025). Closer to this paper, estimated pairwise spillovers among U.S. metropolitan areas have been related to migration, distance and industrial similarity (Malone, 2017), and dynamic correlations of metropolitan prices to migration in the East (Wu et al., 2017). The net spillover positions of Chinese cities have been related to their administrative status, size and income (Yang et al., 2018), and the coastal, low-elasticity states to the national cycle (Prüser and Schmidt, 2021). My raw network agrees with Gabauer et al. (2024) that the Northeast is a net receiver and New York the largest net receiver within it. They find the South rather than the West to be the main source. This paper estimates the network from prices alone. It removes the common factor before measuring the connections, as spatial and forecast-error network studies have done (Bailey et al., 2016; Barigozzi and Hallin, 2017; Huang et al., 2024). It then relates both sides of each market's position and the allocation of its dependence to regulation, labor demand and migration. The combination is what is new. Where Schubert's design fixes the links to identify a response, this one measures how supply and migration align with the network and shows which of those alignments survive the national cycle. What it does not buy is the identification that Schubert's instrument buys. Every result here is an association with measured forecast dependence.

Read together, the results give regulation and migration the different roles the model gives supply and migration. Regulation lines up with how much dependence runs through a market. It does so on the sending side across states in every own-lag network, and on the receiving side

across metropolitan areas where a market's own demand makes its constraint bind. Migration lines up with which markets that dependence comes from among metropolitan areas in different states. The associations follow the model's allocation prediction and, once durable housing is added, the timing its supply multiplier implies. They do not identify causal pass-through, and the regulation index describes a market's land-use regime together with its demand correlates. Across states the sending-side association leans more on the index's state-level components, its ratings of legislative and judicial involvement in land use, than on its survey of local rules. The one-factor purge leaves regional common structure in place, so a factor-adjusted result is structure beyond the national cycle rather than a spillover among markets. Every interval reported here conditions on the estimated networks.

## 2 Data and Measurement

I use the seasonally adjusted monthly Freddie Mac House Price Index (FMHPI) for the 48 contiguous states from January 1975 through December 2025. The index is based on repeat transactions for single-family homes financed with conforming mortgages, those purchased by Freddie Mac or Fannie Mae. It provides long and consistent state coverage but does not represent transactions outside that segment.

State-level house-price growth series provide a complete system of markets covering the United States. Alaska and Hawaii are excluded since they are detached from the mainland. States do not, however, represent single housing markets within the nation. Metropolitan areas or commuting zones are the regions most often used to study housing markets, so a second sample of the largest metropolitan areas by 2000 Census population is used as well. The two systems complement each other. The states cover the whole nation, while the metropolitan areas represent regional market characteristics better.

The penalized VAR could be estimated on several hundred metropolitan series, but small markets' repeat-sales indexes rest on few transactions, so their monthly growth is noisier, and with more and noisier series the penalty would choose among many nearly collinear lags, which makes the retained links close to arbitrary. Therefore, I use a size rule and take the 100 largest core-based statistical areas (CBSAs). They hold 63 percent of the national housing stock, and their price growth weighted by housing stock tracks the national index closely, with a correlation of 0.994 over 1976–2025 (0.998 from 1991 and 0.984 before). The node set is defined by size alone, so it overrepresents the large states and leaves some small states without a sample metropolitan area. The same price series, transformation and estimation apply to this system.[1] The metropolitan system allows asking whether the state-level associations hold across metropolitan areas and, where a state contains several sample areas, within states.

[1] See Appendix D.

### 2.1 Vector autoregression

Measuring dependence across the regional markets requires a model of their joint dynamics, and dependence need not be constant over five decades. I therefore estimate the network in rolling windows, as in the dynamic connectedness analysis of Diebold and Yilmaz (2014), so that every endpoint has its own network summarizing the months before it. The benchmark window is 120 months. When the text refers to connectedness in a year, it means the annual network, estimated on the 120-month window that ends with that year, and the window's last month is the year's endpoint. Because each network summarizes the ten years before its endpoint, movements in connectedness need not reflect developments in the endpoint month alone. The figures and the descriptive statistics use every month's window, and the regressions use the annual networks. If $P_{i,s}$ denotes the index for market $i$ in month $s$, monthly house-price growth is

$$y_{i,s} = \Delta \log P_{i,s}, \tag{1}$$

where log is the natural logarithm. Collect the growth rates of month $s$ in the $N \times 1$ vector $\mathbf{y}_s$, where $N$ is 48 for the states and 100 for the metropolitan areas. Within each estimation window, I use a vector autoregression (VAR) of order $K$,

$$\mathbf{y}_s = A_0 + \sum_{\ell=1}^{K} A_\ell \mathbf{y}_{s-\ell} + \mathbf{u}_s, \qquad E(\mathbf{u}_s) = 0, \quad E(\mathbf{u}_s \mathbf{u}_s') = \Sigma. \tag{2}$$

Here $A_0$ is an intercept vector, $A_\ell$ contains the lag-$\ell$ relationships among markets, and $\mathbf{u}_s$ is the reduced-form disturbance. Following the modern Diebold–Yilmaz formulation, the VAR is a flexible dynamic approximation used to construct forecast-error variance shares (Buchwalter et al., 2026). Its innovations are disturbances in this fitted model, and their covariance $\Sigma$ is estimated from the fitted residuals. Because the number of lagged coefficients is large relative to a rolling sample, I estimate each equation with an elastic-net penalty, which shrinks and selects lag coefficients and leaves the intercept unpenalized. No migration weight, distance or regulation measure enters the VAR or influences which links the estimator retains, so every comparison that follows sets an object estimated from prices alone beside measures built independently of it.

The elastic-net approach follows Demirer et al. (2018), and the monthly lag order, the window length and the penalty rule are choices for this application. The benchmark is a VAR(2) estimated over rolling 120-month windows, with a mixing weight of 0.5 giving equal weight to the lasso and ridge penalty components. Penalties are selected equation by equation by hv-block cross-validation. The 120 months of a window are cut into twelve contiguous blocks of ten, and each block in turn is held out. The months on either side of the held-out block are dropped from the training sample, as many as the largest lag in the equation, so that no training observation shares a lag window with a validation observation. The penalty is the one that minimizes validation loss. Because the fitted residuals are serially correlated, the condition under which Bergmeir et al. (2018) show plain $K$-fold validation to be valid for autoregressions fails. Blocked folds with a buffer around each validation

block follow the $h$-block design of Burman et al. (1994) and its hv-block refinement (Racine, 2000), and Cerqueira et al. (2020) compare such designs for time series. The own lags of each equation carry the same penalty as the cross lags, and none of the lag order, the minimum-loss penalty rule and the equal treatment of own and cross lags optimizes residual whiteness.[2]

Forecast-error variance shares are computed from the moving-average representation of the fitted VAR. The twelve-month decomposition uses only its first twelve coefficient matrices, which exist for any fitted window, but the representation converges, and the fitted system is covariance stationary, only when the system is stable, meaning that every eigenvalue of its companion matrix lies inside the unit circle (Lütkepohl, 2005). Every statistic the paper reports therefore uses stable endpoints only. Selecting at the minimum of the validation loss buys network detail at a cost in stability, and I report the cost rather than remove it.[3] All four benchmark networks, raw and factor-adjusted for the states and for the metropolitan areas, are dynamically stable in every year from 1985 through 2025, and the largest companion-matrix modulus of the raw state network is 0.978. The monthly figures also draw the endpoints at which a network is not stable, six in the raw state network (March through June 2004 and March and April 2005), two in the factor-adjusted state network (March and April 2004), four in the raw metropolitan network (March through June 2004) and none in the factor-adjusted metropolitan network. The windows ending in those months cover most of the run-up to the housing boom's peak, and the connectedness drawn for them lies close to that of the neighboring months, but no reported statistic uses them.

## 2.2 GFEVD and connectedness measures

The fitted VAR supplies the dynamics and innovation covariance needed to measure dependence. For a stable window with fitted mean $\bar{\mathbf{y}}$, its moving-average representation is $\mathbf{y}_s - \bar{\mathbf{y}} = \sum_{h=0}^{\infty} \Theta_h \mathbf{u}_{s-h}$, where $\Theta_0 = I$ and the remaining $\Theta_h$ matrices follow from the VAR coefficients. The generalized $H$-step forecast-error variance decomposition (GFEVD) assigns part of a target market's forecast uncertainty to an innovation in a source market (Koop et al., 1996; Pesaran and Shin, 1998). With target $i$ in the row, source $j$ in the column and $t$ indexing the rolling endpoint, the unnormalized share is

$$\theta^{g}_{ij,t}(H) = \frac{\sigma^{-1}_{jj,t} \sum_{h=0}^{H-1} \left(\mathbf{e}_i' \Theta_{h,t} \Sigma_t \mathbf{e}_j\right)^2}{\sum_{h=0}^{H-1} \mathbf{e}_i' \Theta_{h,t} \Sigma_t \Theta_{h,t}' \mathbf{e}_i}. \tag{3}$$

[2]Section 7.1 re-estimates the state network with the own lags exempt from the penalty, at the benchmark lag order and with eight own lags, and those two alternatives accompany the state results. The metropolitan results rest on the benchmark network alone, because the metropolitan system with its own lags exempt from the penalty is unstable at many endpoints (Appendix D.2). The network diagnostics are in Appendix Table B.1. See Appendix B.1 for the reasons behind each choice.

[3]Appendix B.1 reports the stability diagnostics.

The vector $\mathbf{e}_i$ selects market $i$, and $\sigma_{jj,t}$ is source $j$'s innovation variance in $\Sigma_t$. The generalized decomposition leaves innovations correlated, so the unnormalized shares need not sum to one. Following Diebold and Yilmaz (2012, 2014), I normalize each target row to 100,

$$C_{ij,t}(H) = 100\frac{\theta^g_{ij,t}(H)}{\sum_{k=1}^{N}\theta^g_{ik,t}(H)}, \qquad \sum_{j=1}^{N} C_{ij,t}(H) = 100. \tag{4}$$

Thus $C_{ij,t}$ is the percentage-point contribution assigned to source $j$ in target $i$'s normalized forecast-error decomposition. The forecast horizon is $H = 12$ months throughout, and I suppress $H$ when no ambiguity arises. The shares are invariant to market ordering and measure reduced-form forecast dependence.

The row and column totals summarize different sides of this network. Target $i$'s *external connectedness*, conventionally called FROM connectedness, is the share assigned to all other markets. I denote it by $F_{i,t}$. Source $j$'s TO connectedness sums its contributions to other targets. NET connectedness is the difference between the two (Diebold and Yilmaz, 2012), and the Diebold–Yilmaz Connectedness Index (DYCI) averages the external row totals,

$$\begin{aligned} F_{i,t} &= \sum_{j\neq i} C_{ij,t} = 100 - C_{ii,t}, & T_{j,t} &= \sum_{i\neq j} C_{ij,t}, \\ \mathrm{NET}_{i,t} &= T_{i,t} - F_{i,t}, & \mathrm{DYCI}_t &= \frac{1}{N}\sum_i F_{i,t}. \end{aligned} \tag{5}$$

External connectedness is a market's *receiving side* and TO connectedness its *sending side*, and a market with NET below zero is a *net receiver*. NET itself cannot be logged, so the regressions use $\log \mathrm{NET}_{i,t} = \log(T_{i,t}/F_{i,t})$, which is positive for a net source and negative for a net receiver. Source *allocation* is how external connectedness divides across sources.

### 2.3 The national cycle and the factor-adjusted network

House-price growth across the regional markets shares a strong common component, and its weight has grown, as dynamic-factor and comovement studies of U.S. state and metropolitan prices have found (Del Negro and Otrok, 2007; Kallberg et al., 2014; Cotter et al., 2015; Fairchild et al., 2015; Tidwell et al., 2023). Because that literature finds the common factor central to regional house prices, the paper accounts for it and asks whether what the network measures changes once the factor is removed. In each 120-month window I take the first principal component of the standardized monthly growth rates. A network estimated on the raw series combines dependence that runs through the common factor with dependence among markets beyond it.

Removing common factors before estimating the connections among units has precedents in spatial models of house prices (Bailey et al., 2016) and in forecast-error variance networks (Barigozzi and Hallin, 2017). Here, within each window, each market's monthly growth is regressed on the window's first principal component. The vector autoregression, the penalty selection and the

decomposition are then applied exactly as above to the residuals.[4] Every design choice of the benchmark carries over, and the factor-adjusted cells and totals follow equations (4)–(5). Two properties of the purge govern how it is read. It removes one factor, and correlation among markets of the same Census region survives it, so a factor-adjusted result shows structure beyond the national cycle but cannot separate spillovers from regional common shocks. And the purge removes any within-month dependence among markets that happens to load on the first component, so the factor-adjusted network is a conservative measure of dependence beyond the cycle. Removing two factors, or the number a Bai and Ng (2002) criterion selects, is a stricter version of the same adjustment.[5] The purge also does not separate common shocks from propagation. A disturbance that reaches other markets within a month, or with a lag that the smooth repeat-sales series turn into comovement, also loads on the first component, so a growing common component can reflect propagation as well as common shocks.[6]

Every result is reported for the *raw* network and, where the text or table says so, the *factor-adjusted* one.

## 3 The Connectedness of U.S. Regional Housing Markets

Figure 1 plots the monthly DYCI of the raw networks. The state index drifts up from the high 60s in the late 1980s to about 75 in 2000 and then dips slightly during the housing boom, to 70 in late 2004. The dip is consistent with a boom that was still regionally concentrated. The first metropolitan booms occurred in northern New England and coastal California and spread from there (Ferreira and Gyourko, 2023). In 2002 and 2003 appreciation was in double digits only in the Pacific, Middle Atlantic and New England divisions. The East North Central, East South Central and West South Central divisions grew at 3.5 to 5.1 percent, and the South Atlantic division reached double-digit growth only in 2004.[7]

The index rises as the boom gives way to the Global Financial Crisis, from 71 in early 2006 to 84 at the end of 2008, over about three years and in steps. After the COVID-19 disruption it jumps, from 79 in January 2020 to 88 by July 2020, and reaches its maximum of 93.5 in April 2025. In the Global Financial Crisis prices fell, and a rise in connectedness reads as the systemic-risk signal it is in the Diebold–Yilmaz literature. After COVID-19 prices rose almost everywhere at once, so a systemic-risk reading would be a stretch, since affordability worsened but no housing crisis followed.

[4] Within the cross-validation, the component is estimated on the training months of each fold alone, so the held-out months do not enter the purge that is applied to them.

[5] See Table 5, Panel A.

[6] Appendix C documents the purge, its stability from one window to the next and the diagnostics of the factor-adjusted networks.

[7] The division growth rates are those of Office of Federal Housing Enterprise Oversight (2004). Credit expanded at the same time. Mortgage originations reached what was then a record $3.8 trillion in 2003 on a refinancing wave (Office of Federal Housing Enterprise Oversight, 2004), mortgage credit grew fastest in subprime ZIP codes from 2002 to 2005 (Mian and Sufi, 2009), and the January 2004 federal preemption of state anti-predatory lending laws for national banks raised lending in the affected states (Di Maggio and Kermani, 2017).

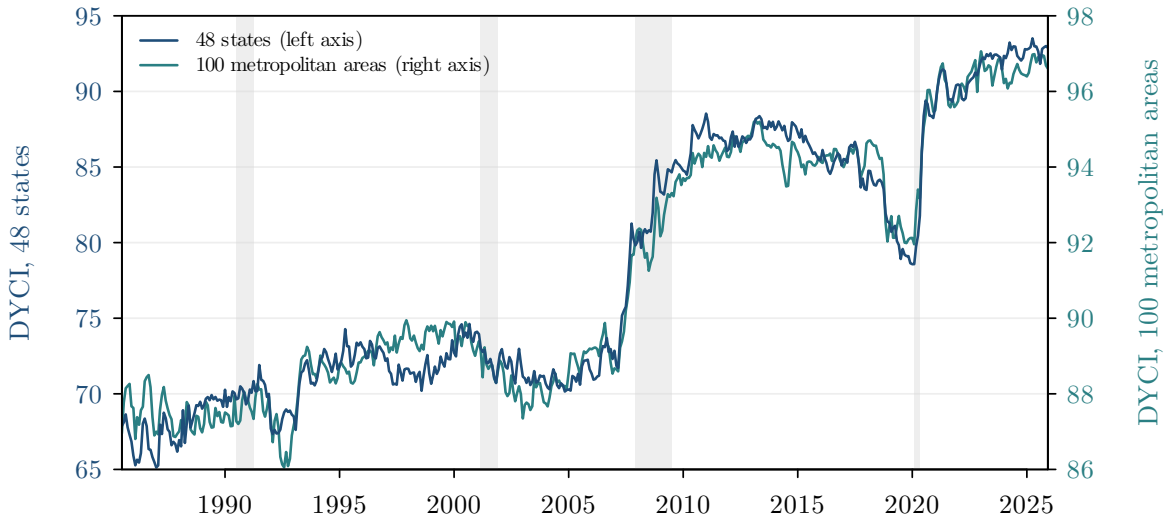



**Figure 1:** Systemwide house-price connectedness, raw networks, 48 states and 100 metropolitan areas, July 1985–December 2025

*Note:* Monthly Diebold–Yilmaz Connectedness Index (DYCI) of the 48-state system (left axis) and of the 100-metropolitan-area system (right axis) from the 120-month rolling elastic-net VAR(2) at the benchmark specification. The two axes differ, so the figure compares how the indexes move rather than their levels. Every month is drawn, including the six state windows (March–June 2004 and March–April 2005) and four metropolitan ones (March–June 2004) whose fitted companion matrix has an eigenvalue outside the unit circle. No statistic in the text uses those windows. Shaded areas denote NBER recessions. The factor-adjusted index is in Figure 4.

The same index estimated on the 100-metropolitan-area system is both higher and much steadier, 91.2 against 78.2 on average over 1985–2025, with about two-fifths of the state index's amplitude. It turns at the same times, with levels correlating at 0.985, and Figure 1 draws it on its own axis to show that the two move together. The level gap is partly mechanical, and the amplitudes of the two indexes should not be compared.[8]

The level of the index depends on how the VAR treats each equation's own lags, but its movements largely do not.[9]

### 3.1 Directional positions

The systemwide index averages over targets, and it hides how unevenly the dependence is spread across them. Because every off-diagonal cell enters one row total and one column total, external connectedness and TO connectedness have the same cross-state mean in every month, 78.2 percentage points on average over 1985–2025. Their dispersion is very different. Pooled across states and months, external connectedness has a standard deviation of 12.6 percentage points and lies between 59.7 and 92.7 from the 10th to the 90th percentile. TO connectedness has a standard deviation

[8]See Online Appendix Section OA.1.4.

[9]With the own lags exempt from the penalty the index averages 82.9 against 87.5 for the benchmark since 2010, and over 1985–2025 the two correlate at 0.985 in levels. With eight own lags it is lower still, at 72.2, and its fitted system is explosive in two-thirds of the windows ending in the 2000s, so it is not compared. See Appendix B.1 and Online Appendix Figure OA.5.

of 45.9 and lies between 32.4 and 135.2, so a state at the 90th percentile accounts for about four times as much of other states' forecast-error variance as one at the 10th. Table 1 reports the same percentiles over the annual networks. States differ far more in how much of the system's dependence is attributed to them than in how much of their own they take from elsewhere, and NET runs from −51.0 at the 10th percentile to 56.4 at the 90th. The metropolitan areas show the same asymmetry more sharply. Pooled across areas and months, external connectedness has a standard deviation of 6.0 percentage points and lies between 83.7 and 96.8 from the 10th to the 90th percentile, and TO connectedness has a standard deviation of 42.7 and lies between 41.8 and 148.8.

Averaged by Census region (Figure 2), the West is a net source, by 8.0 percentage points over 1985–2025 and 17.4 over 2005–2025. The South is close to balance, at 2.6, and the Midwest and the Northeast are net receivers, at −6.2 each.[10] Within the Northeast, New York is the largest net receiver, below zero in 91 percent of the windows and in all but one of those ending from 2005, as in the state network of Gabauer et al. (2024).[11] The most regulated states of the region, Massachusetts, Rhode Island and New Hampshire, are net sources on average. Across all 48 states, California, Georgia, Colorado, Florida and Oklahoma have the highest mean log NET, and North Dakota, New York, Iowa, Vermont and Wyoming the lowest. These are positions in the raw network, and part of their spread is the national cycle, which a market that tracks it closely shares with every other market.

### 3.2 Spatial structure

Pairwise connectedness should have a spatial structure, since nearby markets are more closely linked. Figure 3 reports its spatial gradient estimated month by month by Poisson pseudo-maximum likelihood (PPML), for each window $t$,

$$E\left[C_{ij,t} \mid i, j\right] = \exp\left(\lambda_{i,t}^{T} + \lambda_{j,t}^{S} - \beta_t \log d_{ij}\right), \qquad i \neq j, \tag{6}$$

where $d_{ij}$ is the distance between target $i$ and source $j$ and $\lambda_{i,t}^{T}$ and $\lambda_{j,t}^{S}$ are target and source fixed effects. The log link keeps the zero-valued cells and makes $\beta_t$ an inverse-distance elasticity. The target effects absorb the level of each row but not its weight, so a target with a larger external total weighs more in the estimate. Dividing each row by its external total gives every target the same weight, and the gradients estimated on those shares correlate with the ones drawn here at 0.996 or more, at both scales and in both networks.[12]

The inverse-distance elasticity is positive at every point at which the system is dynamically stable, and at every one of them the 95-percent interval excludes zero. Its mean is 0.278. Geographic proximity is therefore associated with stronger bilateral connectedness throughout the period, but by a diminishing amount, as the mean elasticity falls from 0.410 in the 1990s to 0.117 during 2020–2025.

[10] See Online Appendix Table OA.42.

[11] Their network uses real year-on-year growth for 50 states and the District of Columbia, against nominal monthly growth for the 48 contiguous states here.

[12] See Online Appendix Figure OA.2.

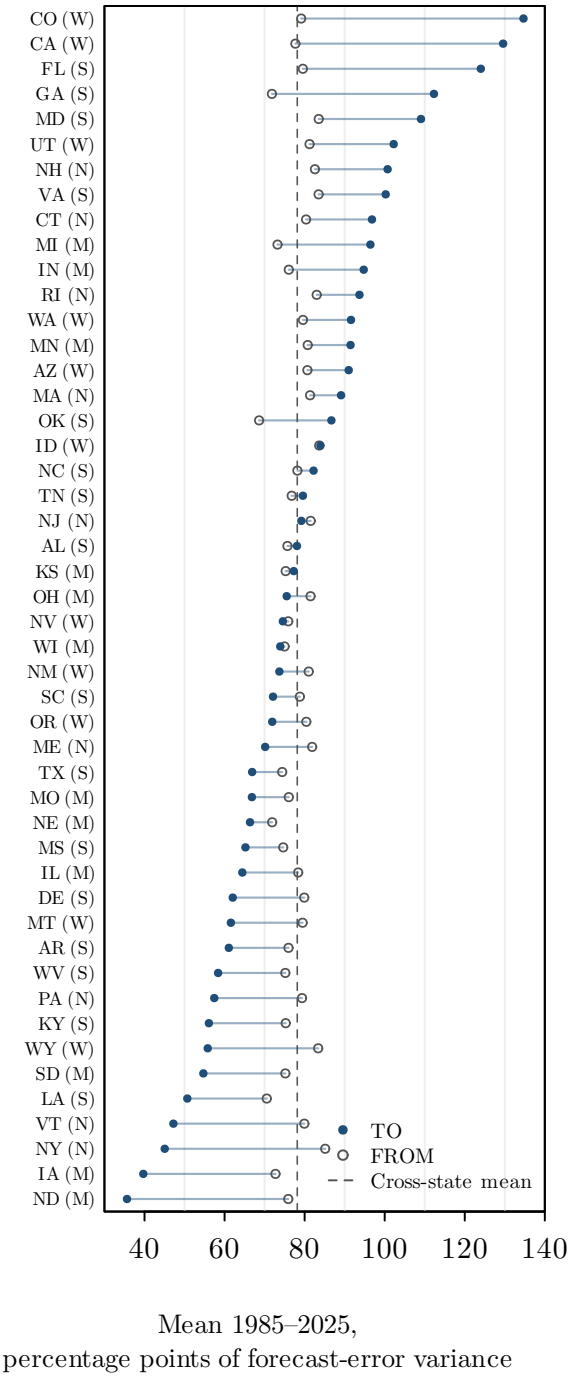



**Figure 2:** Directional positions of the 48 states, raw network, 1985–2025

*Note:* Each state's mean TO connectedness (filled) and external connectedness, FROM (open), over 1985–2025 in the raw benchmark network, in percentage points of forecast-error variance. The segment joining the two is the state's mean NET. States are ordered by their mean TO connectedness, highest at the top. The dashed line is the cross-state mean, which FROM and TO share by construction, 78.2. The letter after each state is its Census region (Northeast, Midwest, South, West). The factor-adjusted positions are drawn beside these in Figure 5.

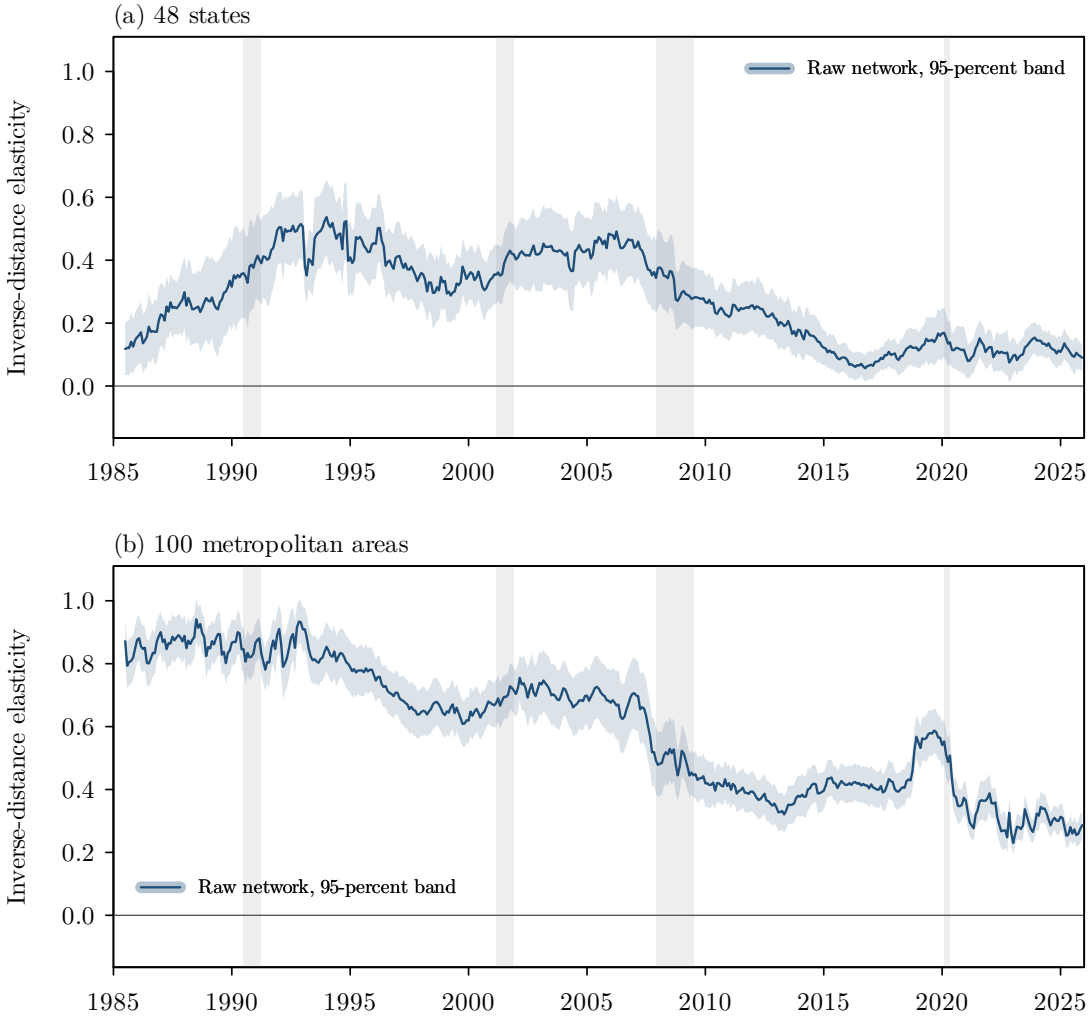



**Figure 3:** The distance gradient in bilateral connectedness, raw networks, July 1985–December 2025

*Note:* Each line reports the coefficient on negative log distance from separate Poisson pseudo-maximum-likelihood regressions of the nonnegative $H = 12$ bilateral connectedness cell, one regression per month, in the raw benchmark network. Panel (a), 48 states, uses interstate distances and panel (b), 100 metropolitan areas, distances between CBSA internal points. Every regression includes source-node and target-node fixed effects. The shaded bands are 95-percent confidence intervals based on two-way source and target clustering, and grey vertical bands denote NBER recessions. Every month is drawn. In the six state and four metropolitan windows whose fitted VAR is not stable the gradient is estimated by the same regression for the figure only, and no statistic in the text uses it. The factor-adjusted gradients are in Figure 6.

Because each of those regressions carries its own covariance matrix, that decline can be described but not tested. Re-estimating them jointly does permit a test. With source and target effects specific to each window the Poisson likelihood is separable by window, so every point estimate is numerically unchanged and the only thing gained is a single covariance matrix. Against the 1990s mean of 0.410, the mean over 2015–2025 is 0.109, a fall of 0.301 with a two-way clustered standard error of 0.045.[13] The decline is not month-to-month noise. The network with unpenalized own lags gives the same picture along a slightly flatter gradient, with a mean of 0.264, positive with an interval excluding zero in every stable window, and the same decline, from 0.348 in the 1990s to 0.125 during 2020–2025. Across the 100 metropolitan areas the gradient is about twice as steep, averaging 0.59, positive with an interval excluding zero in every month, and it declines in the same way, from 0.77 in the 1990s to 0.33 during 2020–2025, correlating 0.69 with the state gradient month by month.

The decline is a fact about house prices rather than an artifact of how the network is estimated. The correlations of monthly price growth among state pairs, computed over the same windows, line up with proximity most strongly in the mid-1990s and least in the mid-2010s, as the estimated gradient does.[14]

The raw networks therefore describe a system in which most of each market's forecast-error variance is attributed to other markets, and increasingly so. Markets differ far more on the sending side than on the receiving side, and dependence falls with distance by a diminishing amount. A common factor that every market loads on could produce each of these features, and the next section asks how much of each it accounts for.

## 4 Connectedness Beyond the National Factor

Removing the national factor changes the picture of Figure 1. Figure 4 sets the factor-adjusted index beside the raw one and the national real house price index. The factor-adjusted index averages 65.7 against 78.2 for the raw index, and it is nearly flat. It averages 66.1 in the windows ending in 1985–1989, 67.6 in the 1990s, 66.7 in the 2000s, 63.8 in the 2010s and 64.0 since 2020, while the raw index rises from 67.8 to 90.5. The gap between the two indexes grows from about 1.7 points in the late 1980s to 26.5 since 2020. The two are shares of different totals, the forecast-error variance of raw growth and of growth beyond the first component, so the gap is a difference between two indexes rather than a decomposition of one. Month by month the factor-adjusted index stays between 56.7 and 71.8, through both housing episodes in which the raw index rises, and the two

[13] See Online Appendix Section OA.1.2.

[14] The check computes, in each window and over the same 120 months, the correlation of monthly price growth for all 1,128 state pairs and asks how strongly those correlations line up with proximity. The two paths diverge twice. Over 2004–2005 the correlation-based measure averages about 0.17 while the estimated gradient stays near a second peak, and over 2022–2024 it averages about 0.32 while the estimated gradient stays near its low. Month by month the two correlate at 0.68. The periods of strongest comovement are mostly ones in which proximity explains least about which states move together, although the recovery of the correlation-based measure after 2020 qualifies that statement for the most recent windows. See Appendix Figure B.1.

indexes are weakly negatively correlated, at −0.20. The rise in systemwide connectedness since the 2000s is therefore almost entirely a rise in the component common to all markets, and dependence orthogonal to it has not grown.

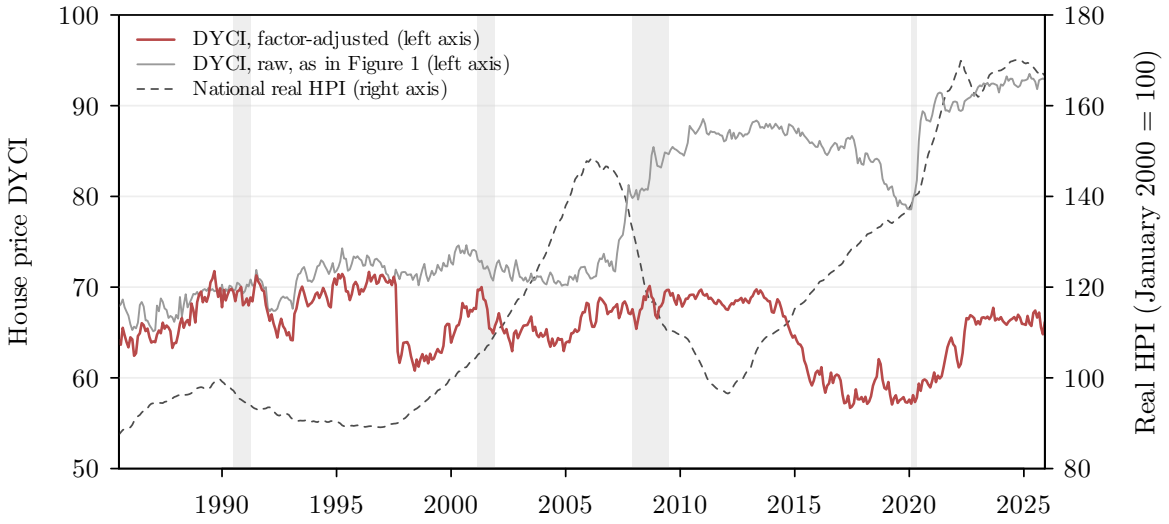



**Figure 4:** Connectedness beyond the national factor and the national real house price index, July 1985–December 2025

*Note:* The red series is the monthly Diebold–Yilmaz Connectedness Index (DYCI) of the factor-adjusted 48-state network, estimated after each window's first principal component of monthly growth is removed, and the grey series is the raw index of Figure 1. Both are read on the left axis. The dashed series is the national real house price index, normalized to 100 in January 2000 and read on the right axis. Appendix A.3 describes its construction, and its October 2025 value is missing. Every month is drawn, including those whose fitted companion matrix has an eigenvalue outside the unit circle, six in the raw network and two in the factor-adjusted one. Shaded areas denote NBER recessions.

The metropolitan system's own first component is purged in the same way across the 100 areas, and the factor-adjusted metropolitan network retains more of its connectedness than the state one. Its index averages 86.2, against 91.2 for the raw metropolitan network, stays between 81.8 and 89.1 in every month and, like the state one, does not share the rise of the raw index.[15]

## 4.1 Directional positions beyond the national cycle

The purge also narrows the directional positions of the raw network, and it narrows the sending side most. Mean external connectedness falls from 78.2 to 65.7 percentage points, so the common factor accounts for about 12 points of each state's external connectedness, while its standard deviation pooled across states and months is almost unchanged, 12.2 against 12.6, and within a year its cross-state spread widens. The standard deviation of TO connectedness falls from 45.9 to 27.0, and that of NET from 46.9 to 29.0, so the purge narrows the spread in net position by about two-fifths. A market that tracks the common factor more closely has more of its own row attributed to other markets and a larger share of their rows, and the purge removes that difference among markets

[15] See Online Appendix Figure OA.3, and Appendix D for the metropolitan networks.

along with the factor's level. At the metropolitan scale the purge narrows the sending side by less. The standard deviation of TO connectedness falls from 42.7 to 33.2 and that of NET from 41.9 to 30.3, reductions of 22 and 28 percent against 41 and 38 percent across the states, while that of external connectedness rises from 6.0 to 7.1. The ordering of states changes but survives in part (Figure 5). Across the 48 states, the rank correlation between a state's mean position in the raw and the factor-adjusted network is 0.77 for TO, 0.63 for external connectedness and 0.74 for log NET. Beyond the national cycle the largest net sources by mean log NET are Washington, Michigan, Colorado, Connecticut and New Jersey, and the largest net receivers Wyoming, North Dakota, West Virginia, New York and South Dakota.

### 4.2 The distance gradient beyond the national cycle

The factor-adjusted networks change the spatial description of the raw networks. The factor-adjusted state gradient averages 0.158, against 0.278 raw in the same months. It is positive in 98 percent of the months, with an interval excluding zero in 80 percent, and it does not decline. It averages 0.11 in the windows ending in 1985–1989, 0.18 in the 1990s, 0.22 in the 2000s, 0.10 in the 2010s and 0.14 since 2020.[16] The factor-adjusted metropolitan gradient averages 0.76, against 0.59 raw, and rises slightly, from 0.73 in the late 1980s to 0.83 since 2020, whereas the raw gradient falls from 0.86 to 0.33. The decline of the distance gradient in the raw networks is therefore associated with the growth of the national cycle, which raises the dependence of distant and nearby markets alike. Once the national factor is removed, the gradient shows no decline across states and a slight rise across metropolitan areas. Figure 6 places the factor-adjusted series beside the raw ones because the two differ materially. The raw estimate lies outside the factor-adjusted interval in about two-thirds of the months at the state scale and four-fifths at the metropolitan one, and from 2007 the raw metropolitan estimate lies below the factor-adjusted interval in all but five months.

Two limits of the purge, that it removes one factor and that it cannot separate common shocks from propagation among markets, bound how far the factor-adjusted networks can be read as structure beyond the national cycle. What the comparison does show is that dependence orthogonal to the national cycle has not grown, that markets still differ in how much of it is attributed to them, and that it still falls with distance. The next two sections ask which characteristics of a market line up with those differences, in the raw network and beyond the national cycle.

[16] Across the annual networks its linear trend, $-0.001$ a year, is not distinguishable from zero, against $-0.008$ raw, with a Newey–West interval with nine lags, because windows up to nine years apart share months. At the metropolitan scale the two annual trends have opposite signs, and the factor-adjusted interval excludes zero.

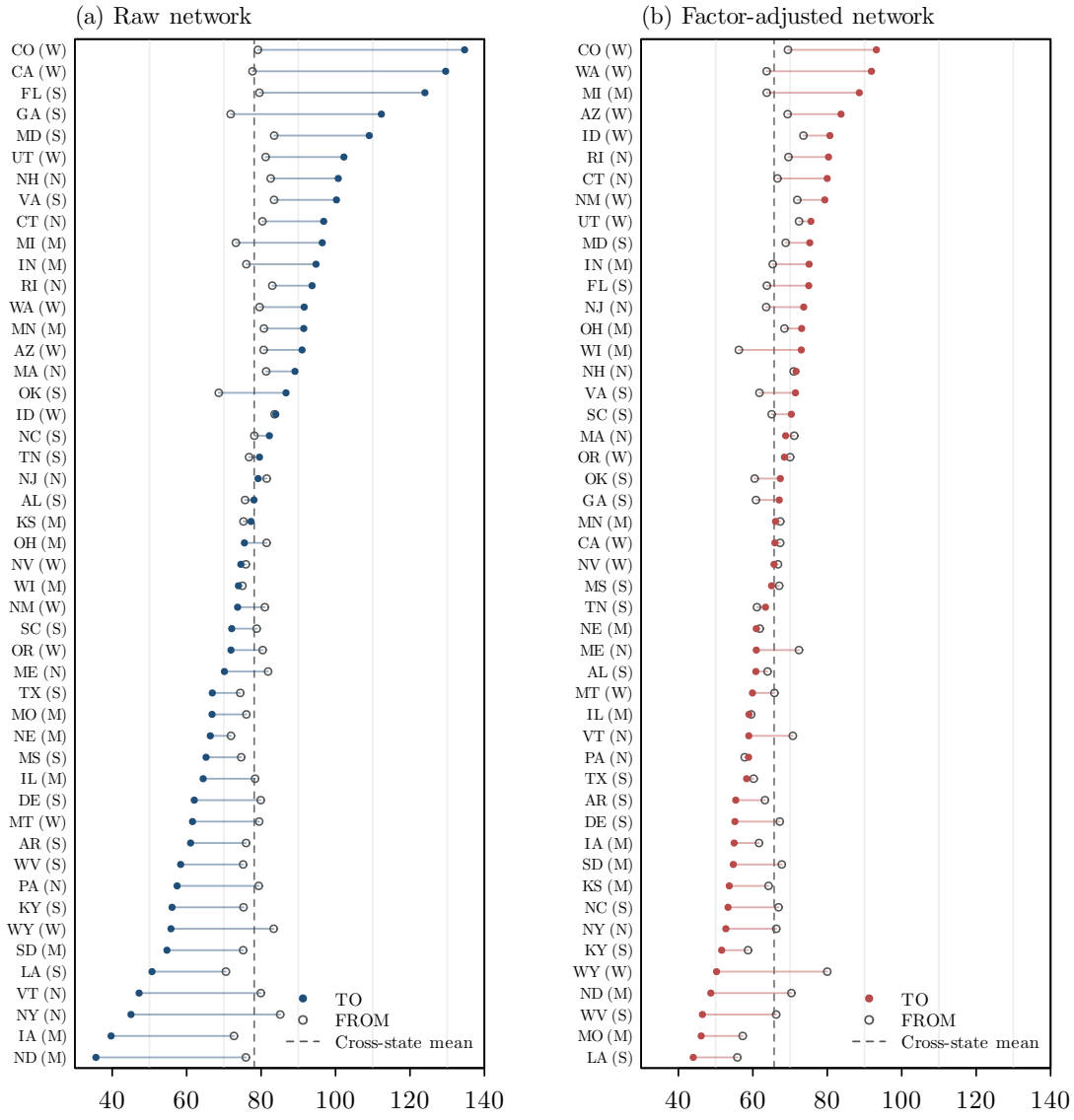



**Figure 5:** Directional positions of the 48 states, raw and factor-adjusted networks, 1985–2025

*Note:* Each state's mean TO connectedness (filled) and external connectedness, FROM (open), over 1985–2025 in the benchmark network, in percentage points of forecast-error variance, in the raw network (a) and the factor-adjusted network (b). The segment joining the two is the state's mean NET. In each panel states are ordered by their mean TO connectedness in that network, highest at the top. The two panels share one horizontal scale. The dashed line is the cross-state mean, which FROM and TO share by construction, 78.2 in the raw network and 65.7 in the factor-adjusted one. The letter after each state is its Census region (Northeast, Midwest, South, West).

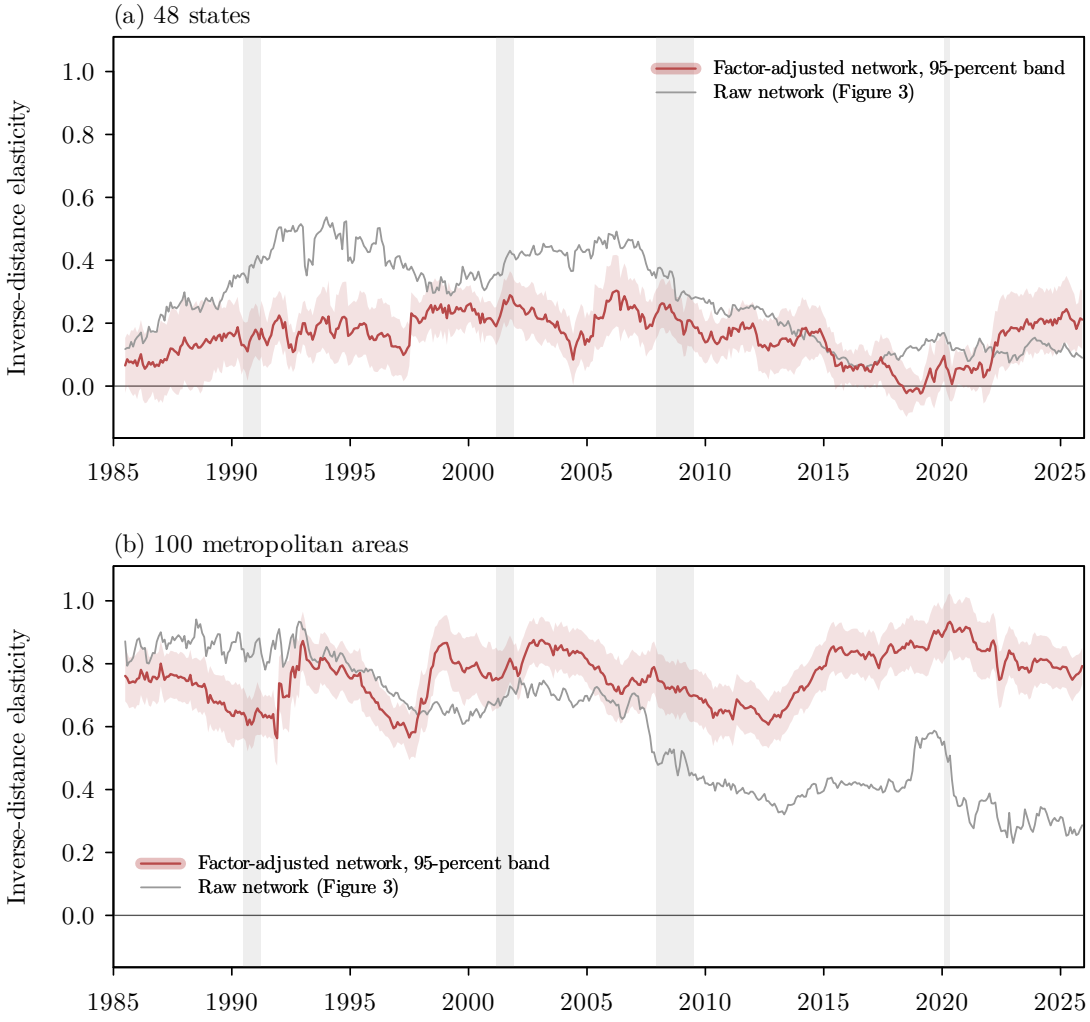



**Figure 6:** The distance gradient in bilateral connectedness, raw and factor-adjusted, July 1985–December 2025

*Note:* Each line reports the coefficient on negative log distance from separate Poisson pseudo-maximum-likelihood regressions of the nonnegative $H = 12$ bilateral connectedness cell, one regression per month, in the factor-adjusted (red, with its 95-percent band) and the raw (grey) benchmark network. The raw series is that of Figure 3. Panel (a), 48 states, uses interstate distances and panel (b), 100 metropolitan areas, distances between CBSA internal points. Every regression includes source-node and target-node fixed effects. The shaded band is a 95-percent confidence interval based on two-way source and target clustering, and grey vertical bands denote NBER recessions. Every month is drawn. In the windows whose fitted VAR is not stable, six raw and two factor-adjusted state windows and four raw metropolitan ones, the gradient is estimated for the figure only.

# 5 Housing Supply, Migration and the Empirical Design

How much of a demand shift shows up in a market's price depends on its housing supply. Where building responds little, higher demand raises prices rather than the stock, so the price of a supply-constrained market moves with the demand of the households who might live there (Glaeser et al., 2008; Saiz, 2010; Paciorek, 2013). The premise is not settled for recent decades. Across U.S. cities, measured constraints do not change how price and quantity growth respond to income growth (Louie et al., 2025), and the comparisons below ask a different question, how forecast-error dependence and its timing line up with regulation. Those households come from other markets, and migration says from which. They can connect two housing markets even when they do not move directly between them. Suppose households from the same origin commonly consider two destinations. When housing in one destination becomes more expensive, some may choose the other, and the destinations are then linked through competition for a common origin population. The location-choice model in Schubert (2024) formalizes both parts. Migration determines which markets a target's price responds to, and the target's housing-supply elasticity determines how strongly it responds.[17]

A household initially located in $k$ chooses among destinations by comparing amenities, wages, housing costs and moving costs. Let $\mu_t^{k\to j}$ denote the share of origin-$k$ households choosing destination $j$, and let $\varphi_t^{i\leftarrow k}$ denote the share of target $i$'s feeder population supplied by origin $k$. Migration-network exposure from source $j$ to target $i$ is

$$\psi_{ij,t} = \sum_{k=1}^{N} \varphi_t^{i\leftarrow k}\mu_t^{k\to j}, \qquad \sum_{j=1}^{N}\psi_{ij,t} = 1. \tag{7}$$

In the data, $\psi_{ij,t}$ is built from the pooled IRS flows of the years $t-15$ to $t-11$.[18] Exposure is high when origins important to target $i$ also assign a large share of their destination choices to source $j$. It incorporates all shared origins and therefore contains more information than the direct flow between $i$ and $j$. Geography matters because moving costs shape the destination shares, but a single distance between the two markets captures only part of the information in the migration network.

For $j \neq i$ exposure splits exactly into two direct terms and a common-origin term,

$$\psi_{ij,t} = \varphi_t^{i\leftarrow i}\mu_t^{i\to j} + \varphi_t^{i\leftarrow j}\mu_t^{j\to j} + \sum_{k\neq i,j}\varphi_t^{i\leftarrow k}\mu_t^{k\to j}. \tag{8}$$

The first term counts residents of $i$ who move to $j$, the second residents of $i$ who came from $j$, weighted by the share of $j$'s residents who stay, and the third the origins the two markets share. With stayers kept on the diagonal of the state flow matrix, as the model requires, the two direct terms dominate. Across states the common-origin term averages 1.3 percent of external exposure in the annual networks from 2005, and exposure is, to a close approximation, direct migration in both

[17] Appendix E provides the full derivation.

[18] Appendix E writes this pre-window matrix as $\psi_{ij,t}^{0}$.

directions, weighted by stayer shares.[19] The metropolitan flow matrices hold movers only, and there the common-origin term is 39 percent. Schubert (2024) builds exposure for commuting zones and drops links within 50 to 150 miles of each zone. That exclusion has no state counterpart, and at the metropolitan scale the paper uses pairs of areas in different states instead, with a radius exclusion as a check.

The diagonal $\psi_{ii,t}$ is the target's own exposure, and $o_{i,t} = 1 - \psi_{ii,t}$ measures its external migration openness. Conditional on choosing an external destination, source $j$'s migration share is $\widetilde{\psi}_{ij,t} = \psi_{ij,t}/o_{i,t}$. In the model, the weight $W_{ij,t}$ that target $i$'s price places on source $j$'s price follows from the target's inverse housing-supply elasticity $\eta_i^H$ and two common parameters, the housing expenditure share $\alpha$ and the dispersion $\nu$ of households' location tastes. With $a_i = (\alpha/\nu)\eta_i^H$,

$$W_{ij,t} = \rho_{i,t}\psi_{ij,t}, \qquad \rho_{i,t} = \frac{a_i}{1 + a_i o_{i,t}}, \tag{9}$$

where $\rho_{i,t}$, the *supply multiplier*, scales every exposure weight of target $i$ and rises with its supply constraint. Summing over sources and dividing by that sum gives the two margins the regressions follow,

$$\sum_{j \neq i} W_{ij,t} = \rho_{i,t} o_{i,t} = \frac{a_i o_{i,t}}{1 + a_i o_{i,t}} \equiv \eta_{i,t}^{nw}, \qquad \frac{W_{ij,t}}{\sum_{r \neq i} W_{ir,t}} = \widetilde{\psi}_{ij,t}. \tag{10}$$

The target's total external weight, $\eta_{i,t}^{nw}$, therefore rises when housing supply is less elastic and when the target is more open to external migration. Conditional on that total, migration shares determine how the weight is allocated across sources. Supply and migration therefore have different jobs in the model. Supply sets how much of a target's price moves with other markets, and migration sets which markets it moves with. These implications motivate separate regressions for a market's directional position and for the allocation of its external connectedness across sources.

The inverse supply elasticity $\eta_i^H$ of the model is a constant, so neither margin depends on the direction of the target's own demand. Durable housing adds that dependence. Supply is inelastic for falling demand in every market, because housing is not torn down when demand falls, but elastic for rising demand where building is unconstrained (Glaeser and Gyourko, 2005). The difference in effective $\eta_i^H$ between constrained and unconstrained targets, and with it the difference in their total external weight, should therefore appear where a target's own demand rises and not where it falls. This timing implication combines the model with durable housing and is not one of the model's comparative statics.

The theoretical weights and the GFEVD cells are not identical. Innovation covariance, indirect paths, the forecast horizon and row normalization also shape measured connectedness. I therefore use the model to organize the empirical comparisons without restricting the VAR by migration weights. In the 48-state system, in the annual networks from 2005, when migration is measured, mean external migration openness is 5.69 percent, against mean external connectedness of 85.29 percent in the raw network. Openness is an input to the model's direct external weights, whose magnitude

[19] See Online Appendix Table OA.33.

also depends on its preference and supply parameters. The much larger measured forecast-error share is therefore not a numerical estimate of those direct weights. The supply multiplier $\rho_{i,t}$ cancels from conditional model weights, so heterogeneity in measured allocation need not be a direct supply comparative static.[20]

### 5.1 Sample, timing and variables

The directional and allocation regressions use different samples because only the second needs migration data. The directional regressions use the annual networks from 1985 through 2025, the full span over which the benchmark network exists and is dynamically stable, for the states and for the metropolitan areas. Their regressors are time invariant or measured from sources that reach back to 1984, so nothing forces the regressions to start later. The allocation regressions use the annual networks from 2005 through 2025. The limit is the migration data. IRS migration flows start with the 1990 vintage, and each network's exposure is built from migration observed before its ten-year price window begins, so that no month of the window can inform the weights it is compared with. The 2005 network is the first whose window leaves room for a five-year pool of such migration.

The variables of the second stage are the empirical counterparts of the model's primitives. The supply constraint enters as the 2006 Wharton Residential Land Use Regulatory Index (Gyourko et al., 2008), with the 2018 index (Gyourko et al., 2021) used for robustness checks where needed. It also enters as the share of land made unavailable by steep slopes from Lutz and Sand (2026), called the steep land share from here on.[21] The migration network enters as annual IRS migration flows, interstate flows for the states and county-to-county flows for the metropolitan areas, from which exposure is built as follows.

I use five-year migration averages to summarize persistent migration relationships before each house-price window.[22] IRS vintages are labeled by the first year of a two-year migration interval, so the 1990 vintage records migration from 1990 to 1991. For the annual network of year $t$, I therefore pool vintages $t-15$ through $t-11$, whose final interval ends in $t-10$, before the 120-month house-price window begins in $t-9$. The earliest available five-year pool, 1990–1994, covers intervals ending in 1991–1995 and supports the 2005 network. The final pool is 2010–2014, for the 2025 network.

The outcomes are target external connectedness $F_{i,t}$ and the conditional connectedness share $\bar{C}_{ij,t}$ defined below. The target regressors are the Wharton index, the steep land share, population

[20] On the unconditional cell the multiplier does not cancel. Online Appendix Table OA.29 shows that the interaction of exposure with the target's regulation there is an accounting identity in the receiving-side gradient and the allocation slope, not separate evidence.

[21] The benchmark land measure retains whole metropolitan and micropolitan areas, including their outlying counties. Each cross-border market is allocated to states by their shares of its 1990 total housing stock, and markets are then weighted within each state by that same housing stock. The Lutz–Sand data cover locations beyond the largest metropolitan markets. Alternative weights use population and the 2000 Census, and a county-to-state aggregate provides a geographic comparison. The primary state WRLURI is the unweighted mean across responding jurisdictions. Appendix A reports coverage, timing and aggregation details.

[22] Schubert (2024) uses averages from an early baseline period to form fixed migration weights. Here, the averaging window advances with the rolling house-price window.

and, for the states, average distance. Regulation and the steep land share are standardized across the markets of each system. Population is the resident population in the calendar year before the network's year and enters in logs. It is the size control, available on a consistent basis for every year, whereas state GDP on a consistent basis does not reach back before 1997.[23] The 2006 index is measured after the early windows. The regulation measures are treated throughout as persistent characteristics of a market's land-use regime rather than as dated treatments, and the regressions compare markets. Bilateral geography consists of log distance, contiguity and a common Census-region indicator.

Table 1 reports the network outcomes and the market and pair characteristics of the state and the metropolitan regressions at their observational levels. The labor-demand shock and the controls specific to the metropolitan demand test are defined with that test.

### 5.2 Directional position

The framework speaks to both sides of a market's position. A constrained target carries a larger inward weight, because its supply multiplier scales every exposure weight in its own row. A market's outward weight, the sum of the weights other markets place on it, depends on those markets' constraints and on how many of their movers weigh it as a destination. What the direct weights imply for the measured decomposition is less definite. The shares are generalized forecast-error shares of a fitted dynamic system, so the innovation covariance, the timing of adjustment and the common cycle all shape them. A constraint that makes a market's own demand shocks show up in its price also makes those surprises larger relative to the responses of other markets. Where a regulated market sits on the sending side is therefore an empirical question.

Each directional outcome is regressed on the same target characteristics. Write $R_i$ for standardized WRLURI and $Y_{i,t}$ for one of $\log T_{i,t}$, the log of TO connectedness, $\log F_{i,t}$, the log of external connectedness, and $\log \mathrm{NET}_{i,t} = \log(T_{i,t}/F_{i,t})$. The model is

$$Y_{i,t} = \delta_t + \beta_R R_i + X'_{i,t}\gamma + \varepsilon_{i,t}, \tag{11}$$

where $\delta_t$ denotes year effects and $X_{i,t}$ contains the steep land share, lagged log population and log average distance. I estimate it by least squares and report $100\beta_R$, the difference in the outcome in log points per standard deviation of the regressor, approximately a percent difference. Because $\log \mathrm{NET} = \log T - \log F$ market by market, its coefficient on every covariate equals the TO coefficient minus the external-connectedness coefficient exactly. External connectedness is strictly positive, so the log is well defined everywhere. TO and NET are far more dispersed than FROM (Table 1), so each coefficient is also read against its outcome's own within-year dispersion.

Every directional regression is estimated twice, on the raw network and on the factor-adjusted one, with the same covariates and sample. The comparison answers the question the raw network

[23] Migration openness exists only from 2005, and entering it would confine the directional regressions to half their span. See Online Appendix Section OA.2.1.

**Table 1:** Summary statistics for connectedness, supply constraints and migration exposure

| Variable | Unit | *N* | Mean | SD | P10 | Median | P90 |
|---|---|---|---|---|---|---|---|
| *Panel A. States, target variables, 41 annual networks 1985–2025* | | | | | | | |
| External connectedness (FROM) | pp | 1,968 | 78.057 | 12.600 | 59.284 | 80.335 | 92.587 |
| TO connectedness | pp | 1,968 | 78.057 | 46.079 | 32.283 | 66.872 | 136.490 |
| Net difference, TO − FROM | pp | 1,968 | 0.000 | 47.162 | -51.156 | -6.959 | 56.350 |
| External connectedness (FROM), factor-adjusted | pp | 1,968 | 65.575 | 12.437 | 49.334 | 65.363 | 82.268 |
| TO connectedness, factor-adjusted | pp | 1,968 | 65.575 | 26.710 | 37.153 | 60.111 | 99.947 |
| Net difference, TO − FROM, factor-adjusted | pp | 1,968 | 0.000 | 28.847 | -34.207 | -2.210 | 35.575 |
| log FROM | Log | 1,968 | 4.343 | 0.176 | 4.082 | 4.386 | 4.528 |
| log TO | Log | 1,968 | 4.197 | 0.578 | 3.475 | 4.203 | 4.916 |
| log NET = log(TO/FROM) | Log | 1,968 | -0.146 | 0.608 | -0.933 | -0.102 | 0.560 |
| Lagged log population | Log | 1,968 | 15.142 | 0.993 | 13.696 | 15.262 | 16.347 |
| log average interstate distance | Log | 48 | 7.392 | 0.221 | 7.154 | 7.330 | 7.724 |
| Steep land share | Percent | 48 | 16.886 | 16.623 | 2.382 | 11.735 | 42.155 |
| WRLURI 2006 | Index | 48 | -0.128 | 0.711 | -0.985 | -0.206 | 0.754 |
| *Panel B. States, directed-pair variables, 21 annual networks 2005–2025* | | | | | | | |
| Conditional connectedness share | pp | 47,376 | 2.128 | 1.913 | 0.308 | 1.609 | 4.601 |
| Conditional migration share (uncentered) | pp | 47,376 | 2.128 | 3.362 | 0.182 | 0.991 | 5.302 |
| log interstate distance | Log | 2,256 | 7.211 | 0.705 | 6.257 | 7.311 | 8.052 |
| Contiguous states | Indicator | 2,256 | 0.095 | 0.293 | 0.000 | 0.000 | 0.000 |
| Same Census region | Indicator | 2,256 | 0.246 | 0.431 | 0.000 | 0.000 | 1.000 |
| *Panel C. Metropolitan areas, target variables, 41 annual networks 1985–2025* | | | | | | | |
| External connectedness (FROM) | pp | 4,100 | 91.156 | 5.975 | 83.795 | 92.718 | 96.758 |
| TO connectedness | pp | 4,100 | 91.156 | 42.386 | 41.816 | 85.838 | 147.377 |
| Net difference, TO − FROM | pp | 4,100 | 0.000 | 41.598 | -47.622 | -4.672 | 53.618 |
| External connectedness (FROM), factor-adjusted | pp | 4,100 | 86.148 | 7.125 | 76.453 | 87.874 | 93.238 |
| TO connectedness, factor-adjusted | pp | 4,100 | 86.148 | 33.242 | 42.023 | 85.295 | 129.561 |
| Net difference, TO − FROM, factor-adjusted | pp | 4,100 | 0.000 | 30.302 | -38.856 | -0.376 | 39.311 |
| log FROM | Log | 4,100 | 4.510 | 0.070 | 4.428 | 4.530 | 4.572 |
| log TO | Log | 4,100 | 4.396 | 0.508 | 3.733 | 4.452 | 4.993 |
| log NET = log(TO/FROM) | Log | 4,100 | -0.114 | 0.500 | -0.745 | -0.053 | 0.454 |
| Lagged log population | Log | 4,100 | 14.028 | 0.826 | 13.164 | 13.788 | 15.272 |
| Steep land share | Percent | 100 | 15.648 | 17.335 | 0.230 | 9.686 | 44.072 |
| WRLURI 2006 | Index | 100 | 0.087 | 0.668 | -0.803 | 0.077 | 1.026 |
| *Panel D. Metropolitan areas, directed-pair variables, 21 annual networks 2005–2025* | | | | | | | |
| Conditional connectedness share | pp | 207,900 | 1.010 | 1.100 | 0.156 | 0.738 | 2.037 |
| Conditional migration share (uncentered) | pp | 207,900 | 1.010 | 2.486 | 0.037 | 0.312 | 2.370 |
| log distance between CBSA internal points | Log | 9,900 | 7.211 | 0.809 | 6.135 | 7.324 | 8.168 |
| Same Census division | Indicator | 9,900 | 0.120 | 0.325 | 0.000 | 0.000 | 1.000 |
| Same principal state | Indicator | 9,900 | 0.037 | 0.188 | 0.000 | 0.000 | 0.000 |

*Note:* Panels A and B cover the 48 states and Panels C and D the 100 metropolitan areas. Panels A and C cover the annual networks 1985–2025, and time-invariant characteristics are reported once for each state or area. The factor-adjusted rows come from the network estimated after each window's first principal component is removed. The land measure is the share made unavailable by steep slopes, aggregated for the states from whole metropolitan and micropolitan markets using 1990 total housing stock (Appendix A.4) and measured over the CBSA polygon for the metropolitan areas. The cross-state standard deviations used to standardize WRLURI 2006 and the steep land share are 0.71054 index units and 0.16623 share units, and the cross-area standard deviations are 0.66845 and 0.17335. TO connectedness sums a market's contributions to the other 47 states or 99 areas and is not bounded by 100; NET, TO minus FROM, sums to zero across markets at every annual network, and $\log \mathrm{NET} = \log(\mathrm{TO}/\mathrm{FROM})$ is positive for a net source and negative for a net receiver. Average distance is not defined for the metropolitan areas. Panels B and D cover the 21 annual networks from 2005 and report conditional shares of the raw network over directed pair-years, with fixed geography once for each directed pair. Each conditional share sums to 100 across the 47 external states or 99 external areas, so its mean is 100/47 or 100/99 by construction. Distance is measured in kilometers before taking logs, between internal points for the metropolitan areas; pp denotes percentage points.

cannot, namely whether a regulated market's position reflects more than heavier comovement with the national cycle, which the decomposition credits to both sides of a market's position at once.

WRLURI, the steep land share and average distance are time invariant, so their coefficients compare markets rather than changes within markets. These regressions describe directional position and do not estimate the model's housing-supply parameter.

The metropolitan system extends the same comparison in two directions. Across the 100 metropolitan areas, Census-division effects absorb the regional composition that 48 states cannot separate from regulation. Within states, the comparison uses each area's *principal state*, the state holding the largest share of its 2010 housing units.[24] It requires at least three sample areas in a state, so that the state contributes a within-state gradient rather than a single pairwise difference, and twelve states qualify, with 64 areas. Principal-state effects absorb everything about a state that does not change over time (its statutes, courts and lenders), and principal-state-by-year effects additionally absorb each state's shocks in every year. What remains are differences among metropolitan areas of one state, of which regulation is one. Coastal location, size and amenities can also differ within a state, and the design does not remove them. The model is equation (11) on the same annual networks, 1985–2025, with lagged log CBSA population as the size control. It adds one control the state system does not need, the outside-node-set share, the share of the population within 200 kilometers that lives outside the node set.[25] The metropolitan Wharton index and the steep land share are built by the rules used for the states, and IRS county-to-county migration exposure uses the same pool, although its flow matrices hold movers only. One construction difference is not a detail. The metropolitan system is open. It retains 61 percent of migrating exemptions, the IRS count that roughly corresponds to persons, whereas the state system retains essentially all of them, so a metropolitan target's migration exposure is measured over a partial pool of movers and its allocation across sources is conditional on that pool.

### 5.3 Allocation across sources

The bilateral analysis asks how a target's external share is distributed across particular sources. Conditional on $F_{i,t} > 0$, define

$$\bar{C}_{ij,t} = 100\frac{C_{ij,t}}{F_{i,t}}, \qquad j \neq i, \qquad \sum_{j \neq i} \bar{C}_{ij,t} = 100. \tag{12}$$

This second normalization allocates 100 percentage points across the 47 external sources. It separates *conditional source allocation* from the target's total external connectedness, since two targets can have the same allocation and different external totals. The unconditional cells $C_{ij,t}$ combine both margins and are retained for descriptive spatial patterns and supporting diagnostics. The full-row

[24] This is the first state named in the area's CBSA title for every multistate area except Washington, D.C., whose largest share lies in Virginia. The assignment is fixed across years and postdates the windows that end before 2010 (Appendix D.1).

[25] Appendix D.3 motivates this control.

denominator in the standard Diebold–Yilmaz normalization cancels from this conditional ratio, which describes the composition of external connectedness without changing its total. The second regression compares the allocation of connectedness and migration within each target's external row. Let

$$x_{ij,t} = 100\widetilde{\psi}_{ij,t} - \frac{100}{47}, \qquad \sum_{j \neq i} x_{ij,t} = 0. \tag{13}$$

Each target has 47 external sources whose conditional migration shares sum to 100 percent. Subtracting their equal-share mean, $100/47 \simeq 2.128$ percentage points, expresses exposure as a deviation from that reference, and $x_{ij,t}$ is called *conditional migration exposure* from here on. A negative value therefore means a below-average share, not a negative migration flow. Centering changes the intercept or fixed effects but not the migration slope. For specification $q$, I estimate

$$\bar{C}_{ij,t} = \beta_{M,q} x_{ij,t} + \boldsymbol{\gamma}_q' \mathbf{G}_{ij} + \lambda_q(i,j,t) + \varepsilon_{ij,t,q}, \tag{14}$$

where $\mathbf{G}_{ij}$ contains bilateral geography when included, the three geography variables and their pairwise interactions, and $\lambda_q(i,j,t)$ denotes the fixed effects of specification $q$. The coefficient $\beta_{M,q}$ is the difference in the conditional connectedness share associated with a one-percentage-point difference in conditional migration exposure.

The specification sequence changes the economic comparison in two steps and one check. The pooled slope compares all directed pairs. Geography accounts for distance, contiguity and common region and lets each one's association depend on the others, so the distance gradient can differ between contiguous and noncontiguous pairs and within and across regions. Source-by-year effects then absorb each source's general pull as a destination and its prominence in the network in each year, so the migration slope compares the sources of a given target and year net of how favored each source is by migrants in general. Target-by-year effects are the check. Because the conditional shares sum to 100 within each target-year and exposure is centered within target and year, time effects are implied by the normalization, and target-by-year effects absorb nothing from the raw outcome or the regressor. Once source-by-year effects are included they absorb only a term of order 1/47, the target's own mean as a source divided by the number of sources. They become a substantive step only where the row constraint is broken, in the metropolitan sample of pairs in different states described below. The last column of the sequence drops geography and keeps both sets of node-year effects, which separates the role of geography from that of the source pull. Two further designs ask narrower questions.[26] Unordered-pair effects absorb fixed symmetric pair attributes while retaining direction and time variation. The *saturated* design, which combines node-year effects (source-by-year and target-by-year effects together) with unordered-pair effects, compares residual allocation differences after removing these additive node-year and symmetric pair components. Persistent pair-specific directional differences can remain, so this specification does not

[26] Both are reported in the Online Appendix.

isolate only annual changes within directed links. Coefficient attenuation across these comparisons does not by itself establish the absence of a time-varying migration mechanism.

At metropolitan scale each target has 99 external sources and exposure is centered at 100/99, and shared Census division replaces contiguity, which has no standard metropolitan counterpart. Divisions nest in regions, so the product of shared division and common region is shared division itself, and the metropolitan geography interacts distance with each of the two. One feature has no analogue in the state panel. Pairs of metropolitan areas with the same principal state share state-level shocks as well as migrants. The metropolitan estimates are therefore reported with and without those pairs.

### 5.4 Inference

For the state system the target regressions compare 48 states, and their regressors vary across states rather than within them, so inference clusters by state and allows arbitrary dependence over time within a state. The standard errors are cluster-robust. With 48 clusters, conventional critical values for cluster-robust $t$ statistics tend to reject too often, so every $p$-value and star in the target tables comes from a wild cluster bootstrap-$t$ that imposes the null hypothesis, with 9,999 draws of Webb's six-point weights, one per cluster (Cameron et al., 2008; MacKinnon and Webb, 2017; Webb, 2023). The 95-percent intervals quoted in the text invert the same test (Roodman et al., 2019). The metropolitan target regressions cluster by metropolitan area, and the within-state columns add the same test with the twelve principal states as clusters, which allows arbitrary dependence among the areas of a state. A difference between the raw and the factor-adjusted coefficient comes from one regression that stacks the two networks, with every regressor and fixed effect specific to its network and both observations of a state or area in one cluster.[27]

Conditional-allocation regressions use shared-node standard errors, which allow two directed observations to be dependent whenever they share a node in either role, with stars against a Student $t$ distribution with one degree of freedom fewer than the number of nodes ($t_{47}$ at state scale). A delete-one-node jackknife provides a separate influence check.[28] The state-cluster inference relies on independence across states for its sampling interpretation, and shared-node uncertainty allows dependence across directed pairs sharing either node rather than arbitrary dependence across disjoint pairs. Year effects and geographic influence checks do not establish these conditions, so the state-clustered precision is not a correction for general spatial dependence. For the sending side, two spatially robust alternatives, a wild cluster bootstrap over the nine Census divisions and Conley standard errors, accompany the robustness checks.[29]

[27] Joint tests in the appendix tables use the cluster-robust covariance with an $F$ reference distribution.
[28] Delete-one-state estimates are reported in the Online Appendix.
[29] See Online Appendix Table OA.41.

# 6 Results

Tightly regulated states account for more of other states' forecast uncertainty, in the raw network and beyond the national cycle. Table 2 reports the directional regressions from 1985 through 2025, the raw network in columns (1)–(3) and the factor-adjusted network in columns (4)–(6). In the raw network a one-standard-deviation increase in the Wharton index is associated with 10.5 log points more TO connectedness (column 1). Once the national factor is removed it is associated with 7.5 log points more (column 4), with a cluster-robust standard error of 2.3 and a 95-percent interval of $[2.7, 12.7]$ that inverts the wild bootstrap test. The two estimates are not distinguishable from each other. Their difference is $-3.0$ with a standard error of 2.6, from one regression that stacks both networks and clusters each state's observations together. Removing the national factor takes about 30 percent off the raw estimate and leaves the gradient estimated more precisely than before. Regulation marks where dependence is attributed and not only how closely a state follows the cycle.

**Table 2:** Regulation and directional connectedness, raw and factor-adjusted networks, 1985–2025

| | Raw | | | Factor-adjusted | | |
|---|---|---|---|---|---|---|
| Dependent variable | log TO | log FROM | log NET | log TO | log FROM | log NET |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| WRLURI 2006 | 10.54** | 2.76*** | 7.78** | 7.50*** | 1.67 | 5.83** |
| | (3.62) | (0.60) | (3.40) | (2.32) | (1.05) | (2.08) |
| Steep land share | 1.19 | 0.72 | 0.47 | −0.71 | −0.19 | −0.51 |
| | (3.43) | (0.46) | (3.25) | (2.39) | (0.87) | (2.28) |
| log average distance | 19.12 | 1.82 | 17.30 | 18.69 | 12.69** | 6.00 |
| | (16.96) | (3.43) | (15.87) | (12.94) | (5.32) | (12.06) |
| Lagged log population | 12.06** | −0.38 | 12.44** | 3.29 | −3.89*** | 7.18** |
| | (3.92) | (0.67) | (4.15) | (2.29) | (0.89) | (2.62) |
| Year fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| State-years | 1,968 | 1,968 | 1,968 | 1,968 | 1,968 | 1,968 |
| Within $R^2$ | 0.097 | 0.056 | 0.071 | 0.072 | 0.083 | 0.053 |

*Note:* 48 states, 41 annual networks 1985–2025. Columns (1)–(3) use the raw network and columns (4)–(6) the factor-adjusted one. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). $\log \text{NET} = \log(\text{TO}/\text{FROM})$ is positive for a net source, and its coefficients equal the TO minus the FROM coefficients exactly. Population is BEA state population in the calendar year before the year of each annual network. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome; on a logged control, it is 100 times the elasticity. Within $R^2$ is computed against the outcome demeaned by whatever the design absorbs, because the unadjusted $R^2$ of these models is mostly the fixed effects. In levels FROM and TO both average 78.06 percentage points in the raw network and 65.58 in the factor-adjusted one. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-$t$ $p$-values that impose the null (9,999 draws). The factor-adjusted minus raw WRLURI coefficient, from one regression stacking both networks, is −3.04 (2.64) for log TO, −1.09 (0.88) for log FROM and −1.95 (3.00) for log NET.

The association is large. Beyond the national cycle, a state at the 90th percentile of regulation accounts for about 20 percent more of other states' forecast-error variance than one at the 10th. Spread over the 47 other states, a standard deviation of regulation is about a tenth of a percentage point of each one's forecast-error variance. The attribution is small for any one partner and sizable

in total. Against the outcome's own dispersion the association is the same in both networks, about a fifth of a standard deviation of TO per standard deviation of regulation.[30]

Regulated states also sit closer to being net sources. Per standard deviation of regulation, log NET, the log ratio of TO to external connectedness, is 7.8 log points higher in the raw network and 5.8 in the factor-adjusted one (columns 3 and 6). In percentage points factor-adjusted NET is 3.8 higher (1.5).[31] The net position is the less precise of the two sending-side results. Its outcome is the more dispersed, and it inherits the imprecision of the receiving side. Size behaves differently. Larger states account for more of other states' variance in the raw network, with an elasticity of TO with respect to lagged population of 0.12 (0.04). Once the national factor is removed the elasticity is 0.03 (0.02). The size association is distinguishable from zero only in the raw network, and the regulation association in both.

I read this as a supply-side fact about how price signals travel. A constraint on building keeps a market's own demand shocks from being absorbed by new housing, so they show up in its price. A price that moves with demand is what households in other markets respond to when they reconsider where to live (Glaeser et al., 2008; Paciorek, 2013). The premise is consistent with Cun and Pesaran (2022). In their estimated model a California productivity shock raises local prices more and draws fewer migrants than a comparable shock in Texas. The framework derives the weight that a constrained market's price places on other markets' prices, not this sending side, so the reading is an informal extension of it.

The result points in the direction of an earlier finding drawn from a few state shocks. In a global VAR of 31 states, Vansteenkiste (2007) finds that a price shock in land-scarce California raises prices across the country while one in Texas does not. From three state shocks the paper suggests that shocks in states with lower land-supply elasticities have stronger spillovers. The comparison here imposes no weights. TO is the share of other markets' forecast-error variance attributed to a market. It combines how large a market's price surprises are with how strongly other markets' forecast errors load on them. Impulse-response studies fix the size of the shock and route it through imposed weights. The estimates do not identify how far a given shock travels. They show that more of other markets' forecast uncertainty is traced to regulated markets beyond the national cycle. The gradient survives removing two common factors but not the larger number the Bai and Ng (2002) criterion selects in each window, so part of it may be regulated states' position within regional clusters.[32]

Across states the sending side leans toward state institutions rather than local rules. Two of the index's eleven components measure how involved a state's legislature and courts are in land use, and they vary only across states. The court index is a rating of Foster and Summers (2005). The

[30] One standard deviation of WRLURI 2006 is 0.71 index points, and the 90th and 10th percentiles lie 2.4 standard deviations apart, so the difference is $100(e^{0.075\times 2.4}-1)\approx 20$ percent. In levels a standard deviation of regulation is 4.9 percentage points of TO ($0.075\times 65.6$, with 65.6 the factor-adjusted mean), or about 0.1 points for each of the 47 partners. The within-year cross-state standard deviation of log TO is 0.56 in the raw network and 0.39 in the factor-adjusted one, so a standard deviation of regulation is associated with 0.19 standard deviations of TO in both.

[31] See Online Appendix Table OA.4.

[32] See Table 5, Panel A.

political-involvement index combines that rating with the state mean of one survey answer (Gyourko et al., 2008). Together they carry 16.5 percent of the index's loading mass. Entered beside the local survey content with the index's own weights, these state-level components carry most of the factor-adjusted TO and log NET associations and the local survey content little of them. The two blocks cannot be ranked with confidence, and each is positively associated with TO when entered alone.[33] If it is the state's legal environment, the margin of regulation that matters for the sending side operates at the scale at which it is measured, the state, and not the town. At the metropolitan scale the blocks are not estimated for TO, because the factor-adjusted metropolitan TO gradient is itself not distinguishable from zero.

### 6.1 Labor demand and the two readings of regulation

The sending side cannot by itself tell a binding supply constraint from larger demand shocks. Regulated places are productive, high-amenity places. Across metropolitan areas, supply-constraint measures are correlated with a range of demand proxies (Davidoff, 2016). High-skill cities are also more regulated, and their local amenities rise with the location choices of high-skill workers (Diamond, 2016). The metropolitan system does not separate the two readings either, because it carries the sending side less securely than the state system. Across the 100 metropolitan areas, with Census-division effects and population controls, a standard deviation of regulation is associated with 6.6 log points more TO connectedness (2.6) in the raw network and 4.6 (3.0) in the factor-adjusted one. The factor-adjusted interval is $[-2.1, 10.6]$.[34] The average gradient beyond the national cycle is not distinguishable from zero at this scale.

The two readings differ in timing. A constraint that turns demand into price binds when demand rises. Durable housing limits the response to falling demand in every market (Glaeser and Gyourko, 2005). A supply reading therefore predicts a steeper sending side where a market's demand is rising, and only there. Larger demand shocks alone would make the gradient rise with the size of the shock in either direction, so they predict that its slopes above and below the average sum to zero.

I measure demand without the price data, as labor demand predicted by industry mix. For each market and estimation window it is a shift-share prediction of wage-bill growth over the window, built from BEA industry earnings with shares from the year before the window and national industry growth that leaves the market out (Bartik, 1991). The prediction tracks actual growth, with $F$ statistics of 16.2 across states and 30.9 across metropolitan areas in regressions of actual on predicted growth. Regulated markets have somewhat stronger predicted demand. The correlation between the index and the prediction averages 0.20 across metropolitan areas and 0.36 across states within a window. The labor-demand shock $S$ is the prediction standardized across markets within each

[33] See Online Appendix Tables OA.37 and OA.38.
[34] See Appendix Table F.2, columns 1 and 6.

window. It says how much more labor demand a market's industry mix predicts than the other markets' over the same ten years.[35]

Two components of demand can be measured without the price data and held fixed.[36] The first is the shock magnitude. It holds fixed long-run relative demand growth, not the size or the volatility of demand surprises within the window. With it held fixed, the factor-adjusted state TO association is 7.2 log points (2.3), against 7.5 in the same years. The second is amenities. The natural-amenity scale of McGranahan (1999) combines climate, topography and water area. It shares its topographic component with the steep land share, so it takes that share's place in the regression rather than joining it. The association is then 7.9 log points (2.3). Amenity-rich states account for more of other states' variance in the raw network, 19.5 log points (6.0) per standard deviation of the scale, but not once the national factor is removed, 2.9 (4.8). Across states, then, amenities are associated with the sending side only in the raw network, as size is. Across metropolitan areas amenity-rich areas account for more of other areas' variance after the purge, 14.9 log points (4.3), and holding amenities fixed lowers the regulation coefficient from 4.6 to 3.2. Holding these two measures fixed leaves the state sending side in place,[37] and the timing of demand gives the sharper test.

The sending-side version of that test, a steeper TO gradient where relative labor demand is strong, is a metropolitan result and a comparison across areas.[38] In the benchmark node set the interaction is 1.9 (0.9) in the factor-adjusted network. Within areas over time it is 0.8 (1.1). Across states it is −1.5 (2.4), at a scale at which the shift-share prediction tracks actual growth less closely. It is absent from the raw network, −0.9 (1.1). Because it is identified between areas, it cannot be separated from the persistent type of place whose industry mix predicts strong growth, the productive, high-amenity places of the demand alternative. It does not survive the change of node set. Neither demand measure covers demand from consumption amenities or from local shocks unrelated to industry mix, so the sending-side result remains consistent with a supply reading without isolating it. I therefore read WRLURI as an observed characteristic of a market's land-use regime, with demand correlates and a state-institutional component, and not as a pure measure of supply constraints.

### 6.2 The receiving side, the national cycle and labor demand

In the raw network regulated states look like receivers of dependence, partly because they follow the national cycle more closely. A standard deviation of regulation is associated with 2.8 log points more external connectedness (Table 2, column 2), or 2.1 percentage points. The first principal component accounts for 8.1 percentage points more of a state's growth variance per standard deviation of the

[35] Appendix F.1 gives the construction, including the shock magnitude used as a control below, the absolute deviation of predicted growth from the year's mean scaled within each industry-classification era.

[36] See Appendix Tables F.1 and F.2.

[37] Online Appendix Table OA.9 adds two amplitude controls built from the price series themselves, each state's first-component share and the standard deviation of its VAR innovation. The factor-adjusted TO association is 6.8 log points (2.3) with them, against 7.5 without.

[38] See Online Appendix Table OA.10. Online Appendix Tables OA.14, OA.15 and OA.16 report the split in percentage points, inference clustered by dominant industry and the distribution of the shock.

index, with a cluster-robust standard error of 1.2. It accounts for 56 percent of that variance on average in the most regulated quarter of states against 35 percent in the least.[39] That is a fact about the price data and needs no network. Where housing supply is inelastic, national conditions show up in prices, and where it is elastic they show up in building. The prices of supply-constrained markets should therefore track the common cycle more closely (Glaeser et al., 2008; Guren et al., 2021; Tidwell et al., 2023). In a row-normalized decomposition a market that tracks the common factor more closely has more of its own row attributed to other markets, whatever its links to particular markets. The raw receiving side of regulated states is therefore partly the national cycle.[40]

Whether regulated states also take more of their variance from other states beyond the cycle depends on how the network treats each state's own lags, and I do not settle it. In the benchmark network the factor-adjusted gradient is 1.7 log points (column 5), with an interval of $[-0.6, 4.0]$ that includes zero. The two networks that shrink the own lags less give gradients distinguishable from zero. In none of the three is the change the purge makes distinguishable from zero.[41] Three things hold in every network. Regulated states load more heavily on the national factor, that loading raises their raw receiving side, and their sending side survives the purge.

The metropolitan system reads the same way, and it can hold a state's institutions fixed. Across the 100 largest metropolitan areas over the same years, with the metropolitan Wharton index and the steep land share standardized within that node set, a standard deviation of regulation is associated with 0.9 log points more external connectedness with Census-division effects and population controls. Among the 64 areas of the twelve principal states with at least three sample areas, under principal-state-by-year effects that absorb every shock common to a state's areas in a given year, it is associated with 0.5 log points more. The wild cluster bootstrap over the twelve states gives that within-state estimate a $p$-value of 0.009. The twelve states hold between three and eleven sample areas. Dropping one at a time leaves the raw estimate between 0.38 and 0.58, although without California, the largest cluster, its $p$-value rises to 0.07. After the purge the two estimates are 0.7 and 0.3, and the within-state $p$-value is 0.33.[42] Differences in regulation among areas of one state hold its statutes, courts and lenders fixed. They go with how much of an area's variance comes from elsewhere in the raw network, and the association is imprecise beyond the national cycle.[43] The steep land share, the second supply proxy, is imprecisely associated with

[39] See Appendix Table C.2.

[40] A disturbance that spreads from one market to others within a month also loads on the first component, so the heavier loading of regulated states could partly be propagation along migration links. A simple diagnostic is not decisive. Holding a state's own regulation fixed, its first-component share is not associated with its migration exposure to regulated markets without further controls, 0.7 percentage points (1.8) per standard deviation of that exposure. With the second-stage controls added the coefficient is 4.8 (1.9), and with migration openness added it is not distinguishable from zero. See Online Appendix Table OA.2.

[41] From one regression that stacks each network's raw and factor-adjusted forms, the change is $-1.1$ (0.9), $-0.6$ (1.1) and $-0.2$ (0.9). The levels are in Table 5, Panel A.

[42] See Appendix Table D.3.

[43] Online Appendix Tables OA.19 to OA.21 split each area's external connectedness by the source's state. Regulated areas take less of it from areas of their own state and more from areas in other states, in both networks, and part of that split rests on the areas that straddle a state line.

every directional outcome of Table 2 and Appendix Table D.3, at both scales and in both networks. Terrain does not do what regulation does.

Where a market's own demand is rising, regulated metropolitan areas take more of their forecast-error variance from other areas. That is the ordering a binding supply constraint implies. In the model a target with less elastic housing supply places more weight on other markets' prices, and with durable housing that difference appears only when the target's own demand rises. A supply reading therefore predicts a receiving-side gradient that is steep where demand is rising and not where it is falling. Larger demand shocks in regulated markets predict a different pattern. They raise the share of a market's forecast-error variance that is its own and so lower its external connectedness, whatever their sign, if an area's own surprises are larger where its predicted demand departs further from the average. Table 3 puts the question to the labor-demand shock across the 100 metropolitan areas. Every column holds fixed the number of an area's same-state sample partners, which is strongly associated with external connectedness, and the climate part of the natural-amenity scale, since regulated markets are also the high-amenity places of the demand alternative. The scale's topography component is left out of that control because it measures terrain, which enters on its own as the steep land share, and its water-area component is left out with it. Adding the full scale, water area included, leaves the interaction in place.[44]

At the average shock the regulation gradient of log FROM beyond the national cycle is 0.51 log points per standard deviation (0.37). It rises with the shock, by 0.76 log points per standard deviation of each (0.21) (column 1). The magnitudes are small because external connectedness varies little across areas. Within a year the standard deviation of log FROM is 8.7 log points, so a standard deviation of the shock moves the regulation gradient by about a tenth of that spread. Split at the average shock, the regulation interaction is 1.24 (0.37) above the average and 0.19 (0.41) below it (column 4). The difference between the two, 1.05 (0.65), is not distinguishable from zero, so the data do not separate a kink at the average from a gradient that rises with relative demand. Larger own shocks would make the slope above the average negative and the two slopes sum to zero. The slope above the average is positive and the sum is 1.42 (0.43).

Within areas over time the interaction keeps its sign, 0.69 (0.28), with a $p$-value of 0.055, so it is not only a difference between persistent types of place. Across states it is positive and imprecise, 1.4 (1.0). It is absent from the raw network, 0.01 (0.16), where the receiving side mostly measures loading on the national factor. It is not produced by a few high-demand places. On a node set of 100 areas apportioned across all 48 states it keeps its sign but is smaller and imprecise.[45] The result rests on the regulation index. Physical and regulatory constraints do not behave alike. The steep land share has no interaction beside regulation, −0.04 (0.18) in column (3), and leaves the regulation interaction at 0.78 (0.22). Split at the average, its slopes are −0.64 (0.29) above and 0.87

[44]The climate index sums the scale's January temperature, January sun, July temperature and July humidity components, averaged over an area's counties with 1990 population weights. It correlates at 0.38 with WRLURI and at 0.42 with the steep land share across the 100 areas, against 0.45 and 0.61 for the full scale. With the full scale and its interaction with the shock added to the climate control, the interaction of column (1) is 0.85 (0.24).

[45]See Table 8 and Online Appendix Table OA.11.

**Table 3:** Regulation, terrain and relative labor demand on the metropolitan receiving side, factor-adjusted network, 1985–2025

| Dependent variable | log FROM | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| WRLURI 2006 | 0.51 | | 0.53 | 0.11 |
| | (0.37) | | (0.38) | (0.50) |
| WRLURI × $S$ | 0.76*** | | 0.78*** | |
| | (0.21) | | (0.22) | |
| WRLURI × $S^+$ | | | | 1.24** |
| | | | | (0.37) |
| WRLURI × $S^-$ | | | | 0.19 |
| | | | | (0.41) |
| Steep land share | | 0.37 | 0.45 | 1.03** |
| | | (0.42) | (0.39) | (0.44) |
| Steep land share × $S$ | | 0.15 | −0.04 | |
| | | (0.20) | (0.18) | |
| Steep land share × $S^+$ | | | | −0.64* |
| | | | | (0.29) |
| Steep land share × $S^-$ | | | | 0.87** |
| | | | | (0.30) |
| Labor-demand shock $S$ | −0.19 | −0.36 | −0.18 | |
| | (0.20) | (0.25) | (0.20) | |
| $S^+$ | | | | −0.25 |
| | | | | (0.34) |
| $S^-$ | | | | −0.09 |
| | | | | (0.34) |
| Controls and same-state sample partners | ✓ | ✓ | ✓ | ✓ |
| Climate amenity index | ✓ | ✓ | ✓ | ✓ |
| Year and Census-division effects | ✓ | ✓ | ✓ | ✓ |
| Areas | 100 | 100 | 100 | 100 |
| Area-years | 4,100 | 4,100 | 4,100 | 4,100 |
| Within $R^2$ | 0.259 | 0.250 | 0.260 | 0.266 |

*Note:* The 100 largest metropolitan areas, 41 annual networks 1985–2025, factor-adjusted benchmark network, estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The dependent variable is log FROM, the log of an area's external connectedness, the share of its forecast-error variance attributed to the other areas. Within a year its standard deviation across areas is 8.7 log points, against 43 for log TO. The labor-demand shock $S$ is predicted labor demand, the shift-share prediction of an area's earnings growth over the window (Appendix F.1), standardized across areas in each annual network, so that it has mean zero and standard deviation one in every window. $S^+ = \max(S, 0)$ and $S^- = \min(S, 0)$. WRLURI 2006 and the steep land share, the share of the area's land with a slope above 15 percent (Lutz and Sand, 2026), are standardized across areas. Controls are the climate amenity index, lagged log population, the outside-node-set share and the number of an area's same-state sample partners, with year and Census-division effects. The climate amenity index is the sum of the four climate components of the USDA natural amenity scale (McGranahan, 1999), January temperature, January sun, July temperature and July humidity, without its topography and water-area components, averaged over an area's counties with 1990 population weights and standardized across areas. In column (4) the WRLURI slope above the average minus that below it is 1.05 (0.65, $p = 0.158$) and their sum is 1.42 (0.43, $p = 0.012$). For the steep land share they are −1.51 (0.46, $p = 0.014$) and 0.22 (0.37, $p = 0.608$). Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in FROM. Within $R^2$ is computed against the outcome demeaned by whatever the design absorbs, because the unadjusted $R^2$ of these models is mostly the fixed effects. Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-$t$ $p$-values that impose the null (9,999 draws). The regressions run in Stata (reghdfe, boottest).

(0.30) below, which sum to 0.22 (0.37) and differ by −1.51 (0.46). That is the pattern larger own shocks in either direction would produce, so steep terrain shows the size pattern rather than the ordering the supply multiplier implies. As on the sending side, the shock measures only the demand that an area's industry mix predicts. Other demand could produce the same ordering, so the result is consistent with the supply multiplier but does not isolate it.

The receiving side leans the other way within the index. Across states the local survey content carries the raw receiving side and the state-level components do not. Within the survey content the local land-use instruments are the one block distinguishable from zero. Beyond the national cycle the survey content also carries the state receiving side, 3.25 (0.86), against −1.79 (1.43) for the state-level components. Across metropolitan areas, where the ratings vary mainly between states, survey content also leads, but the data cannot rank the two blocks, and after the purge every block loses precision.[46] The point estimates place the sending side in state institutions and the receiving side in local rules. Outside the factor-adjusted state receiving side the blocks cannot be ranked with confidence.

### 6.3 Migration and the allocation of dependence

Among metropolitan areas in different states, more of a market's dependence is attributed to the markets its households are linked to by migration. Across states, migration adds nothing to geography. Table 4 builds the comparison on the factor-adjusted networks in two steps and one check, and its column (5) holds the node-year effects without geography. Across states the pooled bivariate slope is 0.058 (column 1). A one-percentage-point increase in conditional migration exposure $x_{ij,t}$ is associated with 0.058 percentage points more conditional connectedness beyond the national cycle. With geography, which here includes the interactions of distance, contiguity and region with one another, it is 0.039 (column 2). Conditional migration exposure has a standard deviation of 3.36 percentage points, so the geography slope corresponds to 0.13 percentage points more conditional connectedness per standard deviation of exposure.

Across states migration adds nothing to geography. The association is imprecise once geography enters and disappears when geography and source-by-year effects are held fixed together. Source-by-year and target-by-year effects without geography leave it at 0.059 (0.017) (column 5), and with geography it falls to 0.006 (column 3). Source-by-year effects absorb each source's general pull as a destination, which is largely its size, since a source's average exposure correlates at 0.84 with its log population. What they leave is the comparison among the sources of a given year. There a target does draw more dependence from the sources its migrants connect it to, but not more than distance, contiguity and region already predict. Moving costs shape both migration and proximity, and at state scale exposure is close to a gravity function of the three. The three alone account for 59 percent of the variance of conditional exposure within source-years and target-years.[47] Once

[46] See Online Appendix Tables OA.37 and OA.38.

[47] Online Appendix Table OA.25 reports the geography benchmarks with and without exposure and the fit that exposure adds in each design.

**Table 4:** Migration exposure and the allocation of connectedness across sources, factor-adjusted networks, 2005–2025

| Dependent variable | Conditional connectedness share (percentage points) | | | | |
|---|---|---|---|---|---|
| | Bivariate | Geography slope | Geography and source-by-year effects | Geography and node-year effects | Node-year effects without geography |
| | (1) | (2) | (3) | (4) | (5) |
| *Factor-adjusted network* | | | | | |
| States | | | | | |
| Conditional migration exposure | 0.058*** | 0.039 | 0.006 | 0.000 | 0.059*** |
| | (0.019) | (0.025) | (0.019) | (0.016) | (0.017) |
| Log distance | | 0.096 | −0.089 | −0.159* | |
| | | (0.080) | (0.068) | (0.089) | |
| Metropolitan areas, out-of-state pairs | | | | | |
| Conditional migration exposure | 0.077*** | 0.069*** | 0.070** | 0.060** | 0.077*** |
| | (0.021) | (0.022) | (0.027) | (0.025) | (0.024) |
| Log distance | | 0.049* | −0.014 | 0.002 | |
| | | (0.027) | (0.038) | (0.041) | |
| Metropolitan areas, all pairs | | | | | |
| Conditional migration exposure | 0.346*** | 0.148*** | 0.184*** | 0.171*** | 0.425*** |
| | (0.066) | (0.046) | (0.057) | (0.056) | (0.072) |
| Log distance | | 0.068** | −0.020 | −0.105 | |
| | | (0.028) | (0.058) | (0.088) | |
| Geography | – | ✓ | ✓ | ✓ | – |
| Source-by-year effects | – | – | ✓ | ✓ | ✓ |
| Target-by-year effects | – | – | – | ✓ | ✓ |
| Directed pair-years, states | 47,376 | 47,376 | 47,376 | 47,376 | 47,376 |
| metropolitan, out-of-state pairs | 200,256 | 200,256 | 200,256 | 200,256 | 200,256 |
| metropolitan, all pairs | 207,900 | 207,900 | 207,900 | 207,900 | 207,900 |

*Note:* The 21 annual factor-adjusted benchmark networks 2005–2025, for 48 target states with 47 sources each and 100 target metropolitan areas with 99 sources each. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The dependent variable is the conditional connectedness share, in percentage points of the target's external connectedness. The regressor of interest is conditional migration exposure, the conditional migration share centered at $100/(N-1)$ within target and year and built from IRS flows pooled over $t-15$ through $t-11$. Conditional shares average $100/47 = 2.13$ percentage points across states and $100/99 = 1.01$ across all metropolitan pairs, and 0.71 across out-of-state pairs. Column (1) is the pooled bivariate slope and column (2) the geography slope. Geography is the log of the great-circle distance in kilometers, contiguity (shared Census division at metropolitan scale), same Census region and their pairwise interactions. Because Census divisions nest in regions, shared division times same region is shared division itself and is not a separate term. Only the exposure and log-distance coefficients are shown. With the interactions, the log-distance coefficient is the distance slope for pairs that are neither contiguous (in the same division) nor in the same region. Column (3) adds source-by-year effects to column (2), and column (4) adds target-by-year effects as well. Column (5) has the source-by-year and target-by-year effects of column (4) without geography. Out-of-state pairs exclude dyads whose members share a principal state, the state holding the largest share of an area's 2010 housing units. No column carries year effects. In the state and all-pairs samples the conditional shares sum to 100 within every target-year and exposure is centered within target and year, so time effects are implied by that normalization, and target-by-year effects absorb nothing from the outcome and regressor. Among out-of-state pairs the shares no longer sum to a constant, and the target-by-year effects of columns (4) and (5) absorb the share of an area's external connectedness that stays inside its own state. Parentheses contain shared-node standard errors, which allow two directed pairs to be dependent whenever they share a node in either role. $^{*}$, $^{**}$ and $^{***}$ denote $p < 0.10$, 0.05 and 0.01 against a Student $t$ distribution with one degree of freedom fewer than the number of nodes. Table 6 repeats the regressions on the raw networks, and its note gives further details.

source-by-year effects enter, the association moves to distance. The distance coefficient for pairs that are neither contiguous nor in the same region goes from 0.10 (0.08) in column (2) to −0.09 (0.07) in column (3). With the interactions this coefficient is the distance slope for pairs in different regions, conditional on exposure. It need not share the sign of the decline of unconditional dependence with distance, which the contiguity, region and exposure terms absorb. Target-by-year effects are a check at this scale, since the normalization fixes the row sums, and they move the estimate to 0.000 (column 4). The raw network, in which the national cycle is part of every cell, starts from a stronger association and shows the same pattern.[48]

Among metropolitan areas in different states the association survives node-year effects and geography. The metropolitan rows repeat the regressions across the 100 metropolitan areas, with exposure built from county-to-county flows under the same pool and with shared Census division replacing contiguity. The county flows hold movers only, so metropolitan exposure is an analog of the model's object rather than the object itself. Pairs of areas with the same principal state share state-level shocks, institutions and lenders as well as migrants. The 3.7 percent of dyads they form carry 19 percent of the variance of exposure, so the metropolitan result is read from pairs of areas in different states. There the association survives the source step beyond the national cycle. It is 0.077 pooled, 0.069 with geography, 0.070 with source-year effects and 0.060 once target-year effects enter. A pair whose exposure is one percentage point above an equal split carries 0.06 points more of the target's external connectedness, about 8 percent of the 0.71-point average share of these pairs. Target-year effects are a substantive step in this sample. Excluding same-state dyads breaks the row constraint, so they absorb the share of an area's external connectedness that stays inside its own state. That share moves against the mean exposure of its out-of-state pairs (a correlation of −0.73 across target-years after the purge), and leaving it out biases the slope upward. Without geography, node-year effects leave 0.077 (0.024) (column 5). Across all metropolitan pairs, same-state dyads included, the slopes are larger at every step, 0.171 in column (4), and the difference comes from those dyads.

What survives among pairs in different states is an association between exposure and the pairwise pattern of dependence across links. Distance, division and region capture only in part the other pair-level affinities that could produce it, such as industrial similarity, social ties and cross-border proximity like that of New York and Bridgeport. The directed test within pairs that follows is null. Nearby pairs carry part of the precision. Dropping the out-of-state pairs whose internal points lie within 100 miles of each other lowers the factor-adjusted estimate of column (4) to 0.047 (0.025), with an interval of $[-0.002, 0.095]$, and within 150 miles to 0.045 (0.028). Most of the point estimate remains without the nearby out-of-state pairs, but it is no longer distinguishable from zero. The data do not show whether the association is stronger among neighbors.[49]

The association is carried by the direct part of exposure, the moves between the two markets. The part that runs through shared origins is too imprecise to include or exclude. Exposure splits

[48] See Table 6.

[49] See Online Appendix Table OA.35.

exactly into two direct terms and a common-origin term, and the common-origin term is what distinguishes the model's exposure from a simple flow measure. Entered together in column (4), the direct part carries the factor-adjusted out-of-state association, 0.064 (0.029), and the common-origin part is 0.013 (0.050).[50] Across states the common-origin part is only 1.3 percent of external exposure, and the state data cannot attribute the association to either part. With both entered and geography held fixed, neither part is distinguishable from zero beyond the national cycle. In the raw network the geography slope loads on the common-origin part, 15.1 (5.6), with a sign that changes across the specifications of Table 6. The common-origin part is positive and distinguishable from zero in the pooled and geography specifications, so the data cannot rule out a role for competition over shared origin populations.

Two further designs ask whether the association has a direction. Unordered-pair effects ask what remains within a pair, the extent to which $j$ weighs more in $i$'s row than $i$ in $j$'s. The saturated design adds source-by-year and target-by-year effects to them. Among metropolitan pairs in different states neither finds an association distinguishable from zero, and across states the saturated evidence is zero or slightly negative. Exposure barely varies within a pair, so the designs have little to work with. The data cannot say that migration makes one market's dependence on another larger than the reverse, and they cannot say that it does not.

Regulation does not change which markets a market's dependence comes from, as the model predicts. The supply multiplier scales every weight of a target's row and cancels when the row is normalized, so the model implies no interaction of exposure with the target's regulation in conditional shares. No such interaction, nor one with the steep land share, is distinguishable from zero across states or among metropolitan areas in different states, on either node set. Among metropolitan pairs in different states the regulation interaction is 0.007 (0.015) against an exposure slope of about 0.06. The data rule out interactions as large as the slope but not ones half its size.[51]

# 7 Robustness

The sending side of regulated states is the most secure result of the paper and the receiving side beyond the national cycle the least, and the pattern of what survives says something about what each side is.

## 7.1 Alternative networks

The network enters every result in this paper, and how it treats each state's own history decides how much of a state's forecast error is attributed to others. The benchmark penalizes own lags exactly as it penalizes cross-state lags, and it leaves the fitted residuals serially correlated. The

[50] See Online Appendix Table OA.34.

[51] Across all metropolitan pairs the regulation interaction is negative, −0.063 (0.021) beyond the national cycle, consistent with regulated areas placing less of their dependence on their same-state partners, the pairs with the highest exposure, and it disappears once those pairs are excluded. See Online Appendix Section OA.5 and Table OA.22.

median absolute lag-one autocorrelation of the in-sample residuals is 0.39, against 0.86 for monthly price growth itself, whose repeat-sales indexes are smooth. I therefore re-estimate the directional results on two networks that differ from the benchmark in the own-lag treatment alone, each in its raw and its factor-adjusted form. The first exempts the two own lags from the penalty, at the same lag order, window, mixing parameter, validation design and path. The second also extends the own lags to eight, the shortest own-lag order at which residual serial dependence becomes weak, and its hv-block buffer widens with the lag order to eight months on either side of each validation block. The second-stage covariates and samples are unchanged, so any difference between the rows of Panel A of Table 5 in a given set of years is attributable to the network alone. The metropolitan results rest on the benchmark network alone, because the metropolitan system with its own lags exempt from the penalty is unstable in many windows.[52] The two alternatives are checks on the benchmark and not candidates for it.[53]

The two alternatives do what they were built to do, at a cost. Freeing the own lags lowers the residual autocorrelation and eight own lags nearly remove it, but each step thins the network and lowers its level, and the eight-own-lag system is not stable in the raw annual networks of 1985 and 2002 through 2009, the boom and bust of the 2000s. Its columns are therefore estimated on its stable years, and the table repeats the benchmark and own-lags-free regressions on those years beside it.[54]

How much of other states' variance is attributed to a state does not depend on how much persistence the network allows each state. In the factor-adjusted networks the TO association is 7.5 log points under the benchmark, 9.1 with the own lags free and 7.1 with eight own lags in that network's stable years, and in the raw networks 10.5, 13.7 and 10.7. Log NET is 5.8, 4.2 and 3.9 in the factor-adjusted networks, each distinguishable from zero at the 5-percent level. How much a state receives does depend on it. The factor-adjusted external-connectedness gradient is 1.7 log points under the benchmark but 5.0 with the own lags free and 3.3 with eight own lags, both distinguishable from zero. Freeing the own lags lets each equation explain more of its own variance and attributes a state's external connectedness to fewer sources, and the cross-state gradient in what remains is steeper, in the raw network as well (5.5 log points against 2.8). In none of the three networks is the change the purge makes to the receiving-side gradient distinguishable from zero. From one regression that stacks each network's two forms, the change is $-1.1$ (0.9), $-0.6$ (1.1) and, in the years in which both forms of the eight-own-lag network are stable, $-0.2$ (0.9). The size of a receiving-side effect is therefore tied to a modeling choice about persistence, and whether the gradient remains beyond the national cycle depends on it, so I leave the question open.

[52] See Appendix D.2.

[53] See Appendix B.1.

[54] On the annual networks from 2005 the median in-sample residual autocorrelation falls from 0.38 to 0.21 and 0.04, Ljung–Box rejects for 37 percent of equations on held-out residuals under eight own lags against 100 percent under the benchmark, and the active cross-state slopes per equation fall from 25.4 to 13.2 and 6.7. The factor-adjusted eight-own-lag network is not stable in three years. See Appendix Table B.1.

**Table 5:** Sending and receiving sides under alternative networks and second-stage specifications, raw and factor-adjusted

| | log TO | | log FROM | | log NET | |
|---|---|---|---|---|---|---|
| | Raw (1) | Factor-adjusted (2) | Raw (3) | Factor-adjusted (4) | Raw (5) | Factor-adjusted (6) |
| *Panel A. Alternative networks* | | | | | | |
| Benchmark network | 10.54** | 7.50*** | 2.76*** | 1.67 | 7.78** | 5.83** |
| | (3.62) | (2.32) | (0.60) | (1.05) | (3.40) | (2.08) |
| Own-lags-free network | 13.73*** | 9.14*** | 5.52*** | 4.95*** | 8.21** | 4.18** |
| | (3.26) | (1.86) | (1.17) | (1.23) | (3.15) | (1.62) |
| Benchmark, eight-own-lag stable years | 9.06** | 8.10*** | 2.19*** | 1.61 | 6.87 | 6.49*** |
| | (3.93) | (2.32) | (0.61) | (1.02) | (3.75) | (2.07) |
| Own-lags-free, eight-own-lag stable years | 13.09*** | 9.56*** | 4.53*** | 5.05*** | 8.56** | 4.51*** |
| | (3.27) | (1.85) | (1.11) | (1.21) | (3.07) | (1.60) |
| Eight-own-lag network, its stable years | 10.69** | 7.15*** | 2.61** | 3.26** | 8.08** | 3.88** |
| | (3.59) | (2.22) | (1.04) | (1.09) | (2.98) | (1.50) |
| Annual networks, rows 1–2 | 41 | 41 | 41 | 41 | 41 | 41 |
| rows 3–5 | 32 | 38 | 32 | 38 | 32 | 38 |
| Benchmark estimator, more common components removed | | | | | | |
| Bai–Ng number of components removed | – | 1.41 | – | −0.50 | – | 1.91* |
| | | (1.14) | | (0.62) | | (0.96) |
| Two components removed | – | 5.36** | – | 1.28 | – | 4.08** |
| | | (2.06) | | (0.79) | | (1.71) |
| *Panel B. Alternative second-stage specifications, benchmark network* | | | | | | |
| Census-region effects | 21.06*** | 11.46*** | 2.17* | 1.92 | 18.89*** | 9.55*** |
| | (5.40) | (3.06) | (1.05) | (1.53) | (5.63) | (3.00) |
| WRLURI 2018 in place of 2006 | 7.44 | 6.30** | 2.15** | 0.90 | 5.30 | 5.40** |
| | (4.01) | (1.86) | (0.71) | (1.05) | (3.78) | (1.87) |
| Saiz unavailable-land share added | 10.93** | 7.70*** | 2.61*** | 1.71 | 8.33** | 5.99** |
| | (3.66) | (2.40) | (0.67) | (1.15) | (3.37) | (2.05) |

*Note:* 48 states. Each cell is 100 times the coefficient on the regulation index, WRLURI 2006 or, where the row says so, WRLURI 2018, in the regressions of Table 2 with the change that the row names. Odd columns use the raw networks and even columns the factor-adjusted networks. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Panel A varies the network. Rows 1–2 use the 41 annual networks 1985–2025, and rows 3–5 the years in which the eight-own-lag network is dynamically stable, 32 in its raw form (1986–2001 and 2010–2025) and 38 in its factor-adjusted form (1986–2002, 2004–2005 and 2007–2025). The last two rows of Panel A remove, in each window and inside each validation fold, the number of principal components that the Bai and Ng (2002) criterion selects on the window's raw growth (a median of 4, between 2 and 8) or two components, before the benchmark network is estimated. They have no raw form. Panel B varies the second-stage specification. It adds Census-region effects, replaces the 2006 index with the 2018 one (each vintage standardized by its own 48-state standard deviation), and adds the Saiz unavailable-land share (the metropolitan-population-weighted state aggregate of Saiz (2010)). Table 7 varies the sample and the timing. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-$t$ $p$-values that impose the null (9,999 draws). The allocation ladder under the own-lag networks is in Online Appendix Table OA.30.

Removing more common structure than one factor weakens the sending side, and what it leaves says what the sending side is. The one-factor purge leaves regional common structure in place, and Census-region effects absorb the regional level of TO but not within-region differences in how strongly states load on a regional component.[55] The Bai and Ng (2002) criterion selects a median of four principal components on the window's raw growth, between two and eight. Re-estimating the network after removing that number, within each window and inside each validation fold, leaves a factor-adjusted TO gradient of 1.4 log points (1.1), with an interval of $[-1.3, 3.7]$, and a log NET gradient of 1.9 (1.0). Removing two components at every window leaves 5.4 (2.1), with an interval of $[0.9, 9.9]$ (Table 5, Panel A).[56] The gradient survives the second factor but not the criterion's count. Two readings fit. Either the factor-adjusted sending side reflects regulated states' position within regional clusters, whose shared shocks are attributed to each member, and not structure among markets, or the larger purge removes real propagation that loads on several factors, which makes it a demanding test. The design cannot choose between them.

### 7.2 Second-stage specifications and inference

Regulated states are not spread evenly across regions. Mean WRLURI 2006 is $-0.430$ in the South, $-0.562$ in the Midwest, 0.785 in the Northeast and 0.040 in the West,[57] so the association could be a regional difference. It is not. Adding Census-region effects (Panel B of Table 5) raises the log TO and log NET estimates, to 21.1 and 18.9 in the raw network and 11.5 and 9.5 in the factor-adjusted one, and moves the external-connectedness estimate little, to 2.17 raw and 1.92 factor-adjusted. The sending-side association therefore appears among states within the same broad region, where it is if anything stronger.

The 2018 vintage of the index weakens the raw network more than the factor-adjusted one. With the 2018 index in place of the 2006 one, each vintage standardized by its own cross-state standard deviation, the factor-adjusted TO association is 6.3 log points and log NET 5.4, both distinguishable from zero. The raw TO association is 7.4 and raw log NET 5.3, neither distinguishable from zero. Adding Saiz's unavailable-land share leaves the factor-adjusted TO association at 7.7. The two state indexes correlate at only 0.561 across states, and the 2018 survey postdates every window ending before 2018, so it measures the regime the early windows faced with more error than the 2006 survey does. Measurement error of that kind attenuates both networks' coefficients, and the raw ones, whose standard errors with the 2018 index are about twice as large, lose their precision first. The vintage check therefore supports the factor-adjusted sending side, the paper's headline, and shows that the raw gradient is the less secure of the two.

How the index is aggregated or conditioned on population changes little, and the factor-adjusted TO association stays between 7.5 and 8.0 log points.[58] Terrain stays unassociated with either side

[55] See Appendix C.

[56] See also Online Appendix Table OA.8.

[57] See Online Appendix Table OA.39.

[58] Weighting jurisdictions by the survey weights gives 7.8 log points (2.2) and weighting them by jurisdiction land area 7.9 (2.4), against 7.5 under the equal-jurisdiction benchmark. Without the population control the association is

of a market's position however it is measured, and the regulation coefficients do not move with the land measure.[59]

Neighboring states share shocks, and allowing for that does not overturn the sending-side gradient. Clustered by the nine Census divisions, the wild bootstrap interval for the factor-adjusted TO coefficient widens to $[1.1, 16.4]$, from $[2.7, 12.7]$ by state, and for log NET it is $[3.0, 11.9]$. Conley standard errors are smaller than the state-clustered ones.[60]

Metropolitan areas that share a dominant industry are exposed to the same demand shocks, so their errors may be correlated, and the metropolitan interaction of Table 3, column (1), survives allowing for that. Clustered by each area's dominant industry in 1975, 16 clusters, its wild bootstrap $p$-value is 0.010, and that of the regulation slope above the average in column (4) is 0.037.[61]

### 7.3 The raw networks and the allocation designs

On the raw networks, in which the national cycle is part of every cell, the state allocation association starts stronger and follows the same pattern (Table 6). Across states the raw association begins at a pooled slope of 0.103 and a geography slope of 0.104, or 0.35 percentage points of conditional connectedness per standard deviation of exposure. Source-by-year and target-by-year effects without geography leave 0.068 (0.014), and with geography it falls to $-0.001$ and $-0.014$, as distance takes it over, with a coefficient that goes from 0.07 (0.15) to $-0.20$ (0.10). Among metropolitan pairs in different states the raw association is weaker than the factor-adjusted one, 0.056 pooled and 0.047 with geography. That is what one would expect if the common factor adds dependence to every pair whether or not its members are linked, so that the purge sharpens the migration association there. Source-year effects take the raw estimate to 0.018 and target-year effects to 0.009, no longer distinguishable from zero, while node-year effects without geography leave 0.046 (0.012). Across all metropolitan pairs the raw slope is 0.060 in column (4).

In the factor-adjusted networks the allocation results move more with the own-lag treatment than the sending side does, plausibly because each step thins the cross-state network and leaves fewer links for migration to explain.[62] The designs with unordered-pair effects, which ask whether dependence

8.0 (2.4), and with 1980 population in place of lagged population 7.5 (2.4). No population-weighted WRLURI series is reported, because the series labeled that way in the aggregation inputs is the equal-jurisdiction mean (Appendix A.4). See Online Appendix Tables OA.36, OA.3 and OA.4.

[59]With the whole-CBSA steep land share aggregated using housing stock or population in 1990 or 2000, or with the county comparator, no steep land share coefficient is distinguishable from zero at the 5-percent level, and the factor-adjusted TO association stays between 7.45 and 7.52 log points. Alternative allocations for the historical Montana housing units and an alternative area denominator for the combined Colorado market leave the regulation coefficient unchanged to two decimal places. See Online Appendix Table OA.36 and Appendix A.4.

[60]The Conley standard errors allow correlation within 500, 1,000 or 1,500 kilometers, and within a state at any lag. See Online Appendix Table OA.41.

[61]With area clusters they are 0.001 and 0.023. The sending-side interaction of Online Appendix Table OA.10 has $p$-values of 0.05 and 0.17 with industry clusters, against 0.05 and 0.19, and is 2.1 (1.0) with the full amenity scale added.

[62]In the raw networks the geography slope is 0.104 under the benchmark and 0.072 with the own lags free, and the unordered-pair-and-year slope 0.093 and 0.054, each in conditional-connectedness percentage points per point of migration exposure, and in the factor-adjusted networks they are smaller still. The saturated comparisons are small

**Table 6:** Migration exposure and the allocation of connectedness across sources, raw networks, 2005–2025

| Dependent variable | Conditional connectedness share (percentage points) | | | | |
|---|---|---|---|---|---|
| | Bivariate | Geography slope | Geography and source-by-year effects | Geography and node-year effects | Node-year effects without geography |
| | (1) | (2) | (3) | (4) | (5) |
| *Raw network* | | | | | |
| States | | | | | |
| Conditional migration exposure | 0.103*** | 0.104** | −0.001 | −0.014 | 0.068*** |
| | (0.027) | (0.044) | (0.021) | (0.020) | (0.014) |
| Log distance | | 0.069 | −0.204** | −0.343*** | |
| | | (0.152) | (0.100) | (0.111) | |
| Metropolitan areas, out-of-state pairs | | | | | |
| Conditional migration exposure | 0.056** | 0.047* | 0.018 | 0.009 | 0.046*** |
| | (0.025) | (0.027) | (0.011) | (0.010) | (0.012) |
| Log distance | | 0.140*** | −0.039 | −0.065** | |
| | | (0.037) | (0.025) | (0.030) | |
| Metropolitan areas, all pairs | | | | | |
| Conditional migration exposure | 0.160*** | 0.071** | 0.067*** | 0.060*** | 0.185*** |
| | (0.036) | (0.027) | (0.023) | (0.022) | (0.033) |
| Log distance | | 0.145*** | −0.042 | −0.097** | |
| | | (0.036) | (0.030) | (0.041) | |
| Geography | – | ✓ | ✓ | ✓ | – |
| Source-by-year effects | – | – | ✓ | ✓ | ✓ |
| Target-by-year effects | – | – | – | ✓ | ✓ |
| Directed pair-years, states | 47,376 | 47,376 | 47,376 | 47,376 | 47,376 |
| metropolitan, out-of-state pairs | 200,256 | 200,256 | 200,256 | 200,256 | 200,256 |
| metropolitan, all pairs | 207,900 | 207,900 | 207,900 | 207,900 | 207,900 |

*Note:* The 21 annual raw benchmark networks 2005–2025. Samples, columns, geography, effects and inference are those of Table 4. Distances are measured between 2000 Census population centers for states and between Census Gazetteer internal points for metropolitan areas. The row under each exposure coefficient gives the log-distance coefficient in the columns that include geography, as in Table 4. Raw conditional shares average 0.88 percentage points across out-of-state pairs. Once source-by-year effects are included, target-by-year effects absorb a term of order 1/47 (1/99 across all metropolitan pairs), the target's own mean as a source divided by 47 (99), in the outcome and in the regressor. Without geography that term moves the exposure coefficient by less than 0.0001, so columns (3) and (4) differ there mainly through the target means of the geography controls. In each annual network a Poisson pseudo-maximum-likelihood regression of the pooled state-to-state flows of the exposure pool (the 2,256 directed pairs of distinct states, stayers excluded) on the three geography variables with origin and destination effects has a McFadden pseudo-$R^2$, $1 - \ell_{\text{model}}/\ell_{\text{constant}}$ with $\ell$ the log likelihood, of 0.926 to 0.936 across the 21 annual networks. The same statistic is 0.623 to 0.672 for origin and destination effects alone, and the within pseudo-$R^2$ of geography, $1 - \ell_{\text{model}}/\ell_{\text{origin and destination}}$, is 0.799 to 0.812. Geography accounts for 59 percent of the variance of state conditional migration exposure within source-by-year and target-by-year effects. The unordered-pair designs, the interactions with target regulation and the steep land share, and delete-one-node jackknife standard errors are in Online Appendix Table OA.23.

has a direction that migration lines up with, have little to work with. More than 98 percent of the variance of exposure lies between links, and unordered-pair-and-year effects retain 26.4 percent of it and the saturated effects 13.6 percent. The designs therefore cannot tell whether migration makes dependence run more in one direction than the other, as when migration's association with metropolitan spillovers is absorbed by origin and destination traits (Malone, 2017).[63] A held-out comparison ranks a target's sources better with migration than with geography but adds little to the fit of their shares. Because neighboring annual windows share most of their months, it describes in-sample ranking rather than predictive performance.[64]

### 7.4 Alternative samples

No single state or region produces the sending-side association. Omitting each of the 48 states in turn leaves the factor-adjusted TO estimate between 6.4 and 8.6 log points, and omitting each Census region in turn between 6.7 (without the South) and 11.1 (without the Northeast). Each of the four region omissions leaves a factor-adjusted TO and log NET association distinguishable from zero at the 5-percent level, whereas in the raw network the TO association without the South, 8.6 (5.1), is not (Table 7, Panel A). With Census-region effects the region omissions give between 9.8 and 13.6. The receiving side is more fragile on this margin, as it is on the estimator, and it leans on the South. Omitting the South lowers the raw external-connectedness estimate from 2.76 to 1.91 log points and the factor-adjusted one from 1.67 to 0.36, and with Census-region effects and the South omitted the two are 0.17 and 0.15. These are influence calculations on fixed national networks, which are not re-estimated after a state or region is removed.[65]

The receiving-side interaction of Table 3, column (1), is not produced by a few high-demand places. It keeps its sign and its precision when areas are left out. It stays between 0.63, without the Mountain division, and 0.94, without the West North Central, when any single Census division is left out, and it is 0.68 (0.22) when San Jose, San Francisco, Seattle, Boston and Austin are left out together.[66]

The metropolitan node set is itself a sample choice, and it bears on the economics. Ranking areas by 2000 population puts eleven of the 100 in California, eight in Florida, seven in Ohio and six each in New York, Pennsylvania and Texas, and leaves nine states without an area, while areas in the

and indistinguishable from zero under all three networks, in both forms and on both samples. See Online Appendix Table OA.30, and Appendix B.3 for the diagnostics of the three networks.

[63]The designs follow the template of Landier et al. (2017), whose banking regressor varies within pairs over time. Across states unordered-pair and year effects give 0.093 raw and 0.026 factor-adjusted, although the fall, −0.066 with a delete-one-node jackknife standard error of 0.050, is not itself distinguishable from zero. The saturated least-squares estimate is −0.010 raw and −0.003 factor-adjusted, its Poisson pseudo-maximum-likelihood counterpart −0.018 (0.009), and among metropolitan pairs in different states it is 0.008 raw and 0.011 factor-adjusted. See Online Appendix Tables OA.23, OA.27 and OA.26.

[64]Adding migration to geographic predictors lowers the mean Kullback–Leibler divergence of predicted from observed conditional shares from 0.346 to 0.340 and the mean root mean squared error from 1.860 to 1.849 share points, and the mean held-out Spearman correlations are 0.050 for geography, 0.228 for migration and 0.200 for their combination. See Online Appendix Tables OA.31 and OA.32.

[65]See Online Appendix Table OA.40.

[66]The sending-side interaction stays between 1.4 and 2.3 and is 1.6 (0.9) without the five areas.

**Table 7:** Sending and receiving sides under alternative samples and timing, raw and factor-adjusted

| | log TO | | log FROM | | log NET | |
|---|---|---|---|---|---|---|
| | Raw (1) | Factor-adjusted (2) | Raw (3) | Factor-adjusted (4) | Raw (5) | Factor-adjusted (6) |
| *Panel A. Alternative samples, benchmark network* | | | | | | |
| Without the Northeast, 39 states | 16.00** | 11.15** | 3.22** | 2.72 | 12.78** | 8.43** |
| | (5.26) | (4.22) | (1.03) | (1.93) | (4.92) | (3.17) |
| Without the Midwest, 36 states | 10.01** | 7.00** | 2.67*** | 1.71 | 7.34* | 5.29** |
| | (3.88) | (2.60) | (0.78) | (1.29) | (3.68) | (2.35) |
| Without the South, 32 states | 8.59 | 6.71** | 1.91** | 0.36 | 6.68 | 6.34** |
| | (5.09) | (2.72) | (0.65) | (0.99) | (4.96) | (2.73) |
| Without the West, 37 states | 13.79*** | 10.42*** | 3.04*** | 1.94 | 10.75** | 8.47*** |
| | (3.90) | (2.40) | (0.78) | (1.41) | (3.69) | (2.07) |
| Each state omitted in turn, range | [9.05, 11.83] | [6.38, 8.57] | [2.52, 3.09] | [0.92, 2.01] | [6.50, 9.07] | [4.99, 7.18] |
| *Panel B. Alternative timing, benchmark network* | | | | | | |
| Annual networks 1985–2004 | 10.50** | 7.43*** | 2.50** | 0.56 | 8.00** | 6.87*** |
| | (3.32) | (2.12) | (1.07) | (1.13) | (3.24) | (1.89) |
| Annual networks 2005–2025 | 10.73* | 7.67** | 3.06*** | 2.75* | 7.66 | 4.92 |
| | (5.92) | (3.43) | (0.58) | (1.37) | (5.94) | (3.38) |
| Annual networks 2010–2025 | 6.84 | 5.38 | 2.23*** | 1.15 | 4.61 | 4.23 |
| | (6.73) | (3.33) | (0.58) | (1.44) | (6.82) | (3.39) |

*Note:* 48 states. Each cell is 100 times the coefficient on the regulation index, WRLURI 2006, in the regressions of Table 2 with the change that the row names. Odd columns use the raw networks and even columns the factor-adjusted networks. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Panel A removes the states of one Census region at a time from the benchmark regressions, on the national networks, which are not re-estimated, and with covariates standardized across all 48 states. Its last row gives the smallest and largest estimate when each of the 48 states is removed in turn, without standard errors. Panel B restricts the annual networks to the named years. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-$t$ $p$-values that impose the null (9,999 draws).

same state share state shocks that the one-factor purge leaves in place. Table 8 re-estimates both networks, with the benchmark specification, on an apportioned node set of 100 areas. It gives each of the 48 contiguous states at least one area and divides the rest by the Huntington–Hill method on the square root of each state's metropolitan population, taking the largest areas first within a state. It shares 78 areas with the benchmark, gives no state more than six, and covers 75 percent of the ranking pool's 2000 population against 77 percent. The allocation result holds. Among pairs in different states the factor-adjusted exposure coefficient of column (4) of Table 4 is 0.052 (0.013), against 0.060 (0.025), and across all pairs it is 0.201 against 0.171. The receiving-side demand interaction keeps its sign and most of its size but not its precision. Column (1) of Table 3 gives 0.60 (0.34) on the apportioned node set, estimated on 99 of its areas because the Wharton survey does not cover Cheyenne, Wyoming, against 0.76 (0.21) on the benchmark, and the slope above the average shock is 0.92 (0.55), against 1.24 (0.37). On a smaller apportioned node set of 75 areas the interaction is 0.82 (0.42).[67] The sending-side interaction does not survive. On the apportioned node set it is 0.8 (1.2), against 1.9 (0.9), and the slope above the average shock falls from 2.6 to 0.8 (1.7), so that interaction is specific to the benchmark node set and I do not lean on it.

[67] See Online Appendix Table OA.12.

**Table 8:** Metropolitan results on the benchmark and an apportioned node set

| | Benchmark, 100 largest | | Apportioned, 100 | |
|---|---|---|---|---|
| | Raw (1) | Factor-adjusted (2) | Raw (3) | Factor-adjusted (4) |
| *Panel A. Table 3,* log *FROM* | | | | |
| WRLURI × $S$, column (1) | 0.01 | 0.76*** | −0.00 | 0.60 |
| | (0.16) | (0.21) | (0.22) | (0.34) |
| WRLURI × $S^+$, column (4) | 0.54** | 1.24** | 0.39 | 0.92 |
| | (0.26) | (0.37) | (0.30) | (0.55) |
| WRLURI × $S^-$, column (4) | −0.63 | 0.19 | −0.20 | 0.35 |
| | (0.30) | (0.41) | (0.44) | (0.69) |
| *Panel B. The same regressions,* log *TO* | | | | |
| WRLURI × $S$, column (1) | −0.87 | 1.86* | −0.96 | 0.76 |
| | (1.09) | (0.85) | (1.41) | (1.17) |
| WRLURI × $S^+$, column (4) | −2.42 | 2.59 | −2.97 | 0.78 |
| | (2.31) | (1.72) | (2.57) | (1.68) |
| WRLURI × $S^-$, column (4) | 0.39 | 0.81 | 0.38 | 0.76 |
| | (2.58) | (1.85) | (2.89) | (2.41) |
| Areas, Panels A and B | 100 | 100 | 99 | 99 |
| *Panel C. Table 4, column (4), migration exposure* | | | | |
| All pairs | 0.060*** | 0.171*** | 0.066*** | 0.201*** |
| | (0.022) | (0.056) | (0.018) | (0.046) |
| Pairs in different states | 0.009 | 0.060** | 0.017*** | 0.052*** |
| | (0.010) | (0.025) | (0.007) | (0.013) |
| Pairs in different states, area-year pairs | 200,256 | 200,256 | 204,204 | 204,204 |
| States with an area | 39 | 39 | 48 | 48 |
| Most areas in one state | 11 | 11 | 6 | 6 |

*Note:* Columns (1)–(2) are the benchmark node set, the 100 largest contiguous-US metropolitan areas by 2000 population. Columns (3)–(4) re-estimate both networks, with the benchmark specification, on an apportioned node set of 100 areas that gives each of the 48 contiguous states at least one area and divides the rest by the Huntington–Hill method on the square root of each state's metropolitan population, taking the largest areas first within a state. It shares 78 areas with the benchmark. Every second-stage variable is rebuilt on the node set's own network, and the supply proxies and the labor-demand shock are standardized within it. Panel A repeats the regressions of Table 3, with the same controls, including the number of same-state sample partners, on the areas with a metropolitan Wharton index value, which the survey does not provide for Cheyenne, Wyoming, an area of the apportioned node set, and Panel B the same regressions with log TO (Online Appendix Table OA.10). Their entries are 100 times the coefficients, with standard errors clustered by area and wild cluster bootstrap-$t$ $p$-values that impose the null (9,999 draws). Panel C repeats column (4) of Table 4, 2005–2025, with its interacted geography controls and with standard errors that allow dependence between pairs sharing an area and $p$-values from $t(G-1)$, $G$ the number of areas. *, ** and *** denote $p < 0.10$, 0.05 and 0.01. Panels A and B and the apportioned columns of Panel C run in Stata, whose version of the Panel C estimator reproduces the benchmark cells.

### 7.5 Alternative timing

Figure 7 estimates the sending-side regressions as a separate 48-state cross-section in every annual network, in both networks. In the factor-adjusted network the TO coefficient is positive in 32 of the 41 years, against 33 raw, and its interval lies above zero in 10, against 12 raw. In no year does the interval lie below zero, in either network or for either sending-side outcome. Averaged by decade, the factor-adjusted coefficient is 7.3 log points over 1985–1989, 9.2 in the 1990s, 9.4 in the 2000s, 6.9 in the 2010s and 2.9 since 2020, and the raw one is about 15 in the 1990s and 2000s and 0.9 since 2020. The sending-side association is therefore present from the first windows, largest in the raw network in the windows that contain the boom and bust of the 2000s, and weakest in both networks in the most recent windows. That is consistent with a national cycle that has come to dominate. The common component accounts for 69 percent of a state's growth variance in the windows since 2020, against 22 percent in those ending in 1985–1989, and a smaller residual variance leaves less room for differences between states.[68] Log NET follows the same profile less precisely. It is positive in 33 factor-adjusted years, with an interval above zero in six, about what a constant gradient would produce given the precision of a single cross-section.[69]

The receiving side has its own profile. In the raw network its coefficient is negative in each year from 1985 through 1988 and largest in the windows that contain the boom and bust of the 2000s. In the windows of the late 1980s, when the common component accounted for only a fifth of a state's growth variance, regulated states took less of their forecast-error variance from elsewhere. In the factor-adjusted network the coefficient averages 1.7 log points against 2.8 raw and correlates 0.49 with the raw profile.[70]

Panel B of Table 7 splits the years. Estimated on the years before 2005 and those from 2005 separately, the factor-adjusted TO association is 7.4 and 7.7, and from 2010 it is 5.4 (3.3). The factor-adjusted net position is 6.9 before 2005, but 4.9 from 2005 and 4.2 from 2010, neither distinguishable from zero. The sending-side association is therefore present in both halves of the period, and the net position mainly in the first.

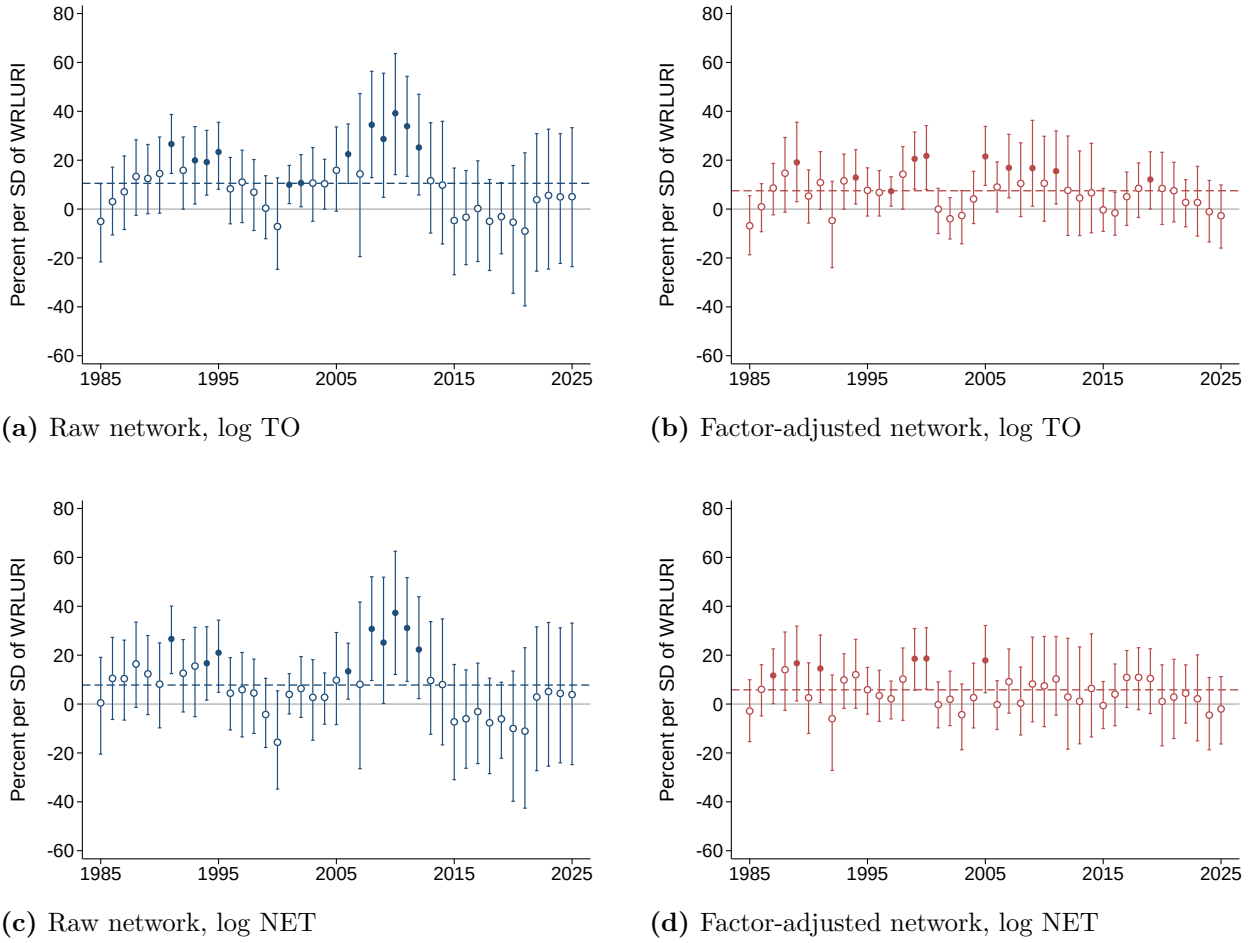
The receiving-side interaction of Table 3, column (1), holds across both industry classifications. It is 0.74 (0.23) in the NAICS period and 0.96 (0.27) in the SIC period, and 0.86 (0.17) outside the windows chained across the SIC–NAICS break. The shock is standardized within each window, so these subsamples use the shock of the full sample.[71]

[68] See Appendix Figure C.1.

[69] With standard errors near 6, a gradient of 5.8 in every year would have an interval above zero in about 7 of the 41 years, and for TO, at 7.5, in about 11. The pooled estimate is more precise than any single year, although the 41 annual networks carry far fewer effectively independent observations than their number, because windows up to nine years apart share months.

[70] The raw coefficient averages −4.8 log points over 1985–1988 and 6.3 log points over 2000–2009, and the factor-adjusted one is positive in 31 of the 41 years, against 36 raw. The early negative raw estimates do not come from the extreme months of 1981–1983 that lie inside every window ending through 1992, since re-estimating the eight networks from 1985 to 1992 with each state's monthly growth winsorized at its 1st and 99th percentiles leaves every 1985–1988 coefficient negative. See Online Appendix Tables OA.5, OA.6 and OA.7.

[71] The sending-side interaction is 2.5 (1.3), 1.9 (1.7) and 1.6 (0.9) in the same periods.




**(a)** Raw network, log TO

**(b)** Factor-adjusted network, log TO

**(c)** Raw network, log NET

**(d)** Factor-adjusted network, log NET

**Figure 7:** The sending-side regulation coefficient year by year, 1985–2025

*Note:* Each point is 100 times the WRLURI 2006 coefficient, approximately percent per standard deviation of WRLURI, from a separate 48-state cross-section regression of log TO (a, b) or log NET = log(TO/FROM) (c, d) on the regressors of Table 2 in one annual network, in the raw (a, c) and the factor-adjusted (b, d) benchmark networks. Bars are 95-percent intervals that invert a wild bootstrap-$t$ test imposing the null (9,999 draws, with each state its own cluster in a cross-section). Filled points are years whose interval excludes zero, empty points years whose interval covers it. The dashed line in each panel is the pooled estimate of Table 2, and the four panels share one vertical scale. See Online Appendix Tables OA.6 and OA.5 for every estimate.

Holding the migration network fixed at its 1990–1994 pool leaves the allocation results in place, so they do not depend on updating the migration network each year.[72]

The timing of the index raises a last question, whether regulation responds to the price histories it is compared with. WRLURI 2006 postdates the windows ending in 1985–2005, and the 2018 survey every window ending before 2018, so regulation could partly respond to the price histories it is compared with, if places that saw fast or volatile price growth later adopted tighter rules. No state measure of regulation before 1990 is available to check this for the early windows. Land-use

[72] In the raw state network the geography slope is 0.105 and the unordered-pair-and-year slope 0.096, against 0.104 and 0.093 under rolling pools, and the saturated estimate is −0.010 (0.018) under either pool. See Online Appendix Table OA.28, and Online Appendix Table OA.24 for the metropolitan counterparts under both pools.

restrictions accumulate over decades, so the 2006 index records a slow-moving regime with a long history and not a recent policy, and the labor-demand literature the paper follows treats the index as a determinant of local housing-supply elasticity (Diamond, 2016). The regressions compare states and not dated treatments, and they claim association, not causation. Regulation may still respond in part to past price growth, so the supply reading is an interpretation of the association and not something the design establishes. Supply elasticities also changed over the period, falling between the booms of the 2000s and the 2010s where regulation tightened (Aastveit et al., 2023), which is one reason the 2018 index is carried as a check.

### 7.6 Inference conditional on the estimated network

Every interval and $p$-value reported in this paper treats the estimated network as data. The wild cluster bootstrap, shared-node and delete-one-node procedures describe sampling variation in the second-stage target and dyadic panels while holding the connectedness shares fixed at their estimated values. They do not describe uncertainty in the first stage. Penalty selection, the elastic-net VAR coefficients, the twelve-month generalized forecast-error variance decomposition that together produce those shares and, for the factor-adjusted network, the extraction of each window's first principal component are all estimated, and none of that estimation error is carried into the reported precision. The estimated network moves with penalty selection from one window to the next. The factor-adjusted state index falls by 7.4 points between the August and September 1997 windows, as the number of cross-state slopes the penalty retains falls from 21.8 to 19.8 per equation.

The comparisons of this section show that the sign of the sending-side association does not depend on one discrete first-stage choice, and that the receiving side's does. A comparison of estimators is not a sampling distribution over networks. Obtaining one would require resampling the monthly panel, repeating penalty selection, the purge and the decomposition in every window, and re-estimating the second stage on each resulting network, all at the adopted specification. Whether and by how much that uncertainty would change the reported intervals is not established here, and the intervals in this paper should be read as conditional statements.

## 8 Conclusion

What connects local housing markets? Measured as a directed network of forecast-error dependence, they are strongly and increasingly connected. In the raw state network about two-thirds of a market's forecast-error variance was attributed to other markets in the late 1980s and more than 90 percent since 2020. Some markets account for several times as much of other markets' variance as others do, and dependence falls with distance, by less over time. Most of this is a common component whose weight has grown since the 2000s. It accounts for almost all of the rise in measured connectedness, for the flattening of the distance gradient and for much of the spread in net positions.

The purge cannot say whether its growth reflects common shocks or propagation among markets. Dependence orthogonal to the component has not grown, and it is not random.

More of other states' forecast uncertainty beyond the national cycle is traced to tightly regulated states, 7.5 log points more per standard deviation of the index in the benchmark network. Regulated states sit closer to being net sources of the system's dependence. The receiving side tells a different story. Regulated states track the national cycle more closely, and in a network estimated on raw price growth that raises their receiving side. Beyond the cycle, regulated metropolitan areas take more of their forecast-error variance from other areas where labor demand predicted from industry mix is strong. That is the ordering a binding supply constraint implies once durable housing is added to the model. The interaction keeps its sign within areas over time and, less precisely, on an apportioned node set. Steep terrain shows no such interaction. Across states, migration exposure adds nothing to distance, contiguity and region once source-by-year effects are held fixed, because at that scale it is close to a gravity function of the three. Across metropolitan areas in different states, migration exposure lines up with which markets a market's dependence comes from, and the association survives node-year effects beyond the national cycle in either metropolitan node set. In the benchmark node set it is carried by direct migration links and loses precision when nearby pairs are dropped. It is an association between exposure and the pairwise pattern of dependence that other pair-level affinities could also produce.

The central results hold up. The pooled sending-side association keeps its sign under every alternative network, sample and timing split. It stays precise under each own-lag treatment of the network, within Census regions and without any single state or region. Its limits say what it is. It is weakest in the windows since 2020, when the national cycle leaves least room for differences between states, and it is imprecise on average across metropolitan areas beyond the national cycle. It does not survive removing the larger number of common factors an information criterion selects in each window, so the factor-adjusted sending side may partly reflect regulated states' position within regional clusters. The net position holds mainly before 2005. Whether regulated markets receive more dependence beyond the cycle depends on how the autoregression shrinks each state's own lags, and the paper leaves it open. The metropolitan demand interaction on the receiving side holds under industry clustering, division and area omissions and both industry classifications. It keeps its sign, with less precision, on an apportioned node set. Its sending-side counterpart holds only in the benchmark node set. The metropolitan migration association among pairs in different states holds under node-year effects, and the data cannot say whether it has a direction within pairs.

These results give an economic reading to a reduced-form housing network. The spatial-equilibrium framework derives two comparative statics, one for a target's external weight and one for the ordering of its allocation shares across sources. The second holds among metropolitan areas in different states. There migration exposure lines up with the markets a market's dependence comes from, and regulation leaves that ordering alone, as the model predicts. The first is found beyond the national cycle only across states and only in the networks that shrink own lags less. What the receiving side follows instead is the timing the model implies once durable housing is

added. Regulated metropolitan areas take more of their variance from elsewhere where their own demand makes the constraint bind. The sending-side regularity, the most secure result of the paper, lies outside the formal model. I read it as a supply constraint that turns a market's own demand shocks into price movements that other markets respond to. The evidence supports that reading without isolating it from larger demand shocks in regulated markets. The variance shares measure forecast-error dependence rather than structural transmission, and the regulation index describes a market's land-use regime together with its demand and state-institutional content. The results are therefore associations that match the model's allocation prediction and its supply multiplier combined with durable housing. Two extensions would sharpen them. A national-plus-regional de-factoring along the lines of Bailey et al. (2016) would say how much of the sending-side pattern is structure among markets rather than regional common shocks. Carrying first-stage estimation uncertainty into the second stage would make the conditional intervals reported here unconditional.

## Data availability

The house-price and migration data are publicly available from Freddie Mac (Freddie Mac, 2026a) and the IRS Statistics of Income (Internal Revenue Service, 2024). The land-unavailability and Wharton regulation measures are available from their original authors (Lutz and Sand, 2026; Gyourko et al., 2008, 2021). State and county population estimates are public releases of the Bureau of Economic Analysis and the Census Bureau Population Estimates Program. The industry wages and earnings behind the labor-demand proxies are BEA regional accounts (U.S. Bureau of Economic Analysis, 2025), the natural-amenity scale is a USDA release (U.S. Department of Agriculture, Economic Research Service, 1999), and metropolitan county composition follows the 2023 Census delineation (U.S. Census Bureau, 2023). The replication package contains the code that estimates the networks and the second-stage regressions, the estimated networks, and the programs that build every table and figure of this paper from them. It also contains the programs that estimate the monthly networks, with their inputs, and the inputs to the comparisons between the state and metropolitan systems. The state steep land share can be rebuilt from the CBSA steep land shares and the Census population and housing-stock counts in the package, including the Colorado grouping and the Montana allocation. Some inputs to the metropolitan comparisons are built by separate programs, documented with their analysis panels. The package also contains the bootstrap draws, input checksums and execution logs of the state-cluster and network-sensitivity runs. Rebuilding the exhibits from the stored networks is separate from re-estimating every network from the raw data.

## References

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# A Data Construction and Aggregation

## A.1 House-price growth

Table A.1 reports monthly growth in the 48-state sample. Average appreciation and volatility differ across states, with many sample extrema occurring during the 1981–82 recession. California and Washington exhibit particularly large monthly declines, of about 10 percent in a month, and Minnesota of about 7 percent. In California the fall is followed within a year by a rise of similar size, and Washington's largest monthly rise is about 6 percent.[73]

[73]These months fall at the peak of mortgage rates—the 30-year fixed rate averaged 18.5 percent in October 1981 (Freddie Mac, 2026b)—when conventionally financed resales nearly stopped. California sales collapsed in 1981 while seller-financed transactions disguised price declines (Case and Shiller, 1987). The FMHPI pairs only sales financed by loans that Freddie Mac or Fannie Mae purchased, so in 1981–82 it rests on a thin and selected set of transactions, and repeat-sales indexes are noisy in thin samples (Goetzmann, 1992). The months are retained in the benchmark, as published. Re-estimating the eight annual networks whose windows contain them, 1985–1992, with each state's monthly growth winsorized at its 1st and 99th percentiles leaves every 1985–1988 coefficient negative (Section 7.5 and Online Appendix Table OA.7).

**Table A.1:** Monthly state house-price log growth, raw and with the national factor removed

| State | Mean | SD | Min | Max | State | Mean | SD | Min | Max |
|---|---|---|---|---|---|---|---|---|---|
| *Panel A. Raw growth, February 1975–December 2025, 611 months* | | | | | | | | | |
| AL | 0.317 | 0.450 | -1.085 | 1.980 | NC | 0.381 | 0.488 | -1.511 | 2.600 |
| AR | 0.300 | 0.538 | -3.510 | 3.145 | ND | 0.336 | 0.498 | -2.375 | 2.699 |
| AZ | 0.406 | 1.006 | -4.360 | 4.448 | NE | 0.344 | 0.411 | -0.707 | 1.648 |
| CA | 0.497 | 1.192 | -10.256 | 10.128 | NH | 0.420 | 0.660 | -1.470 | 2.229 |
| CO | 0.441 | 0.575 | -1.268 | 2.209 | NJ | 0.429 | 0.672 | -1.850 | 2.588 |
| CT | 0.378 | 0.682 | -1.248 | 3.461 | NM | 0.369 | 0.533 | -2.537 | 2.743 |
| DE | 0.390 | 0.507 | -0.848 | 1.672 | NV | 0.406 | 0.973 | -3.508 | 3.898 |
| FL | 0.408 | 1.078 | -4.802 | 6.346 | NY | 0.418 | 0.640 | -2.860 | 4.839 |
| GA | 0.356 | 0.551 | -1.690 | 2.109 | OH | 0.329 | 0.594 | -4.910 | 4.013 |
| IA | 0.319 | 0.403 | -1.484 | 1.744 | OK | 0.317 | 0.516 | -1.585 | 2.300 |
| ID | 0.412 | 0.722 | -1.530 | 3.304 | OR | 0.458 | 0.729 | -1.709 | 2.567 |
| IL | 0.345 | 0.718 | -4.137 | 4.503 | PA | 0.373 | 0.509 | -1.684 | 2.788 |
| IN | 0.328 | 0.547 | -4.140 | 3.492 | RI | 0.458 | 0.747 | -1.623 | 3.593 |
| KS | 0.327 | 0.458 | -1.781 | 1.847 | SC | 0.365 | 0.536 | -1.477 | 3.389 |
| KY | 0.335 | 0.371 | -0.713 | 1.805 | SD | 0.341 | 0.461 | -1.340 | 1.901 |
| LA | 0.313 | 0.473 | -1.636 | 2.047 | TN | 0.370 | 0.535 | -2.312 | 2.638 |
| MA | 0.475 | 0.651 | -1.181 | 2.493 | TX | 0.359 | 0.771 | -3.731 | 5.909 |
| MD | 0.384 | 0.676 | -2.117 | 4.104 | UT | 0.435 | 0.647 | -1.355 | 2.783 |
| ME | 0.422 | 0.633 | -1.398 | 2.377 | VA | 0.396 | 0.532 | -2.109 | 1.838 |
| MI | 0.349 | 0.739 | -4.278 | 5.776 | VT | 0.398 | 0.533 | -1.107 | 2.246 |
| MN | 0.362 | 0.868 | -7.047 | 7.480 | WA | 0.498 | 1.040 | -10.303 | 6.321 |
| MO | 0.346 | 0.697 | -3.482 | 4.558 | WI | 0.367 | 0.573 | -4.328 | 3.167 |
| MS | 0.275 | 0.554 | -3.299 | 2.119 | WV | 0.281 | 0.720 | -3.505 | 3.342 |
| MT | 0.427 | 0.632 | -2.180 | 2.772 | WY | 0.376 | 0.594 | -1.873 | 2.264 |
| *Panel B. First principal component removed, January 1976–December 2025, 600 months* | | | | | | | | | |
| AL | 0.007 | 0.313 | -1.361 | 1.503 | NC | 0.003 | 0.319 | -1.685 | 1.565 |
| AR | 0.005 | 0.333 | -1.198 | 1.497 | ND | 0.030 | 0.396 | -1.766 | 2.124 |
| AZ | 0.040 | 0.569 | -3.100 | 2.242 | NE | -0.006 | 0.328 | -0.941 | 1.148 |
| CA | 0.021 | 0.869 | -8.815 | 7.668 | NH | 0.000 | 0.456 | -1.390 | 1.533 |
| CO | -0.055 | 0.399 | -1.585 | 2.421 | NJ | 0.024 | 0.457 | -2.159 | 1.683 |
| CT | 0.029 | 0.480 | -1.830 | 2.574 | NM | -0.011 | 0.373 | -2.290 | 2.619 |
| DE | 0.001 | 0.326 | -1.022 | 1.039 | NV | -0.008 | 0.526 | -1.606 | 2.448 |
| FL | 0.034 | 0.684 | -3.993 | 5.061 | NY | 0.004 | 0.492 | -3.115 | 3.936 |
| GA | -0.024 | 0.329 | -1.773 | 1.663 | OH | 0.041 | 0.467 | -4.158 | 3.736 |
| IA | 0.023 | 0.299 | -0.979 | 1.292 | OK | -0.032 | 0.403 | -1.508 | 1.524 |
| ID | -0.033 | 0.439 | -1.270 | 1.493 | OR | -0.005 | 0.493 | -1.437 | 2.360 |
| IL | 0.028 | 0.603 | -4.491 | 4.409 | PA | 0.018 | 0.352 | -2.203 | 1.956 |
| IN | 0.057 | 0.390 | -3.011 | 1.875 | RI | 0.044 | 0.482 | -1.382 | 2.741 |
| KS | 0.014 | 0.337 | -1.738 | 1.766 | SC | 0.000 | 0.389 | -2.131 | 2.587 |
| KY | 0.014 | 0.257 | -0.905 | 0.936 | SD | -0.014 | 0.355 | -1.264 | 1.416 |
| LA | -0.027 | 0.363 | -1.623 | 1.101 | TN | 0.018 | 0.378 | -2.176 | 2.274 |
| MA | -0.046 | 0.468 | -1.274 | 1.381 | TX | -0.045 | 0.629 | -3.948 | 5.887 |
| MD | 0.028 | 0.422 | -1.484 | 3.087 | UT | -0.020 | 0.381 | -1.009 | 1.239 |
| ME | 0.035 | 0.459 | -1.976 | 1.499 | VA | 0.028 | 0.291 | -1.793 | 1.105 |
| MI | 0.067 | 0.592 | -5.462 | 4.765 | VT | 0.045 | 0.355 | -0.820 | 1.476 |
| MN | 0.016 | 0.692 | -5.523 | 7.150 | WA | -0.016 | 0.841 | -8.795 | 7.884 |
| MO | 0.005 | 0.523 | -3.073 | 3.110 | WI | 0.024 | 0.463 | -4.103 | 3.647 |
| MS | -0.009 | 0.492 | -3.516 | 2.071 | WV | 0.073 | 0.574 | -1.893 | 2.858 |
| MT | -0.006 | 0.406 | -1.523 | 1.261 | WY | -0.026 | 0.444 | -1.412 | 1.510 |

*Note:* The table reports $100\Delta \log P_{i,t}$ in percentage points. Each state has 611 monthly growth observations from February 1975 through December 2025. The VAR uses the corresponding log-growth series without the factor of 100. Alaska and Hawaii are excluded. The purged panel reports the growth rates on which the factor-adjusted network is estimated (Appendix C). For each annual network's window, these are the residuals from regressing each state's growth on the window's first principal component, kept for the window's final twelve months, and in the first window for all of its months.

## A.2 Migration flows and exposure

For the annual network of year $t$, I sum five annual IRS flow matrices from $t-15$ through $t-11$, using IRS interval-start labels. The final migration interval ends before the first year of the benchmark price-response window. If $Q_{ki,t}$ is the pooled flow from origin $k$ to destination $i$, the origin choice and target feeder shares are

$$\mu_t^{k\to i} = \frac{Q_{ki,t}}{\sum_r Q_{kr,t}}, \qquad \varphi_t^{i\leftarrow k} = \frac{Q_{ki,t}}{\sum_r Q_{ri,t}}. \tag{A.1}$$

Observed same-state exemption counts remain on the diagonal. Both denominators therefore use the full 48-state system. Exposure is $\Psi_t = \Phi_t M_t$, and only after the complete row-stochastic matrix is formed do the regressions remove its diagonal cells. Exposure is measured in percentage points, $100\Psi_t$, and its diagonal in unit-share form gives external openness $o_{i,t} = 1 - \psi_{ii,t}$. Because the stayers are on the diagonal, the direct terms of equation (8) dominate the off-diagonal cells, and the common-origin term averages 1.3 percent of external exposure (Online Appendix Table OA.33).

The 2005 network uses IRS interval-start vintages 1990–1994, and the sequence ends with 2010–2014 for the 2025 network. All objects use the same target-row/source-column orientation.

## A.3 Geography, economic size and the national price overlay

Interstate distance is the great-circle distance in kilometers between 2000 Census population centers. Contiguity denotes a shared state border, and common region denotes membership in the same Census region. Target average distance is the arithmetic mean of distances to the other 47 states, logged after averaging. The state size control is the logged BEA resident population for year $t-1$, available for every year from 1985. It falls inside the price window and supplies a size control, not pre-window identification, and it may itself respond to regulation. Online Appendix Table OA.3 therefore reports the sending-side regressions without it and with the state's 1980 population from the same BEA series in its place. The 1980 value precedes the end of every window, although it lies inside the windows ending in 1985–1989. The factor-adjusted TO association is 8.0 percent without the control, 7.5 with lagged population and 7.5 with 1980 population. GDP enters only the 2005–2025 variant reported in Online Appendix Section OA.2.1, as the logged BEA state series for year $t-1$, real GDP, all-industry total, in millions of chained 2017 dollars (BEA SAGDP9, line 1; equivalently SASUMMARY, line 1 for overlapping years), which does not reach back before 1997.

The metropolitan size controls come from Census Bureau county population estimates for 1980–2024. They combine the 1980s and 2000s intercensal series, the CO-99-8 estimates for the 1990s, and the Vintage 2020 and Vintage 2024 postcensal series, one source per year, put on the 2020 county geography (Dade to Miami-Dade, the Virginia independent cities that merged into their counties, and Shannon to Oglala Lakota). Connecticut's Vintage 2024 figures are for planning regions, so its counties carry their 2020 shares of the state total from 2020 on, and two corrupted 1994 cells of the CO-99-8 release are interpolated. The county sums reproduce every state total in

the releases to within 0.03 percent. A CBSA's population is the sum over its 2020 member counties. The outside-node-set share is the share of the population within 200 kilometers of the CBSA's internal point that lives in counties belonging to no sample area, a county counting as within the radius when its internal point is. Both are for year $t-1$.

The national real-price overlay divides the national (US) column of the same FMHPI file by the BLS seasonally adjusted CPI-U and normalizes the ratio to 100 in January 2000. Like the state network estimates, it runs through December 2025. October 2025 has no published CPI observation, is not interpolated and is drawn as a one-month gap.

### A.4 Housing-constraint aggregation

The constraint proxies originate at different geographic units, so a state-level measure requires an explicit aggregation rule. The target-level models use the Lutz–Sand share unavailable because of steep slopes, aggregated from whole metropolitan and micropolitan areas using 1990 total housing stock and the cross-state allocation described below. The primary Wharton indexes are equal-jurisdiction means among responding jurisdictions in each state,

$$\mathrm{WRLURI}_{s,v} = \frac{\sum_{r\in s} w_{r,v}\mathrm{WRLURI}_{r,v}}{\sum_{r\in s} w_{r,v}}, \qquad w_{r,v} = 1, \tag{A.2}$$

for state $s$ and vintage $v \in \{2006, 2018\}$. Larger values of each proxy denote greater constraint. The Lutz–Sand series and both Wharton vintages cover all 48 contiguous states. The empirical measures are proxies for the model's unobserved inverse supply parameter $\eta_i^H$. The jurisdiction survey carries no population to weight by, so no population-weighted WRLURI series is reported. The Wharton inverse-probability survey weights and Census jurisdiction land area produce genuinely different state measures. The Saiz land measure of Table 5 is genuinely population-weighted. It averages the unavailable-land shares of each state's metropolitan areas with 2000 population weights, each area counted in the state named first in its title. The metropolitan index of Appendix D.1 follows the same equal-jurisdiction rule over the jurisdictions of each metropolitan area. State coverage means that a value is available for every state, not that responding jurisdictions represent its entire housing market. Equal-jurisdiction means may therefore reflect uneven survey coverage. The unchanged 2006 and 2018 state averages have a cross-state correlation of 0.561. Their unstandardized state-index standard deviations are 0.710538 and 0.422642, respectively; each vintage is standardized over the 48 states. This compares survey vintages as well as potentially different sets of responding jurisdictions; the 2018 survey also falls within the estimation period rather than preceding all of it. Table 5 reports the vintage comparison and Online Appendix Table OA.36 the aggregation checks.

**Land constraints over local housing markets.** The benchmark and alternative land measures retain each Lutz–Sand core-based statistical area (CBSA) as a whole, including its outlying counties. The 2020 county-defined geography contains both metropolitan and micropolitan areas. To allocate a market that crosses state borders, let $n_{ms}^{b,v}$ be population or total housing stock (basis $b$) in the

counties of CBSA $m$ that lie in state $s$, measured in Census year $v \in \{1990, 2000\}$. Let $n_m^{b,v} = \sum_s n_{ms}^{b,v}$. The cross-state allocation share and normalized within-state weight are

$$a_{ms}^{b,v} = \frac{n_{ms}^{b,v}}{n_m^{b,v}}, \qquad \omega_{ms}^{b,v} = \frac{a_{ms}^{b,v} n_m^{b,v}}{\sum_k a_{ks}^{b,v} n_k^{b,v}} = \frac{n_{ms}^{b,v}}{\sum_k n_{ks}^{b,v}}. \tag{A.3}$$

The state proxy is $\sum_m \omega_{ms}^{b,v} \ell_m$, where $\ell_m$ is the whole-CBSA share unavailable because of steep slopes. A CBSA entirely within a state has allocation share one; a cross-border CBSA contributes according to that state's share of its population or housing stock. This allocation is applied once, before the within-state weights are normalized. The benchmark uses total housing stock in 1990. All four combinations of weighting basis and Census year are reported.

These measures describe constraints over covered local housing markets, including land outside their inhabited centers, rather than over every county in the state. They use the released unbuffered CBSA polygons. The supplied positive-buffer fields duplicate those values and do not establish an additional surrounding-land measure. The 2020 boundaries provide a common geographic definition, not historical market boundaries for 1990 or 2000. Two historical boundary changes require additional steps. Broomfield County, Colorado, was created after both weighting years from parts of Adams, Boulder, Jefferson and Weld counties. I combine the Denver, Boulder and Greeley CBSAs so that these historical county counts enter one market, without splitting parent counties or adding Broomfield's population twice. The combined slope share is an area-weighted mean of the three released unbuffered shares, using 2020 Census land plus water area. It equals 21.2013 percent; land-area-only weights give 21.1869 percent. Because the release does not supply the raster denominators behind its shares, this is an approximation to the combined footprint's steep land share. The alternative area calculation measures sensitivity to that convention, not a bound on raster error.

For the former Yellowstone county equivalent in Montana, the documented population transfer assigns 52 residents to Park County. The locations of its 46 housing units in 1990 are not observed. I assign them to Park in the benchmark and to Gallatin in the alternative. Park lies outside the retained CBSA geography, whereas Gallatin belongs to Bozeman; the alternative therefore adds 46 units to the covered housing-stock denominator. It changes Montana's state steep land share by 0.000234 percentage points. These two allocations bound the calculation conditional on distributing the units between those counties.

All four weighting rules, both boundary-allocation checks and the county slope comparator cover the same 48 states. Each physical-constraint measure is standardized over those states, and WRLURI keeps its standardization across the 48 states. The county comparator uses the unbuffered county steep land shares and 1990 one-unit housing-stock weights. Online Appendix Table OA.36 reports the resulting associations.

# B VAR and GFEVD Diagnostics

## B.1 Estimator choices and the fitted benchmark

Three choices of the benchmark are not neutral for what the network measures, and this subsection gives the reasons for each. The lag order is the one the Bayesian information criterion (BIC) supports. Computed on the benchmark estimator itself at the 41 annual networks, with the penalized fit's nonzero coefficients as degrees of freedom, BIC rejects one lag by 857 and three lags by 430 on average, and two lags has the lowest criterion in 40 of the 41 years (Online Appendix Table OA.1). With each equation refitted by least squares on its retained slopes, one lag is rejected by 2,005 while three lags lowers the criterion by only 118, so on that count the criterion does not separate two lags from three and the shorter is taken. Two lags is also the order that the information criteria select in the connectedness applications this design follows, the original Diebold–Yilmaz spillover index (Diebold and Yilmaz, 2009) and the regional housing system of Antonakakis et al. (2018). The 120-month window is not varied. It also sets the ten years over which the first principal component is removed and the labor-demand shock is predicted, and the migration pools end before it begins, so a different window would change those objects as well as the network. The penalty is selected at the minimum of validation loss rather than by the one-standard-error rule, because the one-standard-error rule retains only half to two-thirds as many cross-state slopes and the cross-state lag structure is what the measurement exists to recover. The own lags carry the cross-lag penalty for the same reason, since exempting them thins the cross-state block. None of these choices optimizes residual whiteness, which is not a selection criterion here. The own-lag alternatives of Section 7.1, which trade cross-state structure for cleaner residuals, are carried as robustness checks on the benchmark, and that section states the exchange they require rather than presenting it as a repair.

Three diagnostics describe how the fitted system behaves at the 41 annual networks. It is dynamically stable there (Section 2). It leaves no state with an empty equation and no incoming links, since the penalty floor binds for two of the 1,968 annual-network equations and no equation sits at the upper end of the path, where every coefficient would be shrunk to zero. And stability is not the same as well-behaved residuals. The median absolute lag-one autocorrelation of the fitted residuals is 0.39 in sample, against 0.86 for monthly price growth itself over the same windows, so the residuals remain serially dependent at this lag order; in the annual networks from 2005 the in-sample figure is 0.38. On the validation blocks that did not train on each month the median is 0.60. That figure is higher partly because of how the held-out series is assembled, from twelve blocks predicted by twelve fold models, each with its own offset. In five years checked (1985, 1995, 2005, 2015 and 2025), removing each block's mean lowers the held-out median from 0.61 to 0.51, and block means carry 22 percent of the held-out residual variance. Freeing the own lags from the penalty and extending them to eight would weaken the dependence considerably, and the price is stated rather than paid. The eight-own-lag system has no stable variance decomposition in nine of the 41 annual networks, 1985 and 2002 through 2009, and it retains 5.7 active cross-state slopes per equation against the benchmark's 23.5 over all 41 annual networks. Section 7.1 reports both sides

of that exchange, and the rest of this appendix reports the validation design, the penalty path and the full stability and residual diagnostics for the benchmark and the two own-lag alternatives.

## B.2 Validation design and penalty path

The benchmark estimates 48 equations with 96 lagged slopes in a VAR(2) and 120 response months. The ordinary elastic net has a mixing weight of 0.5, a 100-point penalty path with lower ratio $10^{-4}$, and hv-block cross-validation (Burman et al., 1994; Racine, 2000). The 120 response months are cut into twelve contiguous blocks of ten, each block is held out in turn, and the months on either side of it are dropped from the training sample, as many as the maximum lag order of the equation—two for the benchmark and the own-lags-free network, eight for the eight-own-lag network. Plain $K$-fold validation is valid for autoregressions only when the errors are uncorrelated (Bergmeir et al., 2018); with serially correlated residuals, the buffer keeps the months adjacent to a validation block from training on the observations they are then asked to predict. The penalty is chosen equation by equation at the minimum of the cross-validated error. Own lags carry the same penalty as cross-state lags. Two alternatives, which exempt the own lags from the penalty at the benchmark lag order and with eight own lags, are carried beside the benchmark for the state directional and allocation results, the diagnostics of Table B.1 and the descriptions of Figures 4 and 6. The metropolitan results use the benchmark alone (Appendix D.2).

Which own-lag treatment is the benchmark matters for the object this paper measures and for the index that summarizes it. On the 16 years from 2010 in which all three networks are stable, mean systemwide connectedness is 87.4 percent under the benchmark, 83.0 with the own lags free and 72.1 with eight own lags, and in the annual networks from 2005 the networks beneath those levels thin from 25.4 active cross-state slopes per equation to 13.2 and 6.7. The benchmark is chosen by the information criterion for the lag order and by the cross-state structure for the penalty and the own-lag treatment. That structure is what the allocation results use directly. The external-connectedness and directional results use its row and column sums, whose regulation gradients do move with the own-lag treatment (2.8 percent under the benchmark against 5.5 with the own lags free); they are reported under the benchmark for consistency with the allocation analysis, and Section 7.1 carries the alternatives beside them rather than choosing among them.

The benchmark and the own-lags-free network are dynamically stable in all 41 annual networks, with largest companion-matrix radii of 0.978 and 0.986. In those networks the benchmark places no equation at the upper end of the penalty path, while the own-lags-free network places 215 of its 1,968 annual-network equations there, each keeping its unpenalized own lags and at most one cross-state slope. Stability does not extend to every window of the monthly run. The benchmark companion matrix has an eigenvalue outside the unit circle in six of the 486 monthly windows—March through June 2004 and March and April 2005—and the own-lags-free network at twelve, in the same two springs and in 2008. The eight-own-lag network is unstable at 110 of its 484 monthly windows, concentrated in 2002–2012, and in the annual networks of 1985 and 2002 through 2009. A companion matrix with an eigenvalue outside the unit circle has no convergent moving-average representation

and therefore no variance decomposition, so those windows are not estimated networks. They are flagged and left blank in the monthly exhibits rather than patched with a penalty guard, which would remove the instability from view without removing it from the estimator, and the eight-own-lag second stage is estimated on the 32 raw and 38 factor-adjusted annual networks for which that system exists (the 16 from 2010 for the allocation regressions).

The penalty floor binds for two of the 1,968 annual-network equations under the benchmark (one of the 1,008 from 2005) and for none under either alternative, and across the 486 monthly windows it binds for at least one equation in 22 of them under the benchmark and at none under the alternatives, because a free own-lag block absorbs the variance that would otherwise drive the penalty to the floor of its path. Under the benchmark no equation in any annual network sits at the upper end of the path, where every coefficient would be shrunk to zero. Under the two alternatives, whose own lags are not penalized, 215 and 463 of the 1,968 annual-network equations do, and they keep their own lags.

Two qualifications limit the reading of the Ljung–Box rejection rates that accompany this estimator. First, these are residuals after equation-specific elastic-net selection and shrinkage, not residuals from a fixed unpenalized regression (Zou and Hastie, 2005). Shrinkage bias can mechanically leave serial dependence, and the conventional Ljung–Box reference distribution is not calibrated for this post-shrinkage problem. Second, the published FMHPI series is seasonally adjusted using X-12 or an X-12-ARIMA variant (Karamon, 2023); for interior observations these procedures use symmetric moving-average filters with leads and lags (Findley et al., 1998). Such two-sided filtering can itself carry dependence across adjacent observations. The rejection share of 100 percent of equations on the benchmark's held-out residuals is therefore reported as a descriptive warning and should not be read as a literal 48-equation model-failure rate. The two alternatives show what removing the dependence costs. The median absolute lag-one autocorrelation of the held-out residuals is 0.60 under the benchmark, 0.37 with the own lags free and 0.17 with eight own lags, at which point Ljung–Box rejects for 37 percent of equations. The next subsection reports the stability and residual diagnostics for the three networks at the 48-state scale, and Section 7.1 the sensitivity of the headline outcomes to the choice among them.

### B.3 Diagnostics under the own-lag alternatives

The paper reports two alternative specifications that depart from the benchmark in the own-lag treatment alone. On the same validation design, at the same 120-month rolling window, the same cross-state lag order, the same mixing parameter and the same penalty-path floor, the own lags are exempted from the penalty, first at the benchmark lag order and then extended to eight. In every network the buffer around each validation block equals the maximum lag order, two months for the two VAR(2) networks and eight for the eight-own-lag network, so that no training observation shares a lag window with a validation observation. Table B.1 sets the three networks beside each other, in their raw and factor-adjusted forms, on the quantities that decide whether a decomposition exists at all and how it is composed. These are dynamic stability, the density of the fitted network,

the size of the selected penalty, residual dependence in sample and on the validation blocks, and the connectedness index, over all 41 annual networks and over the 21 from 2005, and, at monthly rather than annual frequency, stability and how often penalty selection reaches the floor of the path.

The residual columns summarize first-order dependence rather than a whole autocorrelation profile. The in-sample and held-out entries are medians of the absolute lag-one residual autocorrelation taken across the 48 equations and the annual networks, where the held-out residual of each month is taken from the validation block that did not train on it, at the selected penalty, and the Ljung–Box column is the share of equations rejecting the twelve-lag test at a nominal five percent on those held-out residuals. Freeing the own lags lowers the in-sample median from 0.38 to 0.21 and the held-out median from 0.60 to 0.37, and eight own lags bring them to 0.04 and 0.17, with Ljung–Box rejecting for 37 percent of equations against 100 and 98 percent. Conventional Ljung–Box critical values are not calibrated for equation-specific penalty selection, so these shares describe the residuals rather than test them. By the same token a low rejection count would not establish a serially independent multivariate innovation sequence. What the three networks establish is that the residual dependence of the benchmark can be reduced considerably within the estimator, and what reducing it costs. In the annual networks from 2005, active cross-state slopes per equation fall from 25.4 to 13.2 to 6.7, and in the 16 years from 2010 in which all three are stable mean systemwide connectedness falls from 87.4 to 83.0 to 72.1, as each equation explains more of its own variance before any cross-state term is reached. Over all 41 annual networks the benchmark's in-sample median is 0.39 and its held-out median 0.60, and the own-lags-free and eight-own-lag networks bring the in-sample median to 0.22 and 0.04. Held-out residuals for the raw alternatives are computed only for the annual networks from 2005.

In the factor-adjusted re-estimates, removing each window's national factor lowers systemwide connectedness in every network, from 78.1 to 65.6 at the benchmark, raises the selected penalty, and leaves the eight-own-lag network unstable in three of the 41 annual networks rather than nine. The benchmark's held-out residual dependence is unchanged, 0.60 in both networks.

### B.4 The distance gradient in the raw price data

Whether the flattening of the distance gradient in Figure 3 is a fact about house prices or an artifact of how the network is estimated can be checked without the network. Figure B.1 computes, in each monthly window and over the same 120 months, the Pearson correlation of monthly log price growth for all 1,128 unordered state pairs, and reports how strongly those correlations line up with proximity. No VAR, no penalty and no variance decomposition enters this correlation-based measure. Panel (c) sets it beside the estimated gradient, each standardized over the months both cover, where they correlate at 0.68. The two share their main peak, 0.54 for the raw measure in November 1995 and 0.54 for the estimated gradient in January 1994, and their trough in the mid-2010s, when the correlation-based measure is at or slightly below zero (its minimum is $-0.016$ in November 2016) and the estimated gradient falls to 0.06. They diverge in between and at the end. The correlation-based measure falls to 0.14 in September 2005, when the estimated gradient is

**Table B.1:** Network diagnostics under the own-lag alternatives, raw and factor-adjusted networks

| Network | Stable | Max. radius | Cross/ eq. | Own/ eq. | Rel. $\lambda$ | \|ACF(1)\| in | \|ACF(1)\| out | LB out (%) | DYCI | Floor eq. |
|---|---|---|---|---|---|---|---|---|---|---|
| **Panel A. All 41 annual networks, 1985–2025** | | | | | | | | | | |
| *Raw networks* | | | | | | | | | | |
| Benchmark | 41/41 | 0.978 | 23.5 | 1.62 | 0.029 | 0.387 | 0.602 | 99.9 | 78.1 | 2 |
| Own-lags-free | 41/41 | 0.986 | 11.2 | 2.00 | 0.327 | 0.218 | – | – | 71.6 | 0 |
| Eight-own-lag | 32/41 | 1.074 | 5.7 | 8.00 | 0.628 | 0.041 | – | – | 66.0 | 0 |
| *Factor-adjusted networks* | | | | | | | | | | |
| Benchmark | 41/41 | 0.945 | 21.7 | 1.67 | 0.067 | 0.395 | 0.598 | 100.0 | 65.6 | 0 |
| Own-lags-free | 41/41 | 0.953 | 8.1 | 2.00 | 0.521 | 0.221 | 0.372 | 97.5 | 56.2 | 0 |
| Eight-own-lag | 38/41 | 1.032 | 4.1 | 8.00 | 0.830 | 0.027 | 0.167 | 47.7 | 48.9 | 0 |
| **Panel B. The 21 annual networks 2005–2025, where the allocation regressions sit** | | | | | | | | | | |
| *Raw networks* | | | | | | | | | | |
| Benchmark | 21/21 | 0.978 | 25.4 | 1.65 | 0.018 | 0.381 | 0.597 | 100.0 | 85.3 | 1 |
| Own-lags-free | 21/21 | 0.986 | 13.2 | 2.00 | 0.272 | 0.207 | 0.370 | 98.1 | 80.5 | 0 |
| Eight-own-lag | 16/21 | 1.036 | 6.7 | 8.00 | 0.475 | 0.045 | 0.171 | 36.6 | 72.1 | 0 |
| *Factor-adjusted networks* | | | | | | | | | | |
| Benchmark | 21/21 | 0.945 | 22.0 | 1.68 | 0.067 | 0.385 | 0.586 | 100.0 | 64.8 | 0 |
| Own-lags-free | 21/21 | 0.909 | 8.8 | 2.00 | 0.475 | 0.214 | 0.370 | 99.6 | 55.1 | 0 |
| Eight-own-lag | 20/21 | 1.011 | 4.2 | 8.00 | 0.756 | 0.028 | 0.173 | 44.6 | 43.4 | 0 |
| **Panel C. Monthly windows, July 1985–December 2025** | | | | | | | | | | |
| *Raw networks* | | | | | | | | | | |
| Benchmark | stable in 480 of 486 monthly windows; at least one equation at the penalty floor in 22 | | | | | | | | | |
| Own-lags-free | stable in 474 of 486 monthly windows; at least one equation at the penalty floor in 0 | | | | | | | | | |
| Eight-own-lag | stable in 374 of 484 monthly windows; at least one equation at the penalty floor in 0 | | | | | | | | | |
| *Factor-adjusted network* | | | | | | | | | | |
| Benchmark | stable in 484 of 486 monthly windows; at least one equation at the penalty floor in 1 | | | | | | | | | |

*Note:* In Panels A and B the raw rows use the raw networks and the factor-adjusted rows the factor-adjusted re-estimates of the same three specifications. The factor-adjusted network is re-estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Networks are estimated at the benchmark specification of Section 2; the own-lags-free network exempts the own-lag coefficients from the penalty, and the eight-own-lag network additionally extends each equation's own lags to eight. The columns report the annual networks that are dynamically stable (companion spectral radius below one), the largest radius, the mean active cross-state and own slopes per equation, the median selected penalty relative to the path maximum, the median absolute lag-one residual autocorrelation across equations and annual networks, in sample and on the validation block that did not train on each month, the share of equations rejecting a twelve-lag Ljung–Box test at 5 percent on those held-out residuals, systemwide connectedness (DYCI, percentage points) averaged over the stable annual networks, and the number of annual-network equations whose selected penalty sits at the path floor. The eight-own-lag network is unstable in the annual networks of 1985, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009 (raw) and 1985, 2003, 2006 (factor-adjusted); the benchmark and the own-lags-free network are stable at all 41 in both. Held-out residuals were computed for the raw own-lag alternatives only in the 21 annual networks from 2005, so their Panel A held-out entries are not available; the factor-adjusted runs kept them in all 41. In the monthly windows only the benchmark was estimated in factor-adjusted form, so Panel C has one factor-adjusted row. Ljung–Box critical values are not calibrated for equation-specific penalty selection, so the rejection shares describe the residuals rather than test them.

at a second peak near 0.45, and it recovers to 0.34 in September 2023 before ending the sample at 0.21, while the estimated gradient stays near 0.12. The peak and the trough are therefore properties of the price data. The estimated gradient's plateau through the mid-2000s and its flatness after 2020 are not reproduced by the correlation-based measure.

The second panel draws the distribution of the 1,128 pair correlations in each window—its median, interquartile range and full range—and answers a question the estimated gradient cannot. Comovement and its spatial organization move inversely. In the mid-1980s states barely move together at all—a median pair correlation of 0.19 at December 1985, with an interquartile range of 0.05 to 0.33 and pairs running from −0.47 to 0.84—and what comovement there is has no spatial logic. The most correlated pairs of the 1976–1985 window are Arizona with Indiana at 0.84, Ohio with Washington at 0.81, Michigan with South Carolina at 0.77. The whole distribution moves up between 2007 and 2009. The median rises from 0.21 at December 2005 to 0.68 at December 2009 and the lowest pair correlation turns positive. By the 2016–2025 window the median is 0.73, the interquartile range 0.64 to 0.79, and the tightest pairs are neighbors, Idaho with Utah and Arizona with Utah at 0.95, California with Colorado, Colorado with Utah and Florida with Georgia at 0.93. Yet it is precisely in those late, highly synchronized windows that distance explains least about which states move together, until the correlation-based measure recovers after 2020. For most of the sample, then, the periods of strongest comovement are the ones in which proximity organizes it least.

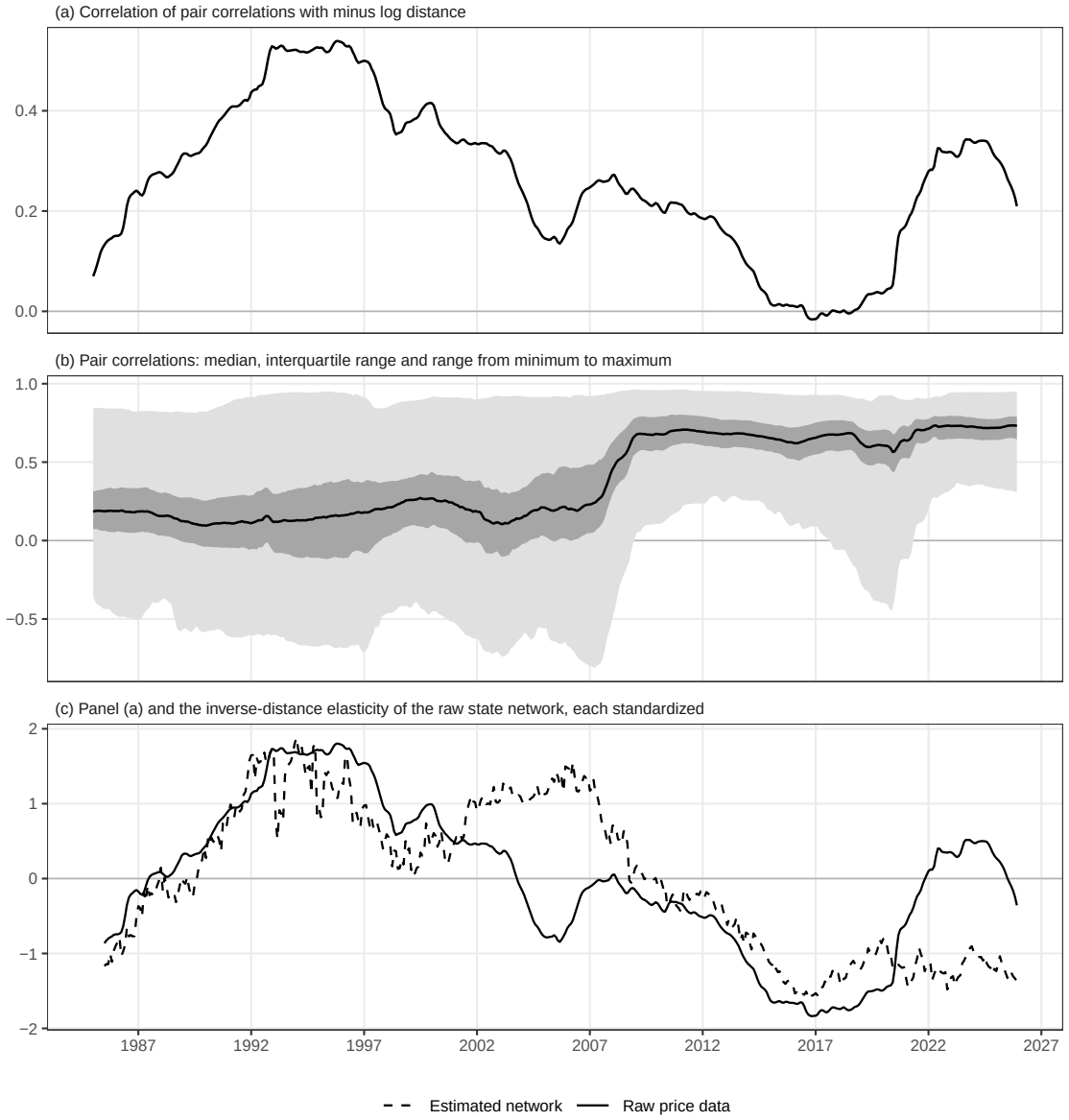



**Figure B.1:** Comovement and its spatial organization in the raw price data, January 1985–December 2025

*Note:* Panels (a) and (b) do not use the estimated network. In each monthly window the Pearson correlation of monthly log house-price growth is computed over the trailing 120 months for all $48 \times 47/2 = 1{,}128$ unordered state pairs. Panel (a) reports the correlation between those 1,128 pair correlations and minus log great-circle distance, the raw-data counterpart of the elasticity in Figure 3. Panel (b) reports the median of the 1,128 correlations (line), their interquartile range (dark band) and their minimum-to-maximum range (light band). Panel (c) overlays panel (a) and the state elasticity of Figure 3, each standardized over the months in which both exist. The windows are 120 months long and overlap, so the series are smooth by construction and successive values are not independent. All three panels are descriptive and no test is reported on them.

# C Factor-Adjusted Networks

## C.1 Construction

The factor-adjusted networks of Section 2.3 apply the benchmark estimator to growth rates from which each window's first principal component has been removed. For a window $t$, let $Y$ be the $120 \times N$ matrix of monthly log growth in the response window. Each column is standardized with the window's own mean and standard deviation, the first right singular vector $r$ of the standardized matrix gives the factor score $f_s = z_s' r$ for month $s$, signed so that it correlates positively with the cross-state mean of growth, and each state's growth is regressed on $f_s$ and a constant over the 120 response months. The residuals are the factor-adjusted series; the lag rows before the window are projected with the same means, standard deviations, singular vector and coefficients. Inside every hv-block fold the four steps are repeated on the fold's training months alone, and the validation block, the buffer months and the lag rows are projected through the fold's own purge, so no validation month informs the purge used to score it. The final fit uses the window-level purge. The metropolitan system purges its own first component across its 100 areas. For each state, 100 times the $R^2$ of its growth on the window's factor score is the share of its growth variance on the first principal component used in Section 2.3, in Figure C.1 and in Table C.2.

The national factor has come to dominate state house-price growth. Averaged across states and windows, the first component accounts for 46 percent of a state's growth variance. The share rises from 22 percent in the windows ending in 1985–1989 to 27 percent in the 1990s, 38 percent in the 2000s, 67 percent in the 2010s and 69 percent since 2020, and in every window it differs widely across states (Figure C.1).

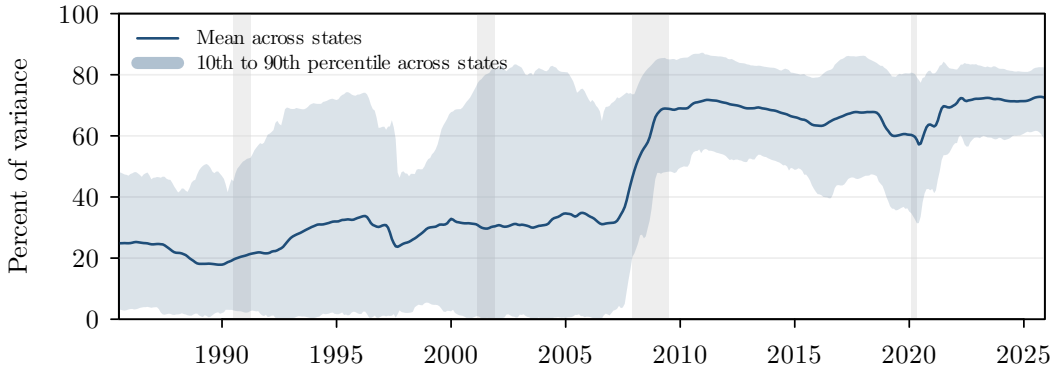



**Figure C.1:** First-component share of state growth variance, July 1985–December 2025

*Note:* In every monthly window, the share of each state's monthly growth variance accounted for by the first principal component of the window's standardized 48-state growth panel. The line is the mean across states, which for standardized growth is the component's share of total variance, and the band the range from the 10th to the 90th percentile across states. Shaded areas denote NBER recessions.

The purge lowers both the first-component share and the mean absolute pairwise correlation of the series in 39 of the 41 annual windows of each network. The two exceptions, 1997 and 2006 for

the states, are windows with a weak common factor, in which the mean absolute correlation still falls. The factor-adjusted state benchmark, the network with free own lags and the metropolitan network are dynamically stable in all 41 annual networks, and the eight-own-lag network in 38. The smallest factor-adjusted external connectedness is 32.5 percentage points, so the log outcomes are well defined, and the within-year standard deviation of log external connectedness is 0.189, against 0.122 raw.

## C.2 What the purge leaves

Table C.1 asks two questions of the purge. The first is whether the rolling first component is the same factor from one window to the next. One month apart its loadings correlate almost perfectly at the median window and by at least 0.83 at every one. Twelve months apart the median correlation is 0.95, but at a few windows with a weak common factor the component changes identity from one year to the next, so in those networks the purge removes a different common direction. The second is what the purge leaves. The first component's share of standardized growth variance falls from a median of 0.33 to 0.21 across the monthly windows, and the number of factors selected by the Bai and Ng (2002) criterion from about four to two. What remains is regional. After the purge the mean pairwise correlation of states in the same Census region is 0.11 at the median window, against $-0.04$ for states in different regions. A single-factor purge therefore leaves regional common structure in place, which is why the text reads a factor-adjusted result as structure beyond the national cycle rather than as spillovers among markets. Online Appendix Table OA.8 re-estimates the state network after removing the number of components the criterion selects on each window's raw growth, after which it selects none at 38 of the 41 windows. The sending-side gradient does not survive that purge (Section 7.1). A national-plus-regional de-factoring that separates the two is not run here.

Table C.2 reports the regression behind the tracking statement of Section 6.2, of a state's share of growth variance on the first principal component on standardized WRLURI and the regressors of Table 2, with year effects and the inference of that table. A standard deviation of WRLURI is associated with 8.07 percentage points more of a state's growth variance on the first component, with a cluster-robust standard error of 1.18 and a 95-percent interval of $[5.42, 10.80]$, against a mean share of 45.9 percent, and lagged log population also enters positively. Panel B describes the share by decade of the window and by quarter of the regulation distribution. The regression is a diagnostic of why the raw receiving-side gradient arises, not a separate finding. It needs no network, only the principal components of each window.

## C.3 Systemwide connectedness and the distance gradient

Online Appendix Figure OA.3 repeats Figure 4 for the 100 metropolitan areas. As at state scale, the raw and factor-adjusted metropolitan indexes are weakly negatively correlated month by month, $-0.33$, so the rise of metropolitan connectedness also runs through the national factor.

**Table C.1:** The first-component purge, its stability across windows and what it leaves

| | Median | P10 | P90 | Minimum |
|---|---|---|---|---|
| *Panel A. Rolling-factor consistency* | | | | |
| Windows one month apart (485 comparisons, 119 shared months) | | | | |
| Loading correlation | 0.999 | 0.997 | 1.000 | 0.828 |
| Score correlation on the shared months | 1.000 | 0.999 | 1.000 | 0.877 |
| Purged-value correlation on the shared months | 1.000 | 0.999 | 1.000 | 0.920 |
| Annual windows twelve months apart (40 comparisons, 108 shared months) | | | | |
| Loading correlation | 0.946 | 0.817 | 0.983 | 0.014 |
| Score correlation on the shared months | 0.999 | 0.971 | 1.000 | 0.060 |
| Purged-value correlation on the shared months | 0.979 | 0.943 | 0.993 | 0.706 |
| *Panel B. What the purge leaves, 486 monthly windows* | | | | |
| First-component share of standardized growth variance | | | | |
| Raw growth | 0.335 | 0.219 | 0.714 | 0.178 |
| After the purge | 0.212 | 0.157 | 0.312 | 0.129 |
| Number of factors, Bai–Ng ICp2 | | | | |
| Raw growth | 4 | 3 | 8 | 2 |
| After the purge | 2 | 2 | 5 | 1 |
| Mean pairwise correlation of growth | | | | |
| Same Census region, raw | 0.342 | 0.232 | 0.739 | 0.191 |
| Different Census regions, raw | 0.221 | 0.089 | 0.692 | 0.065 |
| Same Census region, after the purge | 0.111 | 0.032 | 0.191 | 0.024 |
| Different Census regions, after the purge | -0.037 | -0.069 | 0.162 | -0.077 |
| Loading correlation with the full-sample first component | 0.289 | 0.071 | 0.490 | 0.003 |

*Note:* The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The purge is that of the factor-adjusted state benchmark. Within each window, the first principal component of the standardized 48-state growth panel and an intercept are removed from each state's growth by least squares (Appendix C). Panel A compares the purges of windows one month apart, for every monthly window ending from August 1985 through December 2025, and of consecutive annual windows, twelve months apart. Each comparison reports the absolute correlation of the two windows' loadings, the absolute correlation of their factor scores on the months they share, and the correlation of the purged values of those months. Panel B describes each monthly window from July 1985 through December 2025 by the first principal component's share of the variance of the standardized panel, before and after the purge; the number of factors selected by the ICp2 criterion of Bai and Ng (2002); the mean pairwise correlation of states in the same and in different Census regions; and the absolute correlation of the window's loadings with those of the first component of the whole panel from August 1975 through December 2025. Columns report the distribution of each statistic across the comparisons or windows. No network is estimated for this table; it describes the growth panel on which the factor-adjusted network is estimated.

The factor-adjusted distance gradients of both systems are in the main text, beside the raw series in Figure 6, because the raw estimate lies outside the factor-adjusted interval in about two-thirds of the months at the state scale and four-fifths at the metropolitan one, and because the metropolitan factor-adjusted trend is positive while the raw one falls.

**Table C.2:** Regulation and comovement with the national factor, 1985–2025

*Panel A. Regression of the share of a state's growth variance on the first component (pp)*

| | Estimate | Standard error | 95-percent interval |
|---|---|---|---|
| WRLURI 2006 | 8.07*** | (1.18) | [5.42, 10.80] |
| Steep land share | 0.61 | (1.09) | |
| log average distance | −2.31 | (5.74) | |
| Lagged log population | 2.30** | (0.93) | |
| Year fixed effects | ✓ | | |
| State-years | 1,968 | | |
| Within $R^2$ | 0.164 | | |
| Mean (SD) of the dependent variable | 45.88 (28.84) | | |

*Panel B. Mean share of growth variance on the first component (pp)*

| | All 48 states | Most regulated quarter | Least regulated quarter |
|---|---|---|---|
| Annual networks 1985–1989 | 21.48 | 18.41 | 21.11 |
| Annual networks 1990–1999 | 28.01 | 40.65 | 17.10 |
| Annual networks 2000–2009 | 40.51 | 60.63 | 21.45 |
| Annual networks 2010–2019 | 66.68 | 73.75 | 55.52 |
| Annual networks 2020–2025 | 70.26 | 73.22 | 65.74 |
| All annual networks 1985–2025 | 45.88 | 55.65 | 35.14 |

*Note:* 48 states and 41 annual networks 1985–2025 (1,968 state-years). The dependent variable is 100 times the $R^2$ of a state's monthly growth on the first principal component of the standardized 48-state growth panel in the network's 120-month window, which is the share of its growth variance that the purge of the factor-adjusted network removes. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The share is computed from raw growth and needs no network. Panel A regresses it on the regressors of Table 2 with year effects; coefficients are percentage points per standard deviation of a standardized regressor and per unit of a logged control. Parentheses contain cluster-robust standard errors, clustered by state. $^{*}$, $^{**}$ and $^{***}$ denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-$t$ $p$-values that impose the null (9,999 draws). Brackets contain the 95-percent interval that inverts the same bootstrap test. Intervals are reported for WRLURI only. Panel B averages the share over state-years by period of the network's year, across all states and within the most and least regulated quarters of the WRLURI 2006 distribution (12 states each). The regression is a diagnostic of the raw receiving-side gradient, not a separate finding.

# D Metropolitan-Scale Analysis

This appendix documents the metropolitan system that Section 6 places beside the 48-state analysis. It describes how the node set is built and how representative it is, how the network behaves, and how the receiving-side result depends on the geographic controls and on the removal of the national factor. The sequence of fixed-effect designs, the design variations and the allocation constructions are in Online Appendix Sections OA.4 and OA.5.

## D.1 Node set, covariates and exposure

The metropolitan system is the 100 largest core-based statistical areas of the contiguous United States by Census 2000 population, the counterpart of the 48-state rule. Ranking precedes every screen. A CBSA whose FMHPI series carries a flagged growth month over the estimation span is reported and kept rather than replaced by a smaller market, so the node set is a function of size alone. The price series are the seasonally adjusted Freddie Mac CBSA indexes, transformed exactly as in equation (1). Two coding changes between the 2020 and 2023 CBSA delineations (Cleveland, Poughkeepsie) are reconciled before matching, and metropolitan areas that were split or merged between the 1999 metropolitan definitions used by the Wharton survey and the elasticities of Saiz (2010) and the current CBSAs inherit those measures from every 1999 area of whose 2010 housing stock they hold at least ten percent.

Table D.1 reports the two checks the top-100 rule requires. The average of member growth weighted by 2010 housing stock tracks the national FMHPI with a correlation of 0.994 over 1976–2025, the span the networks use, and a terminal index ratio of 0.973. The fit is closer from 1991 (0.998) than over 1976–1990 (0.984), and weights built from the 1980 or 1990 housing counts leave the early correlation at 0.981 and 0.985. The 100 areas hold 63 percent of the 2010 national housing stock. The system is a large-market system, not a national one, and its connectedness measures are read cross-sectionally for that reason. The selection tests show what the rule selects on. The 100 are more regulated than the other contiguous-US CBSAs, by 0.34 index points or 0.43 of the CBSA-level standard deviation of WRLURI 2006 (0.78 across the 275 CBSAs with the measure), and less elastic, and they are no different in terrain. Selection runs in the direction of the regressor of interest and is stated rather than assumed.

The metropolitan WRLURI 2006 is the equal-jurisdiction mean over the responding jurisdictions of the 1999 metropolitan areas mapped to each CBSA, the rule used for the states (Appendix A.4). The metropolitan values the index authors publish, as carried in the Saiz (2010) file, instead weight jurisdictions by the survey's metropolitan sampling weights. The two constructions correlate at 0.987 across the 256 areas in both files. Terrain is the steep land share over the CBSA polygon, from the same source as the state measure. Saiz's elasticity and unavailable-land share are attached where they exist and, because the elasticity is constructed from regulation and unavailable land, the elasticity is never entered beside either. Each area is assigned a coastal, sunbelt or interior belt. It is coastal if its internal point lies within 100 kilometers of the ocean coastline, otherwise sunbelt

if south of 37°N, and otherwise interior. Migration exposure is built from IRS county-to-county flows aggregated to CBSA pairs with the paper's $t-15$ to $t-11$ pool. Unlike the state matrices, the metropolitan flow matrices hold movers only, so an area's diagonal counts moves between its own counties, its external openness averages about 70 percent, and the common-origin term of equation (8) is 39 percent of external exposure (Online Appendix Table OA.33). Each area is assigned the *principal state* holding the largest share of its 2010 housing units (Washington, D.C., whose largest share lies in Virginia, is the one multistate area not assigned to the first state in its title). The assignment is fixed across years, although it postdates the windows ending before 2010 and falls inside those ending 2010 to 2019, and it is available in all 21 annual networks from 2005. The 100-area system is also open. Only 61 percent of migrating exemptions have both ends inside it, against essentially all for the 48 states, so normalizing flows within the system conditions on staying inside it. Online Appendix Table OA.24 reports the alternative that routes the remainder through an absorbing node.

**Table D.1:** Representativeness and selection of the metropolitan node set

| *Panel A. Tracking of the national index* | 1976–2025 | 1976–1990 | 1991–2025 |
|---|---|---|---|
| Top 100 | | | |
| Correlation with national index growth | 0.994 | 0.984 | 0.998 |
| Slope on national growth | 0.955 | 0.917 | 0.970 |
| Terminal index ratio | 0.973 | 0.998 | 0.975 |
| Maximum absolute index gap (percent) | 3.9 | 2.6 | 3.7 |
| All 383 CBSAs | | | |
| Correlation with national index growth | 0.991 | 0.980 | 0.997 |
| Slope on national growth | 0.902 | 0.859 | 0.924 |
| Terminal index ratio | 0.902 | 0.938 | 0.961 |
| Maximum absolute index gap (percent) | 11.3 | 6.2 | 5.4 |
| *Panel B. Selection on supply constraints* | Top-100 mean | Complement mean | Difference |
| WRLURI 2006 | 0.087 | −0.251 | 0.338*** |
| Saiz supply elasticity | 1.897 | 2.872 | −0.974*** |
| Saiz unavailable-land share | 0.260 | 0.247 | 0.013 |
| Steep land share (Lutz–Sand) | 0.156 | 0.178 | −0.022 |
| log housing units 2010 | 13.246 | 11.286 | 1.961*** |
| *Panel C. Belts.* Coastal 40, sunbelt 28 and interior 32 metropolitan areas | | | |

*Note:* The node set is the 100 largest CBSAs of the contiguous United States by Census 2000 population, ranked before any screen. Three of them (Raleigh, Jackson MS, McAllen) carry one or two flagged growth months over the estimation span and are kept. The top 100 hold 63 percent of the 2010 national housing stock. Panel A compares the housing-stock-weighted average of member growth with the FMHPI national index, both in monthly log growth of the seasonally adjusted indexes, over the 600 months from January 1976 to December 2025, and separately over the 180 months to December 1990 and the 420 months from January 1991. The weights are 2010 housing units in every period, and the comparison index for all 383 contiguous-US CBSAs weights the 377 that carry a 2010 housing stock. The slope is the coefficient of weighted on national growth. The terminal index ratio divides the two cumulated indexes, set to 100 in the month before each period starts, at the period's last month, and the gap is their largest absolute percent difference within the period. With 1980 or 1990 housing units as weights, the 1976–1990 correlation of the top 100 is 0.981 or 0.985 and the terminal index ratio 1.026 or 1.018. Panel B reports Welch two-sample tests of the top-100 mean against the 175–283 other contiguous-US CBSAs with the measure, and stars on the difference use the Welch $p$-value. WRLURI 2006 is in index points. A CBSA's value is the unweighted mean over the responding jurisdictions of every 1999 metropolitan area with at least 10 percent of its 2010 housing stock in the CBSA, the equal-jurisdiction rule used for the states. Across the 275 contiguous-US CBSAs with the measure its standard deviation is 0.78, so the WRLURI difference is 0.43 of a CBSA-level standard deviation. The coastal belt holds the areas whose CBSA internal point lies within 100 km of the ocean coastline, the sunbelt the remaining areas south of 37°N, and the interior belt the rest. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

## D.2 Network estimation and diagnostics

The metropolitan network uses the estimator of Section 2—a rolling elastic-net VAR whose moving-average representation and generalized forecast-error variance decomposition at horizon 12 give the row-normalized connectedness matrix—at the same 41 annual networks and 120-month window. The metropolitan specification chooses the penalty at the minimum of hv-block cross-validation with a buffer around each validation block, on a path floored at $10^{-4}$, the design the state benchmark uses. The two scales share the lag order, window, mixing weight, penalty rule, path floor and fold design, so a difference between them is a difference of node set rather than of estimator. With two lags of all 100 series the design has 200 regressors on 120 observations in every equation, so the elastic net is what makes the system computable and the cross-area block is heavily shrunk. Table D.2 reports the diagnostics that matter for that reading. All 41 annual networks are dynamically stable, and in the 21 from 2005, the years for which the equation diagnostics are computed, the median absolute *in-sample* residual lag-one autocorrelation of 0.33 is below the 0.38 the state network records for the same in-sample object under the same validation design, about 27 cross-area slopes per equation survive shrinkage against 25.4 cross-state slopes at state scale, and one equation of the 2,100 sits at the penalty-path floor. Both autocorrelation figures are in-sample fitted-residual statistics. Neither is comparable with the median absolute *held-out* lag-one residual autocorrelation of the state network, 0.60 (Table B.1), which is assembled from the validation block that did not train on the row and is larger for that reason. Averaged over the 21 annual networks from 2005, systemwide connectedness is 93.9 percent, higher than the 85.3 percent of the state network on the same annual basis, because a row-normalized decomposition assigns more of each node's variance to the other 99 nodes than to the other 47. Levels are therefore only partly comparable across system sizes. Online Appendix Section OA.1.4 reports the same two indexes on the monthly basis, where they average 91.2 and 78.2. The windows from 2005 sit higher at both scales. The same table reports the factor-adjusted metropolitan network. It is stable in all 41 annual networks and keeps about 23 cross-area slopes per equation; its systemwide connectedness averages 86.1 percent against 91.2 raw, and the cross-sectional standard deviation of external connectedness over 2005–2025 more than doubles, from 3.1 to 7.0 points.

Appendix B.3 reports the state network estimated without shrinkage on the own lags as a robustness check on the benchmark. That check does not survive the move to 100 nodes, and the reason is worth recording. Re-estimated on the same 486 monthly windows with the own-lag penalty switched off, the metropolitan system has a companion eigenvalue outside the unit circle at 188 of them against 4 for the benchmark, and in 16 of the 41 annual networks, 10 of the 21 from 2005, against none. Its median companion radius rises from 0.938 to 0.977 and its maximum from 1.058 to 1.184, and the number of cross-area slopes the penalty retains falls from 26.4 to 12.4 per equation. The explosive windows are concentrated in 2001–2011 and 2021–2025, so the monthly index is not interpretable over exactly the stretches the paper reads. With 200 regressors on 120 observations in every equation, shrinking the own lags along with the rest is what keeps the metropolitan system

estimable, and the unshrunk network is reported here as a diagnostic rather than carried into the second stage.

**Table D.2:** Metropolitan network diagnostics, raw and factor-adjusted annual networks

| | Raw network | | Factor-adjusted network | |
|---|---|---|---|---|
| | 1985–2025 | 2005–2025 | 1985–2025 | 2005–2025 |
| Stable annual networks | 41 of 41 | 21 of 21 | 41 of 41 | 21 of 21 |
| Maximum companion-matrix radius | 0.973 | 0.970 | 0.951 | 0.951 |
| Mean systemwide connectedness (DYCI, pp) | 91.2 | 93.9 | 86.1 | 85.6 |
| Mean active cross-metro slopes per equation (of 198) | 26.4 | 27.4 | 23.2 | 23.0 |
| Cross-sectional s.d. of external connectedness (pp) | 4.6 | 3.1 | 6.9 | 7.0 |
| Median absolute in-sample residual lag-1 autocorrelation | – | 0.328 | 0.343 | 0.342 |
| Share of equations rejecting Ljung–Box(12) at 5 percent | – | 0.98 | 0.95 | 0.96 |
| Mean active own slopes per equation (of 2) | – | 1.28 | 1.28 | 1.29 |
| Equations at the penalty-path floor | – | 1 of 2,100 | 0 of 4,100 | 0 of 2,100 |

*Note:* Columns 1–2 use the raw 100-metro benchmark network and columns 3–4 the factor-adjusted one. The factor-adjusted network is re-estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Entries average over the 41 annual networks 1985–2025 and the 21 from 2005 of the 100-metro elastic-net VAR with 2 cross lags, 2 own lags, own-penalty factor 1, $\alpha = 0.5$, 120 response months, penalty chosen at the minimum of purged hv-block validation (2-month purge), penalty-path floor $10^{-4}$, forecast horizon 12, row-normalized GFEVD. The design has 200 regressors on 120 observations in every equation. The cross-sectional s.d. row averages over the annual networks the standard deviation of external connectedness (FROM) across the 100 areas, computed from the networks themselves. The equation-level residual and penalty diagnostics of the raw network were retained only in the annual networks from 2005, so its 1985–2025 column leaves the last four rows blank, while the factor-adjusted run kept them in all 41. The autocorrelation row averages the medians of the annual networks; its $1.96/\sqrt{120}$ band is 0.179. It is an in-sample fitted-residual statistic, not the held-out quantity reported for the state network in Table B.1.

## D.3 External connectedness across fixed-effect structures

Table D.3 reports the metropolitan receiving-side regressions summarized in Section 6.2, in the raw and the factor-adjusted networks. Without the two population controls, a one-standard-deviation increase in metropolitan regulation is associated with 1.3 percent more external connectedness (0.3) in the raw network, 0.9 (0.3) with them, 0.5 (0.1) within states, and 0.5 (0.1) with principal-state-by-year effects, with a wild-cluster $p$-value of 0.009 over the twelve principal states. After the purge the four columns give 0.96, 0.71, 0.35 and 0.34, with standard errors roughly twice as large and wild-cluster $p$-values of 0.31 and 0.33 for the within-state columns; the paired differences from the raw estimates, $-0.17$ to $-0.35$, are not distinguishable from zero.

Two features of the metropolitan comparison need stating. The first is that the two scales do not condition on quite the same controls. Both carry lagged log population as the size control, and the division and state effects do at this scale what average distance does across 48 states. Coefficients at the two scales can be placed on a common footing by dividing each by the within-year cross-sectional standard deviation of log external connectedness in its own node set, 0.122 across the 48 states and 0.054 across the 100 metropolitan areas. On that scale, in the raw networks, a standard deviation of regulation moves a state 0.23 standard deviations of the state distribution and a metropolitan

area 0.16 with the population controls and 0.10 within states, so the metropolitan gradient is about 0.4 to 0.7 of the state one. The comparison is approximate. The regressor is standardized within each node set, and the top-100 node set is more regulated than the other CBSAs (Table D.1).

The second is a control the state panel does not need. The 48 states cover the country and the 100 metropolitan areas do not, and an area whose neighbors lie mostly outside the node set has less of the estimated system to be connected to. The regressions therefore carry the share of the population within 200 kilometers that lives in counties belonging to no sample area, measured in the calendar year before each network's year. It enters negatively, as that reading implies, at $-1.6$ percent per standard deviation with division effects and $-0.7$ within states in the raw network, and entering it with population lowers the regulation association from 1.3 to 0.9 percent under division effects. Online Appendix Table OA.18 shows that neither conclusion depends on the 200-kilometer radius. At 100 and 300 kilometers the raw regulation association is 1.11 and 0.97 percent with division effects and 0.55 and 0.48 within states.

**Table D.3:** Regulation and external connectedness across metropolitan areas, raw and factor-adjusted networks, 1985–2025

| Dependent variable | log external connectedness (FROM) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| *Panel A. Raw network* | | | | |
| WRLURI 2006 | 1.31*** | 0.88** | 0.52*** | 0.51*** |
| | (0.30) | (0.31) | (0.14) | (0.15) |
| Steep land share | 0.24 | 0.18 | −0.23 | −0.23 |
| | (0.47) | (0.43) | (0.16) | (0.18) |
| Lagged log population | | −0.05 | 0.07 | 0.10 |
| | | (0.26) | (0.13) | (0.13) |
| Outside-node-set share | | −1.60*** | −0.71*** | −0.70** |
| | | (0.31) | (0.23) | (0.27) |
| Wild-cluster $p$, principal states | – | – | 0.009 | 0.009 |
| Within $R^2$ | 0.033 | 0.075 | 0.047 | 0.112 |
| *Panel B. Factor-adjusted network* | | | | |
| WRLURI 2006 | 0.96 | 0.71 | 0.35 | 0.34 |
| | (0.61) | (0.65) | (0.29) | (0.32) |
| Steep land share | 0.11 | 0.01 | 0.06 | 0.06 |
| | (0.72) | (0.71) | (0.23) | (0.25) |
| Lagged log population | | −0.91 | 0.24 | 0.25 |
| | | (0.61) | (0.23) | (0.25) |
| Outside-node-set share | | −1.74** | −0.10 | −0.16 |
| | | (0.57) | (0.23) | (0.26) |
| Wild-cluster $p$, principal states | – | – | 0.309 | 0.327 |
| Within $R^2$ | 0.008 | 0.033 | 0.008 | 0.019 |
| WRLURI 2006, factor-adjusted minus raw | −0.35 | −0.17 | −0.17 | −0.17 |
| | (0.45) | (0.45) | (0.26) | (0.28) |
| Year fixed effects | ✓ | ✓ | ✓ | – |
| Census-division fixed effects | ✓ | ✓ | – | – |
| Principal-state fixed effects | – | – | ✓ | – |
| Principal-state-by-year fixed effects | – | – | – | ✓ |
| Sample | All | All | States with $\geq 3$ metros | States with $\geq 3$ metros |
| Metropolitan areas | 100 | 100 | 64 | 64 |
| Metro-years | 4,100 | 4,100 | 2,624 | 2,624 |

*Note:* The 100 largest metropolitan areas of the contiguous United States, 41 annual networks 1985–2025. Panel A uses the raw metropolitan benchmark network and Panel B the factor-adjusted one. The factor-adjusted network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). The dependent variable is the log of the area's external connectedness (FROM). Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome; on a logged control, it is 100 times the elasticity. An area's principal state is the state holding the largest share of its 2010 housing units; columns 3 and 4 use the 64 areas in the 12 principal states with at least three sample areas, and the principal-state-by-year effects of column 4 also absorb each state's shocks in every year. Across the 100 areas, 77 percent of the variance of WRLURI and 34 percent (raw) and 44 percent (factor-adjusted) of that of log external connectedness (net of year means) lie between principal states. WRLURI and the steep land share are standardized within the node set. Population is the CBSA's county population in the calendar year before the year of each annual network. The outside-node-set share is the standardized share of the population within 200 km of the area that lives in counties belonging to no sample area. External connectedness averages 91.16 percentage points in the raw network (92.83 in columns 3 and 4) and 86.15 in the factor-adjusted network (88.97). Parentheses contain cluster-robust standard errors, clustered by metropolitan area. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-$t$ $p$-values that impose the null (9,999 draws). The principal-state row gives the same test with the 12 principal states as clusters (Webb weights, 9,999 draws), allowing arbitrary dependence among the areas of a state. The 12 principal-state clusters differ in size. The areas of columns 3 and 4 are distributed as California 11, Florida 8, Ohio 7, New York 6, Pennsylvania 6, Texas 6, North Carolina 4, Tennessee 4, Connecticut 3, Massachusetts 3, South Carolina 3 and Virginia 3. With one principal state left out at a time, which leaves 11 clusters, the column 4 WRLURI coefficient ranges from 0.38 (without New York) to 0.58 (without North Carolina) in the raw network, with principal-state wild $p$-values from 0.005 to 0.069, and from 0.10 (without Virginia) to 0.58 (without Tennessee) in the factor-adjusted network, with principal-state wild $p$-values from 0.168 to 0.639. The difference row is the factor-adjusted minus the raw WRLURI coefficient from one regression stacking both networks, clustered by area. The belt-effect columns and the other fixed-effect structures of the raw network are in Online Appendix Table OA.17.

# E Spatial-Equilibrium Derivations

The connectedness estimator does not impose a migration or distance matrix on the dynamics of house-price growth. A theory-based representation therefore requires a separate object that has the same directed source–target dimension. The location-choice model in Schubert (2024) provides one. It maps household movements into a row-stochastic migration exposure matrix and then into equilibrium price weights. This appendix derives that mapping, establishes the relevant cross-pair and target-side comparative statics, and shows how the direct weights aggregate to target, source, and system objects. The table lists the symbols of the model, each with one meaning, and the four empirical objects that correspond to them.

| Symbol | Meaning |
|---|---|
| $A_{it}$ | Amenity level of location $i$. |
| $a_{it}$ | Log change in the amenity level. |
| $E_{it}$ | Wage in location $i$. |
| $e_{it}$ | Log change in the wage. |
| $P_{it}$ | House price, the cost of housing in location $i$. |
| $p_{it}$ | Log change in the house price. |
| $L_{it}$ | Population of location $i$. |
| $l_{it}$ | Log change in population. |
| $\alpha$ | Housing expenditure share. |
| $\nu$ | Dispersion of the idiosyncratic location taste. |
| $\kappa_i$ | Housing-supply shifter. |
| $\eta_i^H$ | Inverse housing-supply elasticity of location $i$. |
| $\mu_t^{k\to j}$ | Share of origin-$k$ households choosing destination $j$. |
| $\varphi_t^{i\leftarrow k}$ | Share of target $i$'s feeder population supplied by origin $k$. |
| $\psi_{ij,t}$ | Migration exposure of target $i$ to source $j$; $\psi_{ij,t}^0$ is the pre-window matrix of the empirical design. |
| $o_{i,t}$ | External migration openness, $1-\psi_{ii,t}$. |
| $c$ | The ratio $\alpha/\nu$. |
| $a_i$ | The constant $c\,\eta_i^H$ of target $i$. |
| $\chi_{i,t}$ | The product $a_i o_{i,t}$. |
| $\rho_{i,t}$ | Supply multiplier $a_i/(1+a_i o_{i,t})$. |
| $\eta_{i,t}^{nw}$ | Target $i$'s total external weight $\chi_{i,t}/(1+\chi_{i,t})$. |
| $W_{ij,t}$ | Direct equilibrium price weight from source $j$ to target $i$. |
| $Q_{ki,t}$ | Pooled IRS migration flow from origin $k$ to destination $i$. |
| $F_{i,t}$ | Measured external connectedness of target $i$. |
| $\bar{C}_{ij,t}$ | Conditional connectedness share of source $j$ in target $i$'s row. |
| $x_{ij,t}$ | Conditional migration exposure, the centered regressor of Section 5. |

## E.1 Household location choice and spatial equilibrium

Let $i, j, k \in \{1, \ldots, N\}$ index locations. A worker $\omega$ residing in location $i$ receives indirect utility

$$V_{it}(\omega) = \log A_{it} + \log E_{it} - \alpha \log P_{it} + z_i(\omega) \equiv U_{it} + z_i(\omega), \tag{E.1}$$

where $A_{it}$ is the amenity level, $E_{it}$ the wage, $P_{it}$ the cost of housing or local nontradables, and $\alpha > 0$ the housing expenditure weight. The idiosyncratic taste $z_i(\omega)$ follows a Type-I extreme-value distribution with dispersion $\nu > 0$.

A worker initially in $k$ chooses the destination that maximizes $U_{it} - \tau^{ki} + z_i(\omega)$, where $\tau^{ki}$ is an additive, time-invariant cost of moving from $k$ to $i$. The resulting share of workers from origin $k$ who choose destination $i$ is

$$\mu_t^{k \to i} = \frac{\exp\Big[(U_{it} - \tau^{ki})/\nu\Big]}{\sum_{r=1}^{N} \exp[(U_{rt} - \tau^{kr})/\nu]}. \tag{E.2}$$

The superscript lists the origin first and the destination second. The inverse dispersion $\nu^{-1}$ determines the sensitivity of destination shares to common utility differences. Persistent bilateral moving costs are therefore consistent with a heterogeneous migration network even when local fundamentals change over time.

The inverse supply curve links housing prices to population:

$$P_{it} = \kappa_i L_{it}^{\eta_i^H} \exp(\varepsilon_{it}^H), \tag{E.3}$$

where $L_{it}$ is population, $\kappa_i$ is a time-invariant supply shifter, $\varepsilon_{it}^H$ is a local supply disturbance, and $\eta_i^H > 0$ is the inverse housing-supply elasticity. A larger $\eta_i^H$ denotes a steeper inverse relation between population and price. Destination populations satisfy

$$L_{it} = \sum_{k=1}^{N} \mu_t^{k \to i} L_{k,t-1}. \tag{E.4}$$

An equilibrium at date $t$ is a pair $(P_t, L_t)$ satisfying equations (E.2)–(E.4), given $A_t$, $E_t$, $L_{t-1}$, $\kappa$, $\varepsilon_t^H$, the bilateral moving costs, and the model parameters.

The model is deliberately spare. Workers are myopic, wages are exogenous to this block, and $P_{it}$ is both the housing-services cost entering utility and the observed house price. There is no separate rent, user-cost, tenure, or mortgage sector. Allowing $\eta_i^H$ to vary across locations implements the heterogeneous-supply extension of Schubert's common-elasticity benchmark.

## E.2 Log-linear reallocation and the migration exposure matrix

Lowercase letters denote log changes. In particular, $p_{it} = \Delta \log P_{it}$, $e_{it} = \Delta \log E_{it}$, and $l_{it} = \Delta \log L_{it}$. Define the change in common location utility as

$$q_{it} = a_{it} + e_{it} - \alpha p_{it}. \tag{E.5}$$

Log differentiation of the destination share gives

$$\Delta \log \mu_t^{k \to i} = \nu^{-1} \left[ q_{it} - \sum_{j=1}^{N} \mu_{t-1}^{k \to j} q_{jt} \right]. \tag{E.6}$$

An origin therefore sends a larger share to $i$ when that destination becomes more attractive relative to the other locations available to the same households.

To aggregate these choices at the target, define the share of destination $i$'s feeder population originating in $k$:

$$\varphi_{t-1}^{i \leftarrow k} = \frac{\mu_{t-1}^{k \to i} L_{k,t-2}}{\sum_{r=1}^{N} \mu_{t-1}^{r \to i} L_{r,t-2}} = \frac{\mu_{t-1}^{k \to i} L_{k,t-2}}{L_{i,t-1}}. \tag{E.7}$$

Log differentiation of population accounting yields

$$l_{it} = \sum_{k=1}^{N} \varphi_{t-1}^{i \leftarrow k} \left( \Delta \log \mu_t^{k \to i} + l_{k,t-1} \right). \tag{E.8}$$

Substituting the change in destination shares and collecting terms gives

$$l_{it} = \Delta M_{i,t-1} + \nu^{-1} \left[ (a_{it} - \widetilde{a}_{it}) + (e_{it} - \widetilde{e}_{it}) - \alpha (p_{it} - \widetilde{p}_{it}) \right], \tag{E.9}$$

where

$$\Delta M_{i,t-1} = \sum_{k=1}^{N} \varphi_{t-1}^{i \leftarrow k} l_{k,t-1} \tag{E.10}$$

captures growth in the populations feeding $i$, and the two-step network average of any log change $v$ is

$$\widetilde{v}_{it} = \sum_{k=1}^{N} \varphi_{t-1}^{i \leftarrow k} \sum_{j=1}^{N} \mu_{t-1}^{k \to j} v_{jt}. \tag{E.11}$$

The remaining terms in equation (E.9) compare amenities, wages, and housing costs in target $i$ with the alternatives considered by its feeder origins.

The two-step average defines the migration exposure matrix:

$$\psi_{ij,t-1} = \sum_{k=1}^{N} \varphi_{t-1}^{i \leftarrow k} \mu_{t-1}^{k \to j}, \qquad \sum_{j=1}^{N} \psi_{ij,t-1} = 1. \tag{E.12}$$

Exposure from source $j$ to target $i$ is large when origins that contribute heavily to $i$ also assign a large destination share to $j$. It therefore measures competition for common migrant pools rather than only the direct flow from $j$ to $i$. In the state data, with stayers on the diagonal, the direct terms of equation (8) dominate, and the common-origin term is a small share of external exposure. Using this definition, the two-step average becomes

$$\widetilde{v}_{it} = \sum_{j=1}^{N} \psi_{ij,t-1} v_{jt}. \tag{E.13}$$

The diagonal $\psi_{ii}$ is own exposure. Its complement $o_i \equiv 1 - \psi_{ii}$ is the target's external migration openness. For $j \neq i$, define the composition of that external exposure and the corresponding network house-price change as

$$\widetilde{\psi}_{ij,t-1} = \frac{\psi_{ij,t-1}}{1 - \psi_{ii,t-1}}, \qquad \sum_{j \neq i} \widetilde{\psi}_{ij,t-1} = 1, \tag{E.14}$$

$$p_{it}^{nw} = \sum_{j \neq i} \widetilde{\psi}_{ij,t-1} p_{jt}. \tag{E.15}$$

The conditional weights $\widetilde{\psi}_{ij}$ are useful for deriving the external total. The diagonal-inclusive $\psi_{ij}$ underlies openness, and the main conditional-allocation estimand uses $\widetilde{\psi}_{ij}$.

### E.3 Collecting the own-price terms

Log differentiation of inverse housing supply gives

$$p_{it} = \eta_i^H l_{it} + \Delta\varepsilon_{it}^H. \tag{E.16}$$

Substituting population reallocation, applying the exposure average, and separating target $i$'s own price gives

$$\begin{aligned} p_{it} = {} & \eta_i^H \Delta M_{i,t-1} + \eta_i^H \nu^{-1} \left[ (a_{it} - \widetilde{a}_{it}) + (e_{it} - \widetilde{e}_{it}) \right] \\ & - \alpha \eta_i^H \nu^{-1} p_{it} + \alpha \eta_i^H \nu^{-1} \psi_{ii,t-1} p_{it} + \alpha \eta_i^H \nu^{-1} \sum_{j \neq i} \psi_{ij,t-1} p_{jt} + \Delta\varepsilon_{it}^H. \end{aligned} \tag{E.17}$$

Moving both own-price terms to the left gives

$$\begin{aligned} & \left[ 1 + \alpha \nu^{-1} \eta_i^H (1 - \psi_{ii,t-1}) \right] p_{it} \\ & \quad = \alpha \nu^{-1} \eta_i^H \sum_{j \neq i} \psi_{ij,t-1} p_{jt} + \eta_i^H \Delta M_{i,t-1} \\ & \qquad + \eta_i^H \nu^{-1} \left[ (a_{it} - \widetilde{a}_{it}) + (e_{it} - \widetilde{e}_{it}) \right] + \Delta\varepsilon_{it}^H. \end{aligned} \tag{E.18}$$

Define

$$\zeta_{i,t-1} = \frac{\eta_i^H}{1 + \alpha \nu^{-1} \eta_i^H (1 - \psi_{ii,t-1})}. \tag{E.19}$$

The time-varying solution within this log-linearized system is then

$$p_{it} = \alpha\nu^{-1}\zeta_{i,t-1}\sum_{j\neq i}\psi_{ij,t-1}p_{jt} + \xi_{it}, \tag{E.20}$$

with non-network component

$$\xi_{it} = \zeta_{i,t-1}\Delta M_{i,t-1} + \zeta_{i,t-1}\nu^{-1}\left[(a_{it} - \widetilde{a}_{it}) + (e_{it} - \widetilde{e}_{it})\right] + \frac{\zeta_{i,t-1}}{\eta_i^H}\Delta\varepsilon_{it}^H. \tag{E.21}$$

The empirical implementation associates each annual network $t$ with an exposure matrix $\Psi_t^0$ measured before its house-price window. The matrix is fixed within that window but advances from one year's network to the next. Define

$$\zeta_{i,t}^0 = \frac{\eta_i^H}{1 + \alpha\nu^{-1}\eta_i^H(1-\psi_{ii,t}^0)}, \qquad \eta_{i,t}^{nw} = \alpha\nu^{-1}\zeta_{i,t}^0(1-\psi_{ii,t}^0). \tag{E.22}$$

Equivalently, target $i$'s total external weight is

$$\eta_{i,t}^{nw} = \frac{\alpha\nu^{-1}\eta_i^H(1-\psi_{ii,t}^0)}{1 + \alpha\nu^{-1}\eta_i^H(1-\psi_{ii,t}^0)} \in (0,1). \tag{E.23}$$

Using $\widetilde{\psi}_{ij,t}^0 = \psi_{ij,t}^0/(1-\psi_{ii,t}^0)$, the price equation for date $s$ inside window $t$ becomes

$$p_{i,s} = \eta_{i,t}^{nw}\sum_{j\neq i}\widetilde{\psi}_{ij,t}^0 p_{j,s} + \varepsilon_{i,s}^{nw} = \eta_{i,t}^{nw}p_{i,s}^{nw} + \varepsilon_{i,s}^{nw}. \tag{E.24}$$

The residual retains the parts of the exact solution not represented by the predetermined network-price term:

$$\begin{aligned}\varepsilon_{i,s}^{nw} = {} & \underbrace{\zeta_{i,s-1}\Delta M_{i,s-1}}_{\text{migration access}} + \underbrace{\zeta_{i,s-1}\nu^{-1}\left[(a_{i,s} - \widetilde{a}_{i,s}) + (e_{i,s} - \widetilde{e}_{i,s})\right]}_{\text{relative amenities and wages}} \\ & \underbrace{+ \alpha\nu^{-1}\sum_{j\neq i}\left(\zeta_{i,s-1}\psi_{ij,s-1} - \zeta_{i,t}^0\psi_{ij,t}^0\right)p_{j,s}}_{\text{predetermined-weight approximation}} + \underbrace{\frac{\zeta_{i,s-1}}{\eta_i^H}\Delta\varepsilon_{i,s}^H}_{\text{local supply disturbance}}.\end{aligned} \tag{E.25}$$

The approximation term is current minus predetermined because equation (E.24) replaces the exact coefficient $\zeta_{i,s-1}\psi_{ij,s-1}$ with $\zeta_{i,t}^0\psi_{ij,t}^0$.[74]

Schubert's empirical baseline holds a single 1990–1995 matrix fixed across all outcome years. The algebra requires only that the weights be determined before the outcome window they summarize. The present design implements that timing with 21 advancing pre-window matrices, as described in Section 5.1.

[74]The corresponding approximation in Schubert (2024) presents the difference in the reverse order. Substitution into the preceding derivation yields the current-minus-predetermined order used here.

Let $\chi_{i,t} = \alpha\nu^{-1}\eta_i^H(1-\psi_{ii,t}^0)$. Because $\eta_{i,t}^{nw} = \chi_{i,t}/(1+\chi_{i,t})$,

$$\frac{\partial \eta_{i,t}^{nw}}{\partial(1-\psi_{ii,t}^0)} = \frac{\alpha\nu^{-1}\eta_i^H}{(1+\chi_{i,t})^2} > 0, \qquad \frac{\partial \eta_{i,t}^{nw}}{\partial \eta_i^H} = \frac{\alpha\nu^{-1}(1-\psi_{ii,t}^0)}{(1+\chi_{i,t})^2} > 0. \tag{E.26}$$

The algebraic external total is larger for a more migration-open target and a target with less elastic housing supply. It approaches zero when either openness or the inverse supply elasticity approaches zero, and it approaches one as $\eta_i^H$ or $\nu^{-1}$ becomes arbitrarily large.

## E.4 Model relationships used for the empirical comparison

Set $c \equiv \alpha\nu^{-1}$, $a_i \equiv c\eta_i^H$, and $o_{i,t} \equiv 1-\psi_{ii,t}^0$. For a particular external link, the price equation implies

$$W_{ij,t} = \frac{a_i o_{i,t}}{1+a_i o_{i,t}} \frac{\psi_{ij,t}^0}{o_{i,t}} = \underbrace{\frac{a_i}{1+a_i o_{i,t}}}_{\rho_{i,t}} \psi_{ij,t}^0, \qquad j \neq i.$$

The first representation separates the external total from the composition of the external row. The second uses the diagonal-inclusive exposure retained for openness.

Within a target row, openness $o_{i,t}$ is fixed across sources. Holding it fixed while exposure is reallocated among external links gives

$$\left.\frac{\partial W_{ij,t}}{\partial \psi_{ij,t}^0}\right|_{o_{i,t}} = \rho_{i,t} > 0, \qquad \left.\frac{\partial W_{ij,t}}{\partial \eta_i^H}\right|_{\psi_{ij,t}^0, o_{i,t}} = \frac{c\psi_{ij,t}^0}{(1+a_i o_{i,t})^2} > 0, \qquad \frac{\partial \rho_{i,t}}{\partial \eta_i^H} = \frac{c}{(1+a_i o_{i,t})^2} > 0.$$

Greater diagonal-inclusive exposure therefore receives a larger direct weight. The level gradient between exposure and that weight is also steeper when target supply is more constrained. The model log elasticity, by contrast, is one:

$$\left.\frac{\partial \log W_{ij,t}}{\partial \log \psi_{ij,t}^0}\right|_{o_{i,t}} = 1.$$

An alternative row-stochastic comparison adds mass to link $ij$ entirely by reducing the diagonal, so $do_{i,t}/d\psi_{ij,t}^0 = 1$ while the remaining external links stay fixed. Then

$$\frac{dW_{ij,t}}{d\psi_{ij,t}^0} = \frac{a_i\left[1+a_i(o_{i,t}-\psi_{ij,t}^0)\right]}{(1+a_i o_{i,t})^2} > 0.$$

Monotonicity in the diagonal-inclusive link survives, although its slope differs from the fixed-openness comparison.

Openness requires a separate conditioning statement. If one diagonal-inclusive external link is held fixed while openness rises elsewhere in the row,

$$\left.\frac{\partial W_{ij,t}}{\partial o_{i,t}}\right|_{\psi^0_{ij,t}} = -\frac{a_i^2 \psi^0_{ij,t}}{(1+a_i o_{i,t})^2} < 0.$$

If every external link expands proportionately so that its composition $\psi^0_{ij,t}/o_{i,t}$ is fixed,

$$\left.\frac{dW_{ij,t}}{do_{i,t}}\right|_{\psi^0_{ij,t}/o_{i,t}} = \frac{a_i(\psi^0_{ij,t}/o_{i,t})}{(1+a_i o_{i,t})^2} > 0.$$

The unconditional openness implication applies to the target total:

$$\sum_{j\neq i} W_{ij,t} = \frac{a_i o_{i,t}}{1+a_i o_{i,t}} = \eta^{nw}_{i,t}, \qquad \frac{\partial \sum_{j\neq i} W_{ij,t}}{\partial o_{i,t}} = \frac{a_i}{(1+a_i o_{i,t})^2} > 0.$$

Equivalently, its log odds satisfy

$$\log\left(\frac{\sum_{j\neq i} W_{ij,t}}{1-\sum_{j\neq i} W_{ij,t}}\right) = \log a_i + \log o_{i,t}.$$

Lower own exposure corresponds to a larger target external total. Every pairwise weight shares that ordering only along a proportional expansion that holds the composition of external exposure fixed.

The model's comparative statics therefore concern two objects, target external connectedness and conditional source allocation. The GFEVD row total is a reduced-form counterpart to $\eta^{nw}$, and off-diagonal row shares describe the allocation ordering in $\widetilde{\Psi}$. Bilateral levels combine both margins. The paper's sending-side result is not among these objects. A source's total outward weight, equation (E.32), depends on the other markets' constraints and exposures rather than on the source's own constraint, and the model does not derive how a source's constraint moves its share of other markets' forecast-error variance, so the sending-side reading is an informal extension of the model. Dynamics and innovation covariances keep all measured GFEVD objects distinct from the model weights.

## E.5 Microfoundation of the weighting matrix

The choice probabilities make the direction of the exposure weight transparent. For origin $k$, differentiating the probability of choosing destination $r$ with respect to common utility in destination $j$ gives

$$\frac{\partial \mu^{k\to r}_t}{\partial q_{jt}} = \nu^{-1}\mu^{k\to r}_t\left(\mathbf{1}\{r=j\} - \mu^{k\to j}_t\right). \tag{E.27}$$

The derivatives are positive for a destination's own utility and negative for an alternative destination's utility. Aggregating them with the feeder weights in $\Phi_t$ yields

$$\frac{\partial(l_{it} - \Delta M_{i,t-1})}{\partial q_{jt}} = \begin{cases} \nu^{-1}(1 - \psi_{ii,t-1}), & j = i, \\ -\nu^{-1}\psi_{ij,t-1}, & j \neq i. \end{cases} \tag{E.28}$$

Because $q_{jt}$ contains $-\alpha p_{jt}$, the corresponding price derivatives are

$$\frac{\partial(l_{it} - \Delta M_{i,t-1})}{\partial p_{jt}} = \begin{cases} -\alpha\nu^{-1}(1 - \psi_{ii,t-1}), & j = i, \\ \alpha\nu^{-1}\psi_{ij,t-1}, & j \neq i. \end{cases} \tag{E.29}$$

These derivatives encode the model's allocation comparison. The cross-destination term is proportional to $\psi_{ij}$, and the associated inverse-supply factor varies with the target's constraint parameter.

Using $\rho_{i,t}$ from Section E.4, the direct off-diagonal model weight can be written either with conditional external exposure or with the raw empirical exposure:

$$W_{ij,t} = \eta^{nw}_{i,t}\widetilde{\psi}^{0}_{ij,t} = \rho_{i,t}\psi^{0}_{ij,t}, \qquad i \neq j. \tag{E.30}$$

The two expressions use different normalizations. $\widetilde{\psi}^{0}_{ij,t}$ allocates exposure conditional on choosing an external destination, whereas $\psi^{0}_{ij,t}$ is the full-row weight used for openness. The main conditional allocation uses $\widetilde{\psi}^{0}_{ij,t}$. Their scale factors are $\eta^{nw}_{i,t}$ and $\rho_{i,t}$, respectively.

Summing the direct weights by target, source, and system gives

$$\sum_{j \neq i} W_{ij,t} = \eta^{nw}_{i,t}, \tag{E.31}$$

$$\sum_{i \neq j} W_{ij,t} = \sum_{i \neq j} \rho_{i,t}\psi^{0}_{ij,t}, \tag{E.32}$$

$$\frac{1}{N}\sum_{i}\sum_{j \neq i} W_{ij,t} = \frac{1}{N}\sum_{i} \eta^{nw}_{i,t}. \tag{E.33}$$

These accounting identities connect the pairwise weights to the aggregate comparisons discussed above.

## E.6 System weight operator

To describe simultaneous adjustment across all locations, hold amenities and wages fixed and let $\sigma_{i,t}$ be an unexpected local housing-cost disturbance. The equilibrium price relation is

$$\widehat{p}_{i,t} = \rho_{i,t}\sum_{j \neq i} \psi^{0}_{ij,t}\widehat{p}_{j,t} + b_{i,t}\sigma_{i,t}, \qquad b_{i,t} = \frac{1}{1 + \chi_{i,t}} = 1 - \eta^{nw}_{i,t}. \tag{E.34}$$

Define the window-specific weight and disturbance-loading matrices as

$$(W_t)_{ij} = \begin{cases} \rho_{i,t}\psi^{0}_{ij,t}, & i \neq j, \\ 0, & i = j, \end{cases} \qquad D_t = \operatorname{diag}(b_{1,t}, \ldots, b_{N,t}). \tag{E.35}$$

Stacking these equations gives

$$\widehat{\boldsymbol{p}}_t = W_t\widehat{\boldsymbol{p}}_t + D_t\boldsymbol{\sigma}_t, \tag{E.36}$$

and hence

$$\widehat{\boldsymbol{p}}_t = (I - W_t)^{-1} D_t \boldsymbol{\sigma}_t \equiv \mathcal{A}_t \boldsymbol{\sigma}_t. \tag{E.37}$$

Each row of $W_t$ sums to $\eta_{i,t}^{nw} < 1$, so its spectral radius is below one and the inverse has the Neumann-series representation

$$\mathcal{A}_t = (I - W_t)^{-1} D_t = \left( I + W_t + W_t^2 + W_t^3 + \cdots \right) D_t. \tag{E.38}$$

The first-order term contains the direct migration routes. Higher powers collect indirect paths through intermediate markets. If local disturbances have covariance matrix $\Sigma_{\sigma,t}$, equilibrium prices have covariance

$$\Sigma_{p,t} = \mathcal{A}_t \Sigma_{\sigma,t} \mathcal{A}_t'. \tag{E.39}$$

Theoretical exposure and empirical connectedness can share geography without being identical. Migration exposure represents cross-pair allocation, and the regressions evaluate directional comparisons without imposing identity. A generalized variance decomposition additionally depends on moving-average coefficients and innovation covariance. Connectedness can therefore change with fixed migration weights, while indirect paths can generate a large variance contribution from a small direct equilibrium edge: $C_{ij,t} \neq \psi_{ij,t}$ and $C_{ij,t} \neq W_{ij,t}$.

# F Additional Empirical Results

## F.1 Demand proxies

Tables F.1 and F.2 add two measures of demand to the sending-side regressions, labor demand predicted by industry mix and natural amenities. Both are built without the price data, and neither enters the networks. Column (4) of Table F.2 interacts regulation with the labor-demand shock on the sending side, and Online Appendix Table OA.13 splits that interaction at the year's average shock. Table 3 in Section 6.2 reports the interaction on the receiving side.

Predicted labor demand is a shift-share prediction (Bartik, 1991). For market $j$ and the window ending with year $t$,

$$B_{jt} = \sum_k s_{jk,t-10} \left[\log E_{k,t}^{(-j)} - \log E_{k,t-10}^{(-j)}\right],$$

where $s_{jk,t-10}$ is industry $k$'s share of the market's wage bill in the year before the window, and $E_{k,t}^{(-j)}$ is national industry-$k$ wages in year $t$ excluding the market itself.

- States use BEA wages and salaries by industry (SAINC7S and SAINC7N). Metropolitan areas use BEA county earnings by industry (CAINC5S and CAINC5N), summed over each CBSA's counties.
- Industries are the finest non-overlapping level, 73 SIC and 87 NAICS cells. Construction, real estate and the military are excluded, because house prices move them directly.
- Suppressed cells, 0.8–1.6 percent of the state wage bill and 4–5 percent of metropolitan earnings, are filled top-down within the published industry tree.
- Windows that span the change from SIC to NAICS are chained, with SIC growth up to the break, NAICS growth after it and NAICS shares at the break. The state tables overlap in 1998–2001, and the state shocks of the 2002–2007 networks are chained at 1998. The metropolitan SIC series ends in 2000 and the NAICS series begins in 2001, so the metropolitan shocks of the 2001–2010 networks use SIC growth to 2000 and NAICS growth from 2001, and the one-year 2000–2001 link is omitted. The shock of the 2025 network ends in 2024. The chained and all-SIC shocks agree closely for the windows ending in 1999 and 2000 (correlation 0.99) but not for 2001, so the six state annual networks whose windows cross the break (2002–2007) are left out of the shock-magnitude column; including them gives 7.23. The state interaction of Section 6.1 includes them.

The labor-demand shock is predicted labor demand relative to the year's average across markets, in units of the within-year standard deviation averaged over the classification era $e$,

$$S_{jt} = (B_{jt} - \bar{B}_t)/\bar{\sigma}_{e(t)}.$$

Table 3 and Online Appendix Table OA.10 use instead the prediction standardized across markets in each annual network, $(B_{jt} - \bar{B}_t)/\mathrm{sd}_t(B)$, with mean zero and standard deviation one in every window. The shock magnitude is $|S_{jt}|$, divided by its standard deviation within the era. It measures how far an area's long-run relative demand growth, predicted over the ten years of the window, departs from the year's mean, and not the size or the volatility of demand surprises within the window.

The proxy measures labor demand, not an instrument, so no exclusion restriction is involved. It predicts actual growth over the same window in regressions of actual on predicted growth.

- States: slope 1.39 (0.35) on wage-bill growth, $F = 16.2$.
- Metropolitan areas: slope 1.45 (0.26) on earnings growth, $F = 30.9$.

Its magnitude, the shock magnitude defined above, is nearly uncorrelated with the regulation index ($-0.12$ across states, $-0.08$ across metropolitan areas). The standardized prediction itself is somewhat higher in regulated markets. Its correlation with the index averages 0.36 across states and 0.20 across metropolitan areas within a window, and ranges from 0.06 to 0.62 and from $-0.06$ to 0.38.

Natural amenities are the USDA scale of McGranahan (1999), a county index of winter temperature, winter sun, summer temperature, summer humidity, topographic variation and water area. It is aggregated with 1990 population weights and standardized within each sample. It correlates 0.33 with the regulation index across states and 0.45 across metropolitan areas, where the highest-amenity areas are coastal California. It also correlates with the slope share (0.52 across states, 0.61 across metropolitan areas), which shares its topographic component, so the amenity columns replace the steep land share rather than add to it.

Across states, neither measure moves the factor-adjusted sending side (Table F.1). The association is 7.2 percent with the shock magnitude held fixed and 7.9 with amenities held fixed. In the raw network, amenities predict TO strongly, 19.5 percent (6.0), and the regulation association rises to 12.2 (3.1). The amenity component of the sending side therefore runs through the national cycle.

Across metropolitan areas (Table F.2), amenities predict factor-adjusted TO, 14.9 percent (4.3), and holding them fixed lowers the regulation coefficient from 4.6 to 3.2 (columns 1 and 3).

Column (4) interacts regulation with the labor-demand shock. The factor-adjusted sending side is steeper where relative labor demand is stronger, by 3.0 percent per standard deviation of each (1.1). That is the ordering a supply reading implies. Table F.2 does not hold fixed the number of same-state sample partners, a property of the node set. With it held fixed the interaction is 2.0 (0.9) (Online Appendix Table OA.15, Panel A). In the raw network the interaction is $-0.8$ (1.1) on total TO, and the regulation gradient of raw TO is 6.6 (2.6) (column 6). With the shock standardized within each window, Online Appendix Table OA.10 reports the sending-side interaction, and Section 7 checks the receiving-side interaction of Table 3, column (1), against industry clusters, amenities, leave-outs and the classification periods. In the specification of Table F.2, the comparison

is across metropolitan areas. With area effects the interaction is 1.0 (1.1, clustered by area), not distinguishable from zero, and across states it is −1.5 (2.5).

Online Appendix Table OA.13 splits the interaction at the year's average shock with the number of same-state sample partners held fixed, and Online Appendix Table OA.10, column (4), splits the regulation and steep land share interactions the same way.

- On total TO it is 2.9 (1.5) above and 1.2 (1.6) below. Their difference, 1.7 (2.5), is not distinguishable from zero, and their sum is 4.0 (1.9). Without the partner control the two are 6.7 and −1.2, a difference of 7.9 (3.2) with an interval of $[1.1, 15.0]$.
- A split at the within-year median gives the same pattern.
- By within-year terciles of the shock, the regulation gradient of total TO is 0.9 (2.0), 5.1 (2.2) and 3.0 (2.4) from the bottom tercile to the top. Without the partner control the total gives 2.4 (2.9), 5.5 (3.5) and 6.6 (4.0). The difference between the top and bottom terciles is not distinguishable from zero in any column (Online Appendix Table OA.16).

Durable housing implies an asymmetry (Glaeser and Gyourko, 2005). A constraint limits new construction when demand rises, but when demand falls the housing stock contracts slowly whatever the regulation. With the partner count held fixed the split does not detect that asymmetry. The shock is also relative. Predicted real growth, deflated by the CPI-U over the same ten years, is negative in 0.6 percent of area-windows, from 9 areas, so the split at the year's average compares above- with below-average growth rather than rising with falling demand (Online Appendix Table OA.16).

**Table F.1:** Sending side with demand proxies across states, factor-adjusted and raw networks, 1985–2025

| Dependent variable: | log TO | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| *Panel A. Factor-adjusted network* | | | | |
| WRLURI 2006 | 7.50*** | 7.50*** | 7.20*** | 7.93*** |
| | (2.32) | (2.31) | (2.29) | (2.25) |
| Shock magnitude | | | −2.27 | |
| | | | (1.52) | |
| Natural-amenity scale | | | | 2.91 |
| | | | | (4.76) |
| Steep land share | −0.71 | 0.24 | 0.30 | |
| | (2.39) | (2.31) | (2.16) | |
| log average distance | 18.69 | 19.78 | 20.06 | 4.89 |
| | (12.94) | (12.30) | (11.87) | (23.10) |
| Lagged log population | 3.29 | 2.46 | 2.39 | 2.72 |
| | (2.29) | (2.20) | (2.10) | (2.70) |
| State-years | 1,968 | 1,680 | 1,680 | 1,968 |
| Within $R^2$ | 0.072 | 0.074 | 0.077 | 0.073 |
| *Panel B. Raw network* | | | | |
| WRLURI 2006 | 10.54** | 9.93** | 9.20** | 12.22*** |
| | (3.62) | (3.92) | (3.81) | (3.08) |
| Shock magnitude | | | −5.64 | |
| | | | (3.43) | |
| Natural-amenity scale | | | | 19.48** |
| | | | | (6.04) |
| Steep land share | 1.19 | 2.10 | 2.26 | |
| | (3.43) | (3.63) | (3.68) | |
| log average distance | 19.12 | 20.72 | 21.40 | −56.27* |
| | (16.96) | (17.67) | (17.58) | (27.57) |
| Lagged log population | 12.06** | 11.33** | 11.14** | 7.87 |
| | (3.92) | (4.22) | (4.10) | (4.12) |
| State-years | 1,968 | 1,680 | 1,680 | 1,968 |
| Within $R^2$ | 0.097 | 0.086 | 0.094 | 0.127 |
| Year fixed effects | ✓ | ✓ | ✓ | ✓ |
| Annual networks | 41 | 35 | 35 | 41 |

*Note:* 48 states. Panel A uses the factor-adjusted benchmark network and Panel B the raw benchmark network. The factor-adjusted benchmark network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Column (1) is Table 2, column (4) in Panel A and column (1) in Panel B. Column (2) is the same regression on the 35 annual networks outside the windows that cross the SIC–NAICS change (2002–2007), whose shocks are chained. Column (3) adds the shock magnitude $|S_{jT}|$, the absolute deviation of predicted ten-year labor-demand growth from the year's average across states, in units of its standard deviation within the classification era (Appendix F.1). Column (4) replaces the steep land share with the natural-amenity scale. The first-stage $F$ statistic of predicted labor demand for wage-bill growth is 16.2. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome. On a logged control, an entry is 100 times the elasticity. Parentheses contain cluster-robust standard errors, clustered by state. *, ** and *** denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-$t$ $p$-values that impose the null (9,999 draws).

**Table F.2:** Metropolitan sending side with demand proxies, factor-adjusted and raw networks, 1985–2025

| Dependent variable: log TO | Factor-adjusted network | | | | | Raw network |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| WRLURI 2006 | 4.55 | 4.67 | 3.23 | 4.67 | 3.30 | 6.58** |
| | (2.97) | (2.94) | (2.75) | (2.92) | (2.69) | (2.60) |
| Shock magnitude | | 0.87 | | | | |
| | | (1.28) | | | | |
| Natural-amenity scale | | | 14.87*** | | 15.10*** | |
| | | | (4.28) | | (4.40) | |
| Labor-demand shock | | | | −0.55 | −1.12 | |
| | | | | (1.23) | (1.23) | |
| WRLURI × shock | | | | 3.02** | 2.94* | |
| | | | | (1.13) | (1.39) | |
| Natural-amenity scale × shock | | | | | −0.49 | |
| | | | | | (1.85) | |
| Steep land share | −1.06 | −1.03 | | −0.97 | | 1.29 |
| | (3.48) | (3.47) | | (3.47) | | (3.29) |
| Lagged log population | −4.39 | −4.40 | −3.26 | −4.56 | −3.19 | 0.48 |
| | (2.78) | (2.77) | (2.73) | (2.81) | (2.73) | (2.74) |
| Outside-node-set share | −13.53*** | −13.41*** | −8.18* | −13.78*** | −8.39** | −13.29*** |
| | (3.28) | (3.24) | (3.81) | (3.22) | (3.72) | (3.49) |
| Within $R^2$ | 0.081 | 0.081 | 0.113 | 0.088 | 0.120 | 0.074 |
| Year fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Census-division fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Note:* The 100 largest metropolitan areas, 41 annual networks 1985–2025, 4,100 area-years in every column. The factor-adjusted benchmark network is estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Column (6) repeats column (1) on the raw network. The 95-percent wild bootstrap intervals of the WRLURI coefficient are $[-2.1, 10.6]$ in column (1) and $[1.0, 11.9]$ in column (6). The labor-demand shock $S$ is an area's predicted labor demand relative to the year's average across areas, in units of its within-year standard deviation, and the shock magnitude is $|S|$ (Appendix F.1). Columns (3) and (5) replace the steep land share with the natural-amenity scale. The first-stage $F$ statistic of predicted labor demand for earnings growth is 30.9. Entries are 100 times the coefficients. Per standard deviation of a standardized regressor, an entry is approximately the percent difference in the outcome. On a logged control, an entry is 100 times the elasticity. Parentheses contain cluster-robust standard errors, clustered by metropolitan area. $^{*}$, $^{**}$ and $^{***}$ denote $p < 0.10$, 0.05 and 0.01, from wild cluster bootstrap-$t$ $p$-values that impose the null (9,999 draws).